Torsional multi-DOF model of the alt-azimuth mount: FEM, modal truncation and coupling

hub model: telescope hanging from the hub of the polar axis

Generated by `gimbal_mdof.latex_report`

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Warning. All shaft geometries, materials, joint stiffnesses, rotor and fork inertias and damping values are **placeholders**. Every frequency, transfer function and figure in this document is a *result from placeholder parameters* and must not be read as a characterisation of the real mount (Section 14).

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1 System, conventions and modelling chain

The model is the second link of the chain of increasingly complex models of the alt-azimuth mount. The rigid two-degree-of-freedom model (`gimbal_2dof`, document `modello_matematico.tex`) is extended with the torsional flexibility of the azimuth and polar (elevation) shafts. The computation has four stages, each implemented in a module of `src/gimbal_mdof/`:

1. element properties from the composition of solid and hollow cylinders, or from CAD data (`geometry.py`, Section 4);
2. a torsional FEM of each axis in isolation, with lumped and consistent mass (`fem.py`, Section 6);
3. reduction of each axis by modal truncation in the sense of Gawronski of the full FEM (rigid mode plus the flexible modes in the band of interest; `modal.py`, Section 10);
4. coupling of the two reduced axes through the kinetic energy of the whole system (`coupled.py`, `modal.py`, Sections 8 and 10).

Frames and angles. The frames and sign conventions of the rigid model are kept unchanged; they are collected in Section 2, together with the frame in which every inertia in `config.yaml` must be entered. SI units are used internally.

Numbering and coordinates. Nodes are numbered from 1; element i joins node i to node $i+1$. Azimuth: nodes $1 \dots 9$, node 1 at the bottom (encoder), torque τ_ψ on node 5, node 9 carries the fork (body 1). Polar: nodes $1 \dots 11$, encoder on node 1, torque τ_θ on node 7 (motor rotor; the stator is on the fork), free end on node 11; elements 2 and 4 run in the bearings housed in the fork; the telescope (body 2) sits on element 3 at the node P. ψ_i is the absolute rotation of azimuth node i about $\hat{k}$; θ_i is the rotation of polar node i relative to the fork about $\hat{J}$, which coincides with its absolute rotation about $\hat{J}$ because the fork only rotates about $\hat{k}$. The azimuth encoder reads ψ_1, the polar encoder (stator on the fork) reads θ_1.

Reduced coordinates. $\boldsymbol{\eta} = [\eta_{\psi 0}, \eta_{\psi k}, \dots, \ \eta_{\theta 0}, \eta_{\theta k}, \dots]^{\mathsf{T}}$, with η_{a0} the rigid rotation of axis a and η_{ak} the amplitudes of the retained flexible modes (Section 10). The full FEM coordinates are the rotations of all nodes, $\mathbf{q}_\psi = [\psi_1, \dots, \psi_9]^{\mathsf{T}}$ and $\mathbf{q}_\theta$ (polar nodes, with P).

2 Reference frames

The frames follow the convention of the De Donatis thesis: an inertial frame and one body frame for each rotating body. The origins of the body frames lie on the rotation axes and *not* at the centres of mass, so that the eccentricity of each centre of mass with respect to its centre of rotation is a separate, explicit quantity. The data are in the section `frames` of `config.yaml`; the frames are built by `gimbal.params.costruisci_terne` and the inertia tensors are transported and rotated by `gimbal.inertia.trasporta` and `gimbal.inertia.ruota`, called by `gimbal_mdof.payload.tensore_da_config`.

Table 1: Frames of the model.

frame	origin	unit vectors	role
N, inertial	O_N, centre of the base of the azimuth shaft	$\hat{x}, \hat{y}, \hat{z}$: axes of the Onshape assembly, $\hat{z}$ vertical upwards along the azimuth axis	reference for all CAD exports
1, body 1 (azimuth, fork)	o_1, on the azimuth axis	$\hat{\imath}$ horizontal projection of the pointing direction, $\hat{k} = \hat{z}$, $\hat{\jmath} = \hat{k} \times \hat{\imath}$ (along the polar axis)	rotates with ψ_L about $\hat{k}$
2, body 2 (polar, telescope)	o_2, centre of rotation of the polar shaft	$\hat{K}$ pointing direction (positive towards the optics), $\hat{I}$ upwards at $\theta = 0$, $\hat{J} = \hat{K} \times \hat{I} = -\hat{\jmath}$ polar axis	rotates with θ_p about $\hat{J}$ relative to body 1

Why $\hat{J} = -\hat{\jmath}$. Body 2 is fixed by three requirements: $\hat{I}$ coincides with $\hat{k} = \hat{z}$ when the telescope is horizontal, $\hat{K}$ points out of the optics, and $\theta > 0$ raises the pointing. A right-handed frame then leaves no choice for the third axis: $\hat{J} = \hat{K} \times \hat{I}$, which at $\theta = 0$ ($\hat{K} = \hat{\imath}$, $\hat{I} = \hat{k}$) is $\hat{\imath} \times \hat{k} = -\hat{\jmath}$. A positive rotation θ about this $\hat{J}$ gives $\hat{K} = \cos\theta\,\hat{\imath} + \sin\theta\,\hat{k}$, i.e. it raises the pointing, as required. $\hat{\jmath}$ is fixed independently by body 1 ($\hat{\jmath} = \hat{k} \times \hat{\imath}$, with $\hat{\imath}$ the horizontal projection of the pointing), so it points the opposite way. $\hat{J} = +\hat{\jmath}$ would require either $\theta > 0$ to lower the pointing (the convention of the two-DOF model) or a left-handed frame. The sign only affects how θ and the tensor components are written; the physics is unchanged. O_N, o_1 and o_2 all lie on $\hat{z}$ and differ only by a translation along $\hat{z}$; o_2 is the intersection of the azimuth and polar axes:

$$\mathbf{r}_{o_1}\big|_N = [0,\ 0,\ 59.695]^{\mathsf{T}}\ \text{mm}, \qquad \mathbf{r}_{o_2}\big|_N = [0,\ 0,\ 548.865]^{\mathsf{T}}\ \text{mm}, \tag{2.1}$$

$$\mathbf{R}_{N1} = \mathbf{R}_z(\psi_L), \qquad \mathbf{R}_{12} = \mathbf{R}_y(-\theta_p)\,\mathbf{\Pi} = \begin{bmatrix} -\sin\theta_p & 0 & \cos\theta_p \\ 0 & -1 & 0 \\ \cos\theta_p & 0 & \sin\theta_p \end{bmatrix}, \qquad \mathbf{R}_{N2} = \mathbf{R}_{N1}\mathbf{R}_{12}, \tag{2.2}$$

$$\mathbf{R}_z(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{R}_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}. \tag{2.3}$$

with $\mathbf{\Pi} = \left[\begin{smallmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{smallmatrix}\right]$ the constant rotation that takes body 1 to body 2 at $\theta = 0$ ($\hat{K} = \hat{\imath}$, $\hat{I} = \hat{k}$, $\hat{J} = -\hat{\jmath}$). $\mathbf{R}_{XY}$ maps components in frame Y to components in frame X; a tensor changes frame as $\mathbf{I}|_Y = \mathbf{R}_{XY}^{\mathsf{T}}\,\mathbf{I}|_X\,\mathbf{R}_{XY}$. ψ is positive counter-clockwise about $\hat{k}$ seen from above. θ is positive by the right-hand rule about $\hat{J}$, so $\theta > 0$ *raises* the pointing: θ is the elevation ($\theta = 0$ horizon, $\theta = 90°$ zenith). This is the opposite of the rigid two-degree-of-freedom model, where $\theta > 0$ lowered the pointing; internally the code keeps the pointing frame of that model and uses $\theta_{2dof} = -\theta$.

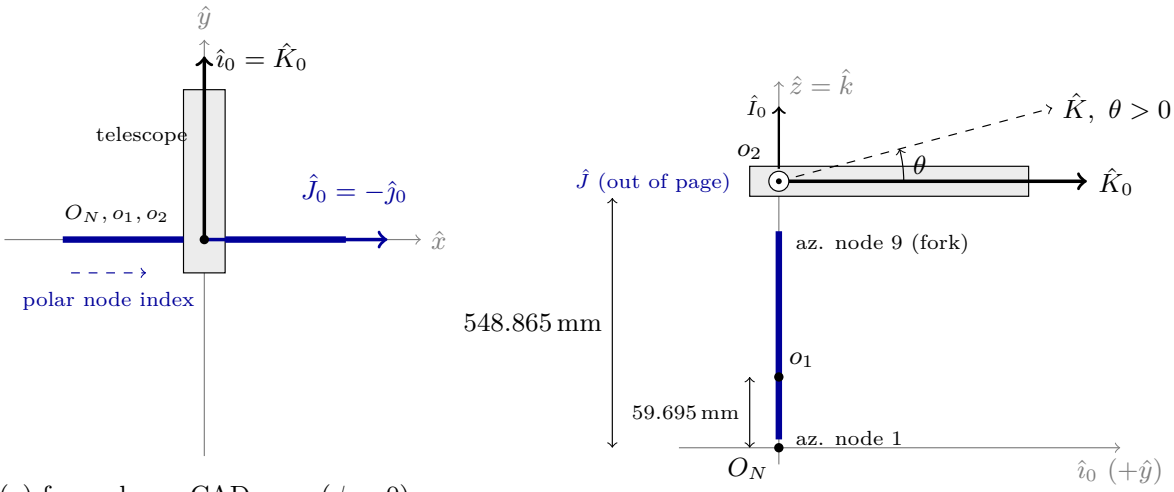
CAD pose. The CAD assembly is a snapshot of the mount at $\psi = 0$ and $\theta = \theta_{cad}$. Given the pointing direction $\hat{\imath}_0$ in N (`pointing_cad`),

$$\hat{\jmath}_0 = \hat{z} \times \hat{\imath}_0, \qquad \mathbf{R}_{N1}^{(0)} = \big[\,\hat{\imath}_0 \;\big|\; \hat{\jmath}_0 \;\big|\; \hat{z}\,\big], \qquad \mathbf{R}_{N2}^{(0)} = \big[\,\hat{I}_0 \;\big|\; \hat{J}_0 \;\big|\; \hat{K}_0\,\big] = \mathbf{R}_{N1}^{(0)}\,\mathbf{R}_{12}(\theta_{cad}), \tag{2.4}$$

so that $\hat{K}_0$ is the pointing direction, $\hat{I}_0$ points upwards when the telescope is horizontal and $\hat{J}_0 = \hat{K}_0 \times \hat{I}_0 = -\hat{\jmath}_0$. Internally the code works in the pointing frame of the rigid model (first axis along $\hat{K}$, second along $\hat{\jmath}$) and $\mathbf{\Pi}$ converts every component printed in this document. The code checks that $\hat{J}_0 = -\hat{\jmath}_0$ agrees with `polar_axis_cad`. Currently $\hat{\imath}_0 = +\hat{y}$, $\hat{J}_0 = +\hat{x}$, $\theta_{cad} = 0°$:

$$\mathbf{R}_{N1}^{(0)} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{R}_{N2}^{(0)} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}. \tag{2.5}$$

In words: in the CAD pose the telescope points along $\hat{K}_0 = +\hat{y}$ (towards the optics), the polar axis is $\hat{J}_0 = +\hat{x}$ and $\hat{I}_0 = +\hat{z}$.



(a) from above, CAD pose ($\psi = 0$)

(b) side view, plane $(\hat{\imath}_0, \hat{z})$

Figure 1: CAD pose of the mount, drawn from the data of `frames`. (a) View from above: telescope along $\hat{\imath}_0 = \hat{K}_0$, polar shaft along $\hat{J}_0 = -\hat{\jmath}_0$ (blue) with the direction of increasing polar node index; O_N, o_1, o_2 project onto the same point. (b) Side view in the vertical plane of the pointing direction: origins O_N, o_1, o_2 on $\hat{z}$, azimuth shaft from node 1 to node 9 (fork), $\hat{I}_0$ upwards, $\hat{J}$ out of the page, $\theta > 0$ raising the pointing.

2.1 Eccentricities

Because the body origins are on the rotation axes, the centre of mass of each body has an eccentricity with respect to its centre of rotation:

$$\mathbf{e}_1 = \mathbf{r}_{G_1} - \mathbf{r}_{o_1} \text{ (fork, in frame 1)}, \qquad \mathbf{d}_2 = \mathbf{R}_{N2}^{(0)\mathsf{T}}\left(\mathbf{r}_{G_2} - \mathbf{r}_{o_2}\right) \text{ (telescope, in frame 2)}. \tag{2.6}$$

$\mathbf{d}_2$ enters the dynamics through the tensor $\mathbf{I}_{p,o_2}|_2$ (Huygens–Steiner) and the gravitational energy $m_p g\,(d_K \sin\theta_p + d_I \cos\theta_p)$ (height of G_2 above o_2); d_J lies along the polar axis and never changes the height of the centre of mass. The fork turns only about $\hat{k}$, so in the current model its eccentricity enters only through its moment of inertia about the azimuth axis, $J_1 = I_{kk,G_1} + m_1\,(e_{1,x}^2 + e_{1,y}^2)$; keeping o_1 on the axis prepares the extension in which $\mathbf{e}_1$ acts explicitly.

2.2 Coordinates along the shafts

- **Azimuth shaft:** along $\hat{k} = \hat{z}$, node index increasing upwards (node 1 encoder at the bottom, node 9 fork at the top). ψ_i is the absolute rotation of node i about $\hat{k}$. The azimuth shaft turns only about its own axis, so only the axial inertias of its parts enter.
- **Polar shaft:** along $\hat{J} = -\hat{\jmath}$. θ_i is the rotation of node i about $\hat{J}$ relative to body 1, which is also its absolute rotation about $\hat{J}$. The polar node index increases along $+\hat{J}$ (`nodes_increase_along`), so the centre of mass of a polar piece at abscissa x from node 1 sits at

  $$y_c = y_1 + s\,x, \qquad s = +1, \tag{2.7}$$

  from o_2 along $\hat{J}$, where y_1 is `y_node1_from_o2_mm`. Currently y_1 is not given (`null`, Q7): the code then sets $y_c = 0$ for every piece, i.e. no Steiner term in J^{ax}. y_c enters only J^{ax} (Section 8).
- **Telescope (hub model):** element 3 has no shaft; its nodes coincide with the hub centre H, and the telescope is the node P joined to H by the attachment spring k_b (Section 5); only the telescope inertia (`payload`) is attached to P.
- **Element data** (`mode: manual`): J about the shaft's own axis; I_t about a transverse axis through the element's own centre of mass; `y_c_mm` measured from node i in the direction of increasing node index.

2.3 In which frame each inertia is entered

Telescope (`payload`). The dynamics needs the tensor of the telescope about o_2 in frame 2, $\mathbf{I}_{p,o_2}|_2$, with components $I_{II}, I_{JJ}, I_{KK}, I_{IJ}, I_{IK}, I_{JK}$ along $\hat{I}, \hat{J}, \hat{K}$, the mass m_p and the eccentricity $\mathbf{d}_2 = [d_I, d_J, d_K]$. All tensors are in tensor convention, $I_{ab} = -\int a\,b\,dm$ for $a \neq b$. Two input frames are accepted (key `frame`):

1. `frame: cad_cg` (recommended). Enter the tensor *exactly as Onshape exports it with no mate connector selected*: about the centre of mass G_2 of the telescope, with axes parallel to $\hat{x}, \hat{y}, \hat{z}$ (`I_kgm2`, $\mathrm{kg\,m^2}$), and the centre of mass $\mathbf{r}_{G_2}|_N$ measured from O_N (`r_G_mm`, mm). The code applies the sign convention (`products_sign`), then

$$\mathbf{r} = \mathbf{r}_{G_2} - \mathbf{r}_{o_2}, \qquad \mathbf{I}_{p,o_2}|_N = \mathbf{I}_{p,G_2}|_N + m_p \left(\|\mathbf{r}\|^2 \mathbf{I} - \mathbf{r}\,\mathbf{r}^{\mathsf{T}} \right), \tag{2.8}$$

$$\mathbf{I}_{p,o_2}|_2 = \mathbf{R}_{N2}^{(0)\mathsf{T}}\, \mathbf{I}_{p,o_2}|_N\, \mathbf{R}_{N2}^{(0)}, \qquad \mathbf{d}_2 = \mathbf{R}_{N2}^{(0)\mathsf{T}}\, \mathbf{r}. \tag{2.9}$$

   With the current CAD pose this amounts to (components of $\mathbf{I}_{p,o_2}|_N$ on the right, $\Delta x, \Delta y, \Delta z$ the components of $\mathbf{r}$ in N):

$$I_{II} = I^N_{zz}, \quad I_{JJ} = I^N_{xx}, \quad I_{KK} = I^N_{yy}, \tag{2.10}$$

$$I_{IJ} = I^N_{xz}, \quad I_{IK} = I^N_{yz}, \quad I_{JK} = I^N_{xy}, \tag{2.11}$$

$$d_I = \Delta z, \quad d_J = \Delta x, \quad d_K = \Delta y. \tag{2.12}$$

2. `frame: body2_o2`. Enter $\mathbf{I}_{p,o_2}|_2$ directly (about o_2, axes $\hat{I}, \hat{J}, \hat{K}$) and $\mathbf{d}_2$ in mm (`d_mm`); no transformation is applied.

Checklist for an Onshape export with `frame: cad_cg`:

- select only the telescope (there is no shaft through it) and keep `subtract_shaft_parts: false`;
- select no mate connector, so that the tensor is about the centre of mass with axes parallel to $\hat{x}, \hat{y}, \hat{z}$, and the centre of mass is measured from O_N;
- export in the CAD pose of `frames`: telescope pointing along $+\hat{y}$ at $\theta_{cad} = 0°$; if the assembly is in another pose, update `pointing_cad`, `polar_axis_cad` and `theta_cad_deg`;
- convert to $\mathrm{kg\,m^2}$ and mm, and set `products_sign` (`tensor` if the off-diagonal terms are $-\int a\,b\,dm$, `positive` if they are $+\int a\,b\,dm$).

The code then checks that the tensor is symmetric, positive definite and satisfies the triangle inequality, and warns (`PAYLOAD_ASSE_SOSPETTO`) if the smallest centroidal moment is not about $\hat{K}$: for an optical tube this usually reveals a wrong pose or a wrong frame. Current input: `source: manual`, `frame: cad_cg`, `subtract_shaft_parts: false`.

Current telescope data. As entered in `config.yaml` (Onshape, Telescopio LAMERA <1>, without mate connector):

$$\mathbf{I}_{p,G_2}\big|_N = \begin{bmatrix} 0.805643 & -1.679\times10^{-6} & 7.51\times10^{-7} \\ -1.679\times10^{-6} & 0.284157 & -0.0128698 \\ 7.51\times10^{-7} & -0.0128698 & 0.797637 \end{bmatrix} \ \mathrm{kg\,m^2}, \tag{2.13}$$

$$m = 16.61 \text{ kg}, \qquad \mathbf{r}_{G_2}\big|_N = [-0.0004821,\ 67.392,\ 544.07] \text{ mm}. \tag{2.14}$$

Values used by the dynamics: $m_p = 16.61\,\mathrm{kg}$, eccentricity $\mathbf{d}_2 = [-4.79,\ -0.0004821,\ 67.39]\,\mathrm{mm}$ (components along $\hat{I}, \hat{J}, \hat{K}$),

$$\mathbf{I}_{p,o_2}\big|_2 = \begin{bmatrix} 0.87307 & 7.1264\times10^{-7} & -0.0075079 \\ 7.1264\times10^{-7} & 0.88146 & -1.1393\times10^{-6} \\ -0.0075079 & -1.1393\times10^{-6} & 0.28454 \end{bmatrix} \ \mathrm{kg\,m^2}. \tag{2.15}$$

Fork (body 1, azimuth node 9). A scalar: the moment of inertia of the fork alone about the azimuth axis $\hat{k}$. From an export about the centre of mass with axes parallel to N, $J_1 = I_{zz,G_1} + m_1\,(x^2_{G_1} + y^2_{G_1})$ (the azimuth axis passes through O_N, so the horizontal components of $\mathbf{r}_{G_1}$ are those of the eccentricity $\mathbf{e}_1$).

Rotors and other node inertias. J about the axis of the shaft they sit on; for the polar axis also m and the transverse I_t about their own centre of mass, located at the node.

3 Input geometry

The figures are generated from `config.yaml`. Each segment is drawn as a rectangle whose width is proportional to its length and whose height is proportional to its outer diameter; the bore of hollow cylinders is hatched, torque-carrying steel layers are blue, aluminium light blue and non-carrying layers (spacers) orange; elements given as CAD data (`mode: manual`) are cross-hatched. Dimensions are in mm.

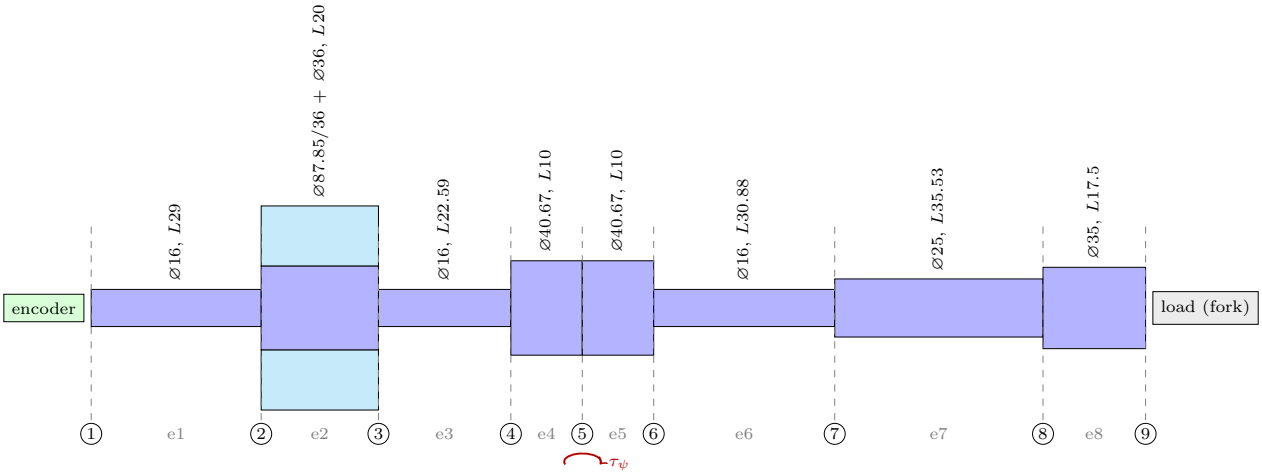


Figure 2: Azimuth shaft as entered in `config.yaml` (drawn horizontally; node 1 is at the bottom). Elements shorter than the readable minimum are stretched: lengths are to scale within each element, not across elements. Vertical scale: proportional to the diameter.

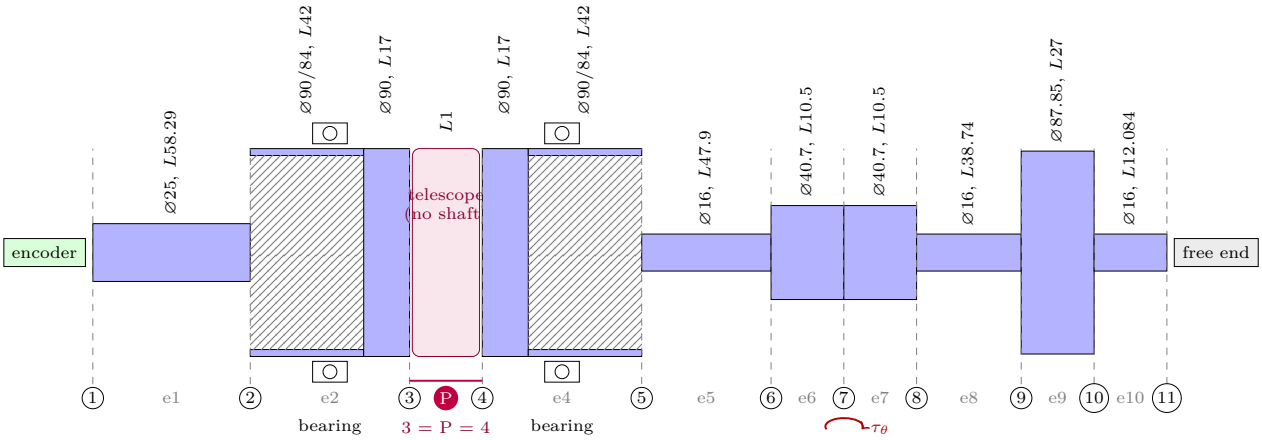


Figure 3: Polar shaft as entered in `config.yaml`; elements 2 and 4 are the side hubs (tube and flange) and element 3 has no shaft: its nodes coincide with the hub centre H, from which the telescope hangs through k_b (the support ring is in Figure 4). Elements shorter than the readable minimum are stretched: lengths are to scale within each element, not across elements. Vertical scale: proportional to the diameter.

4 Element properties

For an annular layer of length L, outer and inner diameters D_o, D_i ($D_i = 0$ for a solid cylinder), density ρ and shear modulus G:

$$m = \rho \frac{\pi}{4} \left(D_o^2 - D_i^2\right) L, \qquad J = \frac{\pi \rho L}{32} \left(D_o^4 - D_i^4\right) = \frac{m}{8} \left(D_o^2 + D_i^2\right), \tag{4.1}$$

$$I_t = m \left[\frac{D_o^2 + D_i^2}{16} + \frac{L^2}{12}\right], \qquad J_p = \frac{\pi}{32} \left(D_o^4 - D_i^4\right). \tag{4.2}$$

J is the axial mass moment of inertia, I_t the transverse one about the layer's own centre of mass and J_p the polar second moment of area. Coaxial layers of a segment add their masses and inertias, and the torque-carrying layers act in parallel; the segments of an element add their inertias and act in series, together with

an optional joint compliance:

$$k_s = \frac{1}{L_s} \sum_{\ell \text{ carrying}} G_\ell J_{p,\ell}, \qquad \frac{1}{k_e} = \sum_s \frac{1}{k_s} + \frac{1}{k_{joint}}, \qquad J_e = \sum_s \sum_\ell J_{s\ell} + \sum J_e^x. \tag{4.3}$$

J_e^x are concentrated extra inertias attached to the element. An element given as CAD data enters with its J_e and k_e directly. The transverse inertia of the element about its own centre of mass follows from the layers by the parallel-axis theorem along the shaft.

Table 2: Element properties, azimuth axis: layer contributions and element totals (m in g, inertias in kg m^2, k in N m/rad).

el.	seg.	lay.	shape	D_o	D_i	L	mat.	carr.	m	J	I_t	k
1	1	1	solid	16	0	29	steel	yes	46.6	1.49×10^{-6}	4.02×10^{-6}	1.71×10^{4}
1	total (geometry)					29			46.65	1.493×10^{-6}	4.015×10^{-6}	1.708×10^{4}
2	1	1	solid	36	0	20	steel	yes	163	2.64×10^{-5}	1.86×10^{-5}	6.35×10^{5}
2	1	2	hollow	87.85	36	20	aluminium	yes	283	0.000319	0.000169	7.64×10^{6}
2	total (geometry)					20			446.3	0.0003457	0.0001877	8.278×10^{6}
3	1	1	solid	16	0	22.59	steel	yes	36.3	1.16×10^{-6}	2.13×10^{-6}	2.19×10^{4}
3	total (geometry)					22.59			36.34	1.163×10^{-6}	2.127×10^{-6}	2.193×10^{4}
4	1	1	solid	40.67	0	10	steel	yes	104	2.15×10^{-5}	1.16×10^{-5}	2.07×10^{6}
4	total (geometry)					10			103.9	2.149×10^{-5}	1.161×10^{-5}	2.068×10^{6}
5	1	1	solid	40.67	0	10	steel	yes	104	2.15×10^{-5}	1.16×10^{-5}	2.07×10^{6}
5	total (geometry)					10			103.9	2.149×10^{-5}	1.161×10^{-5}	2.068×10^{6}
6	1	1	solid	16	0	30.88	steel	yes	49.7	1.59×10^{-6}	4.74×10^{-6}	1.6×10^{4}
6	total (geometry)					30.88			49.67	1.589×10^{-6}	4.742×10^{-6}	1.604×10^{4}
7	1	1	solid	25	0	35.53	steel	yes	140	1.09×10^{-5}	2.01×10^{-5}	8.31×10^{4}
7	total (geometry)					35.53			139.5	1.09×10^{-5}	2.013×10^{-5}	8.311×10^{4}
8	1	1	solid	35	0	17.5	steel	yes	135	2.06×10^{-5}	1.38×10^{-5}	6.48×10^{5}
8	total (geometry)					17.5			134.7	2.063×10^{-5}	1.375×10^{-5}	6.482×10^{5}

Table 3: Element properties, polar axis: layer contributions and element totals (m in g, inertias in kg m^2, k in N m/rad).

el.	seg.	lay.	shape	D_o	D_i	L	mat.	carr.	m	J	I_t	k
1	1	1	solid	25	0	58.29	steel	yes	229	1.79×10^{-5}	7.38×10^{-5}	5.07×10^{4}
1	total (geometry)					58.29			228.9	1.788×10^{-5}	7.375×10^{-5}	5.066×10^{4}
2	total (hub)					214			0	0	0	1805
3	total (telescope)					1			0	0	0	$--$
4	total (hub)					214			0	0	0	1805
5	1	1	solid	16	0	47.9	steel	yes	77	2.47×10^{-6}	1.6×10^{-5}	1.03×10^{4}
5	total (geometry)					47.9			77.05	2.466×10^{-6}	1.596×10^{-5}	1.034×10^{4}
6	1	1	solid	40.7	0	10.5	steel	yes	109	2.26×10^{-5}	1.23×10^{-5}	1.98×10^{6}
6	total (geometry)					10.5			109.3	2.263×10^{-5}	1.232×10^{-5}	1.976×10^{6}
7	1	1	solid	40.7	0	10.5	steel	yes	109	2.26×10^{-5}	1.23×10^{-5}	1.98×10^{6}
7	total (geometry)					10.5			109.3	2.263×10^{-5}	1.232×10^{-5}	1.976×10^{6}
8	1	1	solid	16	0	38.74	steel	yes	62.3	1.99×10^{-6}	8.79×10^{-6}	1.28×10^{4}
8	total (geometry)					38.74			62.31	1.994×10^{-6}	8.79×10^{-6}	1.279×10^{4}
9	1	1	solid	87.85	0	27	steel	yes	1.31×10^{3}	0.00126	0.000711	1.67×10^{7}
9	total (geometry)					27			1309	0.001263	0.0007111	1.668×10^{7}
10	1	1	solid	16	0	12.084	steel	yes	19.4	6.22×10^{-7}	5.48×10^{-7}	4.1×10^{4}
10	total (geometry)					12.084			19.44	6.22×10^{-7}	5.475×10^{-7}	4.1×10^{4}

5 Hub and compliant telescope attachment (hub model)

Purpose and switch. This document is generated with `polar.hub.enabled: true` (file `config_hub.yaml`, results in `results_hub/`). The original model, with the telescope rigidly attached to the shaft node P, is kept in `config.yaml` and `results/`; the flag switches between the two. The hub model follows the antiresonance measured at the elevation encoder (Section 12): a zero at an encoder on the load side requires a large inertia outside the encoder–motor path, so the telescope is no longer rigid on the shaft but hangs from the hub through a compliance k_b.

Topology. The two side hubs and the support ring form the hub, which carries the torque between the two polar bearings; its centre H (at o_2) carries the inertia of the hub. The two halves of the hub replace polar elements 2 and 4 as massless springs, element 3 (no shaft) collapses into H, and the telescope is a node P off the chain:

$$1 \overset{e_1}{\longleftrightarrow} 2 \overset{\text{half hub}}{\longleftrightarrow} \text{H} \overset{\text{half hub}}{\longleftrightarrow} 5 \overset{e_5}{\longleftrightarrow} 6 \cdots 11, \qquad \text{H} \overset{k_b}{\longleftrightarrow} \text{P (telescope)}. \tag{5.1}$$

Geometry. Each side hub is a single steel body: a solid flange fixed to the ring and a tube that seats the bearings. The support ring (aluminium) holds the telescope, which is slid into it, so its axis is the telescope axis $\hat{K}$, perpendicular to the polar axis; the side hubs are fixed to it at two diametrically opposite points. The materials follow from the mass: the nominal geometry gives $7.643 \times 10^5\,\text{mm}^3$ and 3.63 kg with an aluminium ring and steel side hubs, against 3.534 kg in Onshape.

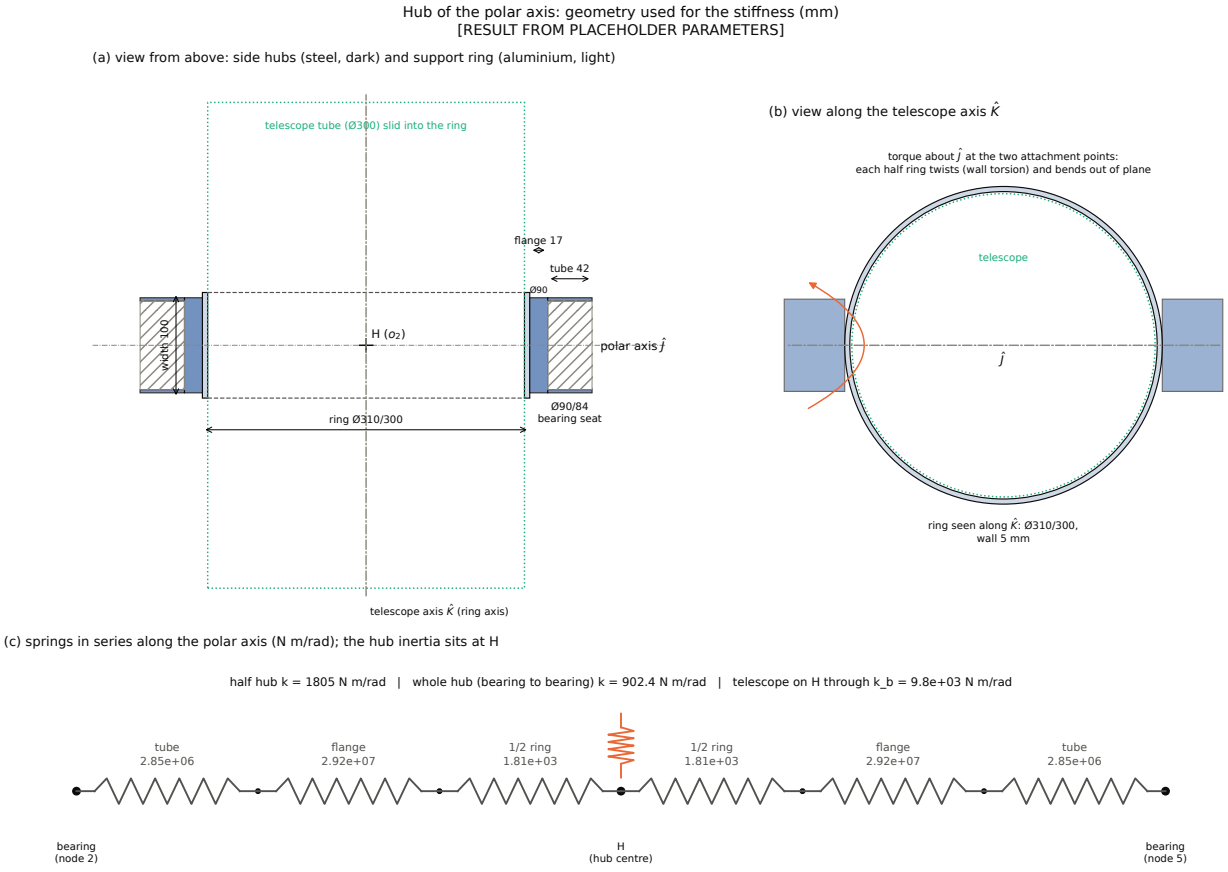


Figure 4: Hub of the polar axis as used for the stiffness: (a) view from above, with the polar axis horizontal and the telescope (ring) axis vertical; (b) view along the telescope axis; (c) springs in series from bearing to bearing, with the telescope attached to the hub centre H through k_b. Dimensions in mm, stiffnesses in N m/rad. *Result from placeholder parameters.*

Stiffness. The parts are springs in series. The tube and the flange are shafts in torsion, $k = GJ_p/L$. The ring, loaded by a couple M about $\hat{J}$ at each attachment point, works as two half rings in parallel; along each

half the moment is the constant vector $\frac{1}{2}M\hat{J}$, which splits into an out-of-plane bending moment $\frac{1}{2}M\cos\varphi$ about the radius and a torque $\frac{1}{2}M\sin\varphi$ about the tangent. The complementary energy (Castigliano) gives

$$\theta = \frac{\pi R}{4}\left(\frac{1}{EI_r}+\frac{1}{GJ_t}\right)M, \qquad k_{ring} = \frac{4}{\pi R}\left(\frac{1}{EI_r}+\frac{1}{GJ_t}\right)^{-1}, \tag{5.2}$$

$$I_r = \frac{t\,b^3}{12}, \qquad J_t = \beta\, b\, t^3, \quad \beta = \tfrac{1}{3}\big(1-0.63\,\tfrac{t}{b}+0.052\,(\tfrac{t}{b})^5\big), \qquad E = 2G(1+\nu), \tag{5.3}$$

with the mean radius $R = 152.5\,\mathrm{mm}$, the wall thickness $t = 5\,\mathrm{mm}$, the width $b = 100\,\mathrm{mm}$ and $\nu = 0.33$: $EI_r = 2.98\times10^4\,\mathrm{N\,m^2}$ and $GJ_t = 109\,\mathrm{N\,m^2}$. The thin wall twists easily, so the ring dominates the series. The ring is taken alone, without the stiffening of the telescope tube slid into it (user's choice): its value is a lower bound.

Table 4: Hub parts, geometry (mm) and torsional stiffness about the polar axis (N m/rad). *Result from placeholder parameters.*

part	geometry	k
side hub: bearing-seat tube	90/84 x 42, steel	2.848×10^6
side hub: flange	90 x 17 (solid), steel	2.918×10^7
support ring (two half rings)	310/300 x 100, aluminium	903
half hub (tube, flange, half ring: $2k_{ring}$)	in series	1805
whole hub, bearing to bearing	in series	902.4
telescope attachment k_b	placeholder (see text)	9800

Inertia. From the Onshape export of the three parts (Onshape: Ruota di trasmissione <1>, <2> + T-02 Anello di registrazione del telescopio <1>), mass 3.534 kg, centre of mass at 0 mm from o_2, tensor about the centre of mass in the axes of N ($\mathrm{kg\,m^2}$)

$$\mathbf{I}_{hub,G}|_N = \begin{bmatrix} 0.01881 & -1.5\times10^{-8} & 2\times10^{-9} \\ -1.5\times10^{-8} & 0.09604 & 2.438\times10^{-6} \\ 2\times10^{-9} & 2.438\times10^{-6} & 0.08237 \end{bmatrix}. \tag{5.4}$$

Transported to o_2 and rotated into the frame of body 2, it gives $J_{hub} = 0.01881\,\mathrm{kg\,m^2}$ about the polar axis, at node H, and $0.08237\,\mathrm{kg\,m^2}$ about $\hat{z}$ at $\theta = 0$, added to J^{ax} because the azimuth load exported from Onshape does not contain the hub (`in_azimuth_load: false`).

Telescope attachment. The spring k_b joins H to the telescope node P, which carries only the telescope (payload section). Its value is not computed from a geometry: it is a placeholder, the two-mass estimate from the antiresonance and resonance measured at the elevation encoder. With the shafts, the reduced polar tree has $k_{\theta,1} = 1743$ (encoder–H), $k_{\theta,2} = 1535$ (H–motor), $k_{\theta,3} = 9694$ (motor–rear end) and $k_{\theta,4} = k_b = 9800\,\mathrm{N\,m/rad}$ (H–P).

6 Finite element model of each axis

Element matrices. The two-node torsional element has

$$\mathbf{K}_e = k_e\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \qquad \mathbf{M}_e^{\ell} = \frac{J_e}{2}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{(lumped)}. \tag{6.1}$$

For the consistent mass, the inertia per unit length $\bar{J}(x)$ is piecewise constant (one value $\bar{J}_s = J_s/L_s$ per segment) and the shape functions are linear, $N_1 = 1 - x/L_e$, $N_2 = x/L_e$:

$$(\mathbf{M}_e^{c})_{ab} = \sum_s \bar{J}_s \int_{x_s}^{x_s+L_s} N_a N_b\, dx = \sum_s \bar{J}_s\,\frac{L_s}{6}\Big(2N_a^0N_b^0 + N_a^0N_b^1 + N_a^1N_b^0 + 2N_a^1N_b^1\Big) + \frac{J_e^x}{2}\,\mathbf{I}, \tag{6.2}$$

where N_a^0, N_a^1 are the values of N_a at the ends of segment s: the formula is exact because the integrand is the product of two linear functions on each segment. For one uniform segment it gives $\mathbf{M}_e^c = \frac{J_e}{6}\left[\begin{smallmatrix}2&1\\1&2\end{smallmatrix}\right]$. We write a_e, b_e, c_e for the entries $(1,1)$, $(1,2)$, $(2,2)$ of $\mathbf{M}_e^c$.

Nodes H and P (hub model). Element 3 has no shaft: its nodes 3 and 4 coincide with the hub centre H, which carries the hub inertia J_H^n. Elements 2 and 4 are the two halves of the hub, massless springs (their inertia is in J_H^n). The telescope node P is an extra node, last in the ordering, joined to H by the massless spring k_b: it carries no inertia of its own in $\mathbf{M}$, the telescope enters through the payload (Section 5). The matrices are therefore those of a tree, not of a chain.

Assembly. The global matrices are assembled by summing on the shared nodes; the concentrated inertias J^n (rotors, fork) are added on the diagonal and the generalised force is $\mathbf{Q} = \tau\,\mathbf{e}_p$ with p the motor node. In symbolic form:

Azimuth axis (node order 1, 2, 3, 4, 5, 6, 7, 8, 9):

$$\mathbf{K} = \begin{bmatrix} k_1 & -k_1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -k_1 & k_1+k_2 & -k_2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -k_2 & k_2+k_3 & -k_3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -k_3 & k_3+k_4 & -k_4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -k_4 & k_4+k_5 & -k_5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -k_5 & k_5+k_6 & -k_6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -k_6 & k_6+k_7 & -k_7 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -k_7 & k_7+k_8 & -k_8 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -k_8 & k_8 \end{bmatrix}$$

$$\mathbf{M}^\ell = \begin{bmatrix} \frac{J_1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{J_1}{2}+\frac{J_2}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{J_2}{2}+\frac{J_3}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{J_3}{2}+\frac{J_4}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{J_4}{2}+\frac{J_5}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{J_5}{2}+\frac{J_6}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{J_6}{2}+\frac{J_7}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{J_7}{2}+\frac{J_8}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & J_9^n+\frac{J_8}{2} \end{bmatrix}$$

$$\mathbf{M}^c = \begin{bmatrix} a_1 & b_1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ b_1 & a_2+c_1 & b_2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & b_2 & a_3+c_2 & b_3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & b_3 & a_4+c_3 & b_4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & b_4 & a_5+c_4 & b_5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & b_5 & a_6+c_5 & b_6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & b_6 & a_7+c_6 & b_7 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & b_7 & a_8+c_7 & b_8 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & b_8 & J_9^n+c_8 \end{bmatrix}$$

Numerically (SI units, consistent mass, shafts and node inertias only):

$$\mathbf{K} = \begin{bmatrix} 1.71\times10^4 & -1.71\times10^4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1.71\times10^4 & 8.29\times10^6 & -8.28\times10^6 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -8.28\times10^6 & 8.3\times10^6 & -2.19\times10^4 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -2.19\times10^4 & 2.09\times10^6 & -2.07\times10^6 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2.07\times10^6 & 4.14\times10^6 & -2.07\times10^6 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -2.07\times10^6 & 2.08\times10^6 & -1.6\times10^4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1.6\times10^4 & 9.92\times10^4 & -8.31\times10^4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -8.31\times10^4 & 7.31\times10^5 & -6.48\times10^5 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -6.48\times10^5 & 6.48\times10^5 \end{bmatrix}$$

$$\mathbf{M}^c = \begin{bmatrix} 4.98\times10^{-7} & 2.49\times10^{-7} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2.49\times10^{-7} & 0.000116 & 5.76\times10^{-5} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 5.76\times10^{-5} & 0.000116 & 1.94\times10^{-7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1.94\times10^{-7} & 7.55\times10^{-6} & 3.58\times10^{-6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3.58\times10^{-6} & 1.43\times10^{-5} & 3.58\times10^{-6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3.58\times10^{-6} & 7.69\times10^{-6} & 2.65\times10^{-7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2.65\times10^{-7} & 4.16\times10^{-6} & 1.82\times10^{-6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1.82\times10^{-6} & 1.05\times10^{-5} & 3.44\times10^{-6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3.44\times10^{-6} & 1.25 \end{bmatrix}$$

Polar axis (node order 1, 2, H, 5, 6, 7, 8, 9, 10, 11, P):

$$\mathbf{K} = \begin{bmatrix} k_1 & -k_1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -k_1 & k_1+k_2 & -k_2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -k_2 & k_2+k_4+k_b & -k_4 & 0 & 0 & 0 & 0 & 0 & 0 & -k_b \\ 0 & 0 & -k_4 & k_4+k_5 & -k_5 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -k_5 & k_5+k_6 & -k_6 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -k_6 & k_6+k_7 & -k_7 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -k_7 & k_7+k_8 & -k_8 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -k_8 & k_8+k_9 & -k_9 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -k_9 & k_{10}+k_9 & -k_{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -k_{10} & k_{10} & 0 \\ 0 & 0 & -k_b & 0 & 0 & 0 & 0 & 0 & 0 & 0 & k_b \end{bmatrix}$$

$$\mathbf{M}^\ell = \begin{bmatrix} \frac{J_1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{J_1}{2}+\frac{J_2}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & J_H^n+\frac{J_2}{2}+\frac{J_4}{2}+\frac{J_b}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{J_4}{2}+\frac{J_5}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{J_5}{2}+\frac{J_6}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{J_6}{2}+\frac{J_7}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{J_7}{2}+\frac{J_8}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{J_8}{2}+\frac{J_9}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{J_{10}}{2}+\frac{J_9}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{J_{10}}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{J_b}{2} \end{bmatrix}$$

$$\mathbf{M}^c = \begin{bmatrix} a_1 & b_1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ b_1 & a_2+c_1 & b_2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & b_2 & J_H^n+a_4+a_b+c_2 & b_4 & 0 & 0 & 0 & 0 & 0 & 0 & b_b \\ 0 & 0 & b_4 & a_5+c_4 & b_5 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & b_5 & a_6+c_5 & b_6 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & b_6 & a_7+c_6 & b_7 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & b_7 & a_8+c_7 & b_8 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & b_8 & a_9+c_8 & b_9 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & b_9 & a_{10}+c_9 & b_{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & b_{10} & c_{10} & 0 \\ 0 & 0 & b_b & 0 & 0 & 0 & 0 & 0 & 0 & 0 & c_b \end{bmatrix}$$

Numerically (SI units, consistent mass, shafts and node inertias only):

$$\mathbf{K} = \begin{bmatrix} 5.07\times10^4 & -5.07\times10^4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -5.07\times10^4 & 5.25\times10^4 & -1.8\times10^3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1.8\times10^3 & 1.34\times10^4 & -1.8\times10^3 & 0 & 0 & 0 & 0 & 0 & 0 & -9.8\times10^3 \\ 0 & 0 & -1.8\times10^3 & 1.21\times10^4 & -1.03\times10^4 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1.03\times10^4 & 1.99\times10^6 & -1.98\times10^6 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1.98\times10^6 & 3.95\times10^6 & -1.98\times10^6 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1.98\times10^6 & 1.99\times10^6 & -1.28\times10^4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1.28\times10^4 & 1.67\times10^7 & -1.67\times10^7 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1.67\times10^7 & 1.67\times10^7 & -4.1\times10^4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -4.1\times10^4 & 4.1\times10^4 & 0 \\ 0 & 0 & -9.8\times10^3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 9.8\times10^3 \end{bmatrix}$$

$$\mathbf{M}^c = \begin{bmatrix} 5.96\times10^{-6} & 2.98\times10^{-6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2.98\times10^{-6} & 5.96\times10^{-6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.0188 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 8.22\times10^{-7} & 4.11\times10^{-7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4.11\times10^{-7} & 8.36\times10^{-6} & 3.77\times10^{-6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3.77\times10^{-6} & 1.51\times10^{-5} & 3.77\times10^{-6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3.77\times10^{-6} & 8.21\times10^{-6} & 3.32\times10^{-7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 3.32\times10^{-7} & 0.000422 & 0.000211 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.000211 & 0.000421 & 1.04\times10^{-7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1.04\times10^{-7} & 2.07\times10^{-7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Checks. Both matrices are symmetric, $\mathbf{M}$ is positive definite, $\mathbf{K1} = \mathbf{0}$ (one rigid mode per axis), and $\mathbf{1}^{\mathsf{T}}\mathbf{M}\mathbf{1} = \sum_e J_e + \sum J^n$ for both mass models.

Lumped versus consistent mass. For one element,

$$\mathbf{M}_e^{\ell} - \mathbf{M}_e^{c} = \begin{bmatrix} b_e + \delta_e & -b_e \\ -b_e & b_e - \delta_e \end{bmatrix}, \qquad \delta_e = \frac{c_e - a_e}{2}, \qquad \det\left(\mathbf{M}_e^{\ell} - \mathbf{M}_e^{c}\right) = -\delta_e^2, \tag{6.3}$$

using $a_e + 2b_e + c_e = J_e$. Hence the difference is positive semidefinite *if and only if* $\delta_e = 0$, i.e. when the inertia of the element is symmetric about its mid-span (in particular for uniform elements, where $v^{\mathsf{T}}(\mathbf{M}^{\ell} - \mathbf{M}^{c})v = \sum_e \frac{J_e}{6}(v_i - v_{i+1})^2$). In that case $\mathbf{M}^{\ell} \succeq \mathbf{M}^{c}$ and, by the Courant–Fischer min-max characterisation,

$$\omega_k^2 = \min_{\dim S = k} \max_{v \in S} \frac{v^{\mathsf{T}}\mathbf{K}v}{v^{\mathsf{T}}\mathbf{M}v} \quad \Longrightarrow \quad \omega_k^{\ell} \le \omega_k^{c} \quad \forall k, \tag{6.4}$$

because a larger mass lowers every Rayleigh quotient. With non-symmetric elements the ordering is *not* guaranteed and may fail for individual modes; the code reports it as a warning (Section 14).

Damping. $\mathbf{C} = \beta\mathbf{K} + \mathbf{C}_b$, with $\beta = 2\zeta/\omega_1$ tuned on the first flexible mode of each isolated axis ($\zeta = 0.01$, placeholder; $\beta_\psi = 4.232\times10^{-6}\,\mathrm{s}$, $\beta_\theta = 2.908\times10^{-5}\,\mathrm{s}$). The mass-proportional part is deliberately omitted: it would damp the rigid rotation of a free shaft. $\mathbf{C}_b$ is the viscous friction of the bearings, $\frac{b}{2}\,\mathrm{diag}(1,1)$ on the two nodes of a bearing element.

7 Natural frequencies and mode shapes

7.1 The eigenvalue problem

Free, undamped torsional vibration of an axis obeys $\mathbf{M}\ddot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{0}$. Looking for harmonic solutions $\mathbf{q} = \boldsymbol{\varphi}\, e^{j\omega t}$ gives the generalised symmetric eigenvalue problem

$$\left(\mathbf{K} - \omega^2\mathbf{M}\right)\boldsymbol{\varphi} = \mathbf{0} \quad \Longleftrightarrow \quad \mathbf{K}\boldsymbol{\varphi} = \lambda\,\mathbf{M}\boldsymbol{\varphi}, \qquad \lambda = \omega^2, \tag{7.1}$$

whose non-trivial solutions are the roots of the characteristic polynomial $\det(\mathbf{K} - \lambda\mathbf{M}) = 0$, of degree n in λ. Because $\mathbf{K}$ is symmetric positive semidefinite and $\mathbf{M}$ symmetric positive definite, all eigenvalues are real and non-negative, and the eigenvectors can be chosen mass-orthonormal:

$$\boldsymbol{\varphi}_i^{\mathsf{T}}\mathbf{M}\boldsymbol{\varphi}_j = \delta_{ij}, \qquad \boldsymbol{\varphi}_i^{\mathsf{T}}\mathbf{K}\boldsymbol{\varphi}_j = \omega_i^2\delta_{ij}, \qquad \lambda_i = \frac{\boldsymbol{\varphi}_i^{\mathsf{T}}\mathbf{K}\boldsymbol{\varphi}_i}{\boldsymbol{\varphi}_i^{\mathsf{T}}\mathbf{M}\boldsymbol{\varphi}_i}\ \text{(Rayleigh quotient)}. \tag{7.2}$$

Since $\mathbf{K1} = \mathbf{0}$, every free axis has exactly one rigid mode, $\lambda_1 = 0$, $\boldsymbol{\varphi}_1 \propto \mathbf{1}$; the others are the torsional modes, with $f_i = \sqrt{\lambda_i}/2\pi$. When the gravitational stiffness k_g is added at the telescope node (isolated polar axis) the matrix $\mathbf{K} + k_g\mathbf{e}_P\mathbf{e}_P^{\mathsf{T}}$ may have a *negative* eigenvalue, reported by the code with a negative sign and interpreted as the static instability of the unbalanced telescope rather than as a mode. With the stiffness-proportional damping $\mathbf{C} = \beta\mathbf{K}$ the modal equations decouple, $\ddot{\eta}_i + 2\zeta_i\omega_i\dot{\eta}_i + \omega_i^2\eta_i = \boldsymbol{\varphi}_i^{\mathsf{T}}\mathbf{Q}$ with $\zeta_i = \beta\omega_i/2$, which is why β is tuned on the first flexible mode.

7.2 Analytical solutions

Uniform shaft. A uniform free–free shaft obeys the wave equation $\partial_t^2\vartheta = c^2\partial_x^2\vartheta$, $c = \sqrt{G/\rho}$, with free ends $\partial_x\vartheta = 0$ (no torque), so

$$\varphi_n(x) = \cos\frac{n\pi x}{L}, \qquad \omega_n = \frac{n\pi}{L}\sqrt{\frac{G}{\rho}}, \qquad n = 0, 1, 2, \ldots \tag{7.3}$$

independent of the diameter. The FEM of N equal elements has the exact discrete spectrum $\theta_n = n\pi/N$ and

$$\omega_n^{\ell} = \frac{2c}{h}\sin\frac{\theta_n}{2}, \qquad \omega_n^{c} = \frac{c}{h}\sqrt{\frac{6\,(1-\cos\theta_n)}{2+\cos\theta_n}}, \qquad h = L/N, \tag{7.4}$$

for lumped and consistent mass respectively; expanding, $\omega_n^{\ell,c} \approx \omega_n\,[1 \mp (n\pi/N)^2/24]$, so the two models bound the exact frequency from below and above and converge with second order.

Table 5: Uniform free–free steel bar ($L = 0.3\,\mathrm{m}$, $D = 20\,\mathrm{mm}$, $\rho = 8000\,\mathrm{kg/m^3}$, $G = 77\,\mathrm{GPa}$, $c = 3102.4\,\mathrm{m/s}$): exact frequencies $f_n = nc/2L$ against the torsional FEM with N equal elements. The errors follow the closed forms $\mp(n\pi/N)^2/24$.

N	n	f_n [Hz]	lumped [Hz]	error	consistent [Hz]	error
4	1	5170.7	5038.82	−2.55 %	5304.41	2.59 %
	2	10341.4	9310.53	−9.97 %	11403	10.3 %
	3	15512.1	12164.8	−21.6 %	18530.4	19.5 %
16	1	5170.7	5162.4	−0.161 %	5179.01	0.161 %
	2	10341.4	10275.1	−0.641 %	10408	0.644 %
	3	15512.1	15288.8	−1.44 %	15737.2	1.45 %
64	1	5170.7	5170.18	−0.01 %	5171.22	0.01 %
	2	10341.4	10337.2	−0.0402 %	10345.5	0.0402 %
	3	15512.1	15498.1	−0.0903 %	15526.1	0.0904 %

7.3 Numerical solution

The code never solves the characteristic polynomial: finding the roots of a polynomial whose coefficients span many decades (here λ ranges from about 10^0 to $10^{10}\,\mathrm{s}^{-2}$) is ill-conditioned. It uses the following linear-algebra tools instead.

1. **Torsional modes of an axis** (`fem.modi`). `scipy.linalg.eigh(K, M)` calls the LAPACK driver `dsygvd` for the generalised symmetric-definite problem: Cholesky factorisation $\mathbf{M} = \mathbf{L}\mathbf{L}^{\mathsf{T}}$, reduction to the standard symmetric problem $\mathbf{L}^{-1}\mathbf{K}\mathbf{L}^{-\mathsf{T}}\mathbf{y} = \lambda\mathbf{y}$, Householder tridiagonalisation, a divide-and-conquer eigensolver, and back-transformation $\boldsymbol{\varphi} = \mathbf{L}^{-\mathsf{T}}\mathbf{y}$, which returns mass-orthonormal modes. The cost is $O(n^3)$ on matrices of size at most 12. The rigid mode comes out as $|\lambda| \sim 10^{-16}\lambda_{max}$ and is set to zero below the threshold $10^{-12}\lambda_{max}$; frequencies are returned in Hz.

2. **Modal basis** (`modal.modi_con_rigido`). The same `eigh` on the full FEM with the telescope frozen, $(\mathbf{K}, \bar{\mathbf{M}})$; the column of the null eigenvalue is checked to be proportional to $\mathbf{1}$ and replaced by $\mathbf{1}$, the others are kept mass-normalised. The retained columns are a Ritz basis that contains the exact modes, so their frequencies are exact (Section 10).

3. **Coupled linearised system** (`ModelloLineare.poli`). With gyroscopic and centrifugal terms the problem is no longer self-adjoint, so the poles are the eigenvalues of the state matrix $\mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{M}_0^{-1}\mathbf{K}_0 & -\mathbf{M}_0^{-1}\mathbf{C}_0 \end{bmatrix}$ computed with `numpy.linalg.eigvals` (LAPACK `dgeev`: balancing, Hessenberg reduction, shifted QR). A pair $\lambda = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$ gives $f_n = |\lambda|/2\pi$ and $\zeta = -\mathrm{Re}\,\lambda/|\lambda|$; `tf.classifica` labels each root as null, real stable, real unstable or complex. The azimuth integrator gives a double null pole, a defective eigenvalue that `dgeev` returns scattered at $\sim 10^{-9}|\lambda|_{max}$; the code treats anything below that threshold as zero.

4. **Zeros.** The invariant zeros of each SISO channel are the finite generalised eigenvalues of the Rosenbrock pencil $\left(\begin{bmatrix} \mathbf{A} & \mathbf{b}_j \\ \mathbf{c}_i & 0 \end{bmatrix}, \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & 0 \end{bmatrix}\right)$, computed with `scipy.linalg.eig` (QZ algorithm, `dggev`) in homogeneous form, discarding the infinite eigenvalues ($\beta \approx 0$).

5. **Frequency response.** $\mathbf{G}(j\omega) = \mathbf{C}_q\,\mathbf{P}(j\omega)^{-1}\mathbf{E}$, with $\mathbf{P}(s) = \mathbf{M}_0 s^2 + \mathbf{C}_0 s + \mathbf{K}_0$, is evaluated directly by a batched `numpy.linalg.solve` over the whole frequency grid, not through the polynomial coefficients, and cross-checked against $\mathbf{C}(j\omega\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}$.

Verification. The test suite checks the numerical solution against the analytical one: the shaft convergence above (lumped from below, consistent from above), the identity $\mathbf{v}^{\mathsf{T}}(\mathbf{M}^{\ell} - \mathbf{M}^{c})\mathbf{v} = \sum_e J_e(v_i - v_{i+1})^2/6$, the exactness of the retained modal frequencies, the equality of the poles with the roots of $\det \mathbf{P}(s)$, the alternation of poles and zeros on the collocated channels, and the equality of the zeros of the collocated channel with the natural frequencies of the structure with the motor node clamped.

8 Coupled model

Coordinates. The coupled model is written first with *all* FEM degrees of freedom: $\mathbf{q}_\psi$ (9 azimuth nodes, absolute rotations about $\hat{k}$) and $\mathbf{q}_\theta$ (11 polar nodes 1, 2, H, 5, 6, 7, 8, 9, 10, 11, P, rotations about $\hat{J}$ relative to the fork). The fork rotates with $\psi_L = \psi_9$ (rigid fork) and the telescope with θ_p, the rotation of node P: $\mathbf{q}_p = [\psi_L,\ \theta_p]^{\mathsf{T}} = \mathbf{E}\,\mathbf{q}$, with $\mathbf{E}$ selecting the two attachment nodes. This is the reference model (exact within the torsional FEM) of which the reduced model of Section 10 is the projection $\mathbf{q} = \mathbf{\Phi}\boldsymbol{\eta}$.

Kinematics. Every body of the polar axis has angular velocity $\boldsymbol{\omega}_j = \dot{\psi}_L\hat{k} + \dot{\theta}_j\hat{J}$. For the telescope, Poisson's formula $[\boldsymbol{\omega}]_\times = \mathbf{R}^{\mathsf{T}}\dot{\mathbf{R}}$ with $\mathbf{R} = \mathbf{R}_{N2}$ gives, with components along $\hat{I}, \hat{J}, \hat{K}$ (derived with SymPy and checked against the closed form),

$$\boldsymbol{\omega}_2\big|_2 = \begin{bmatrix} \dot{\psi}_L \cos(\theta_p) \\ \dot{\theta}_p \\ \dot{\psi}_L \sin(\theta_p) \end{bmatrix}. \tag{8.1}$$

Energies and dissipation.

$$T = \tfrac{1}{2}\dot{\mathbf{q}}_\psi^{\mathsf{T}}\mathbf{M}_\psi\dot{\mathbf{q}}_\psi + \tfrac{1}{2}J^{ax}\dot{\psi}_L^2 + \tfrac{1}{2}\dot{\mathbf{q}}_\theta^{\mathsf{T}}\mathbf{M}_\theta\dot{\mathbf{q}}_\theta + \tfrac{1}{2}\boldsymbol{\omega}_2|_2^{\mathsf{T}}\,\mathbf{I}_{p,o_2}|_2\,\boldsymbol{\omega}_2|_2, \tag{8.2}$$

$$V = \tfrac{1}{2}\mathbf{q}_\psi^{\mathsf{T}}\mathbf{K}_\psi\mathbf{q}_\psi + \tfrac{1}{2}\mathbf{q}_\theta^{\mathsf{T}}\mathbf{K}_\theta\mathbf{q}_\theta + m_p g\,(d_K\sin\theta_p + d_I\cos\theta_p), \qquad \mathcal{D} = \tfrac{1}{2}\dot{\mathbf{q}}^{\mathsf{T}}\mathbf{C}\,\dot{\mathbf{q}}, \tag{8.3}$$

with $\mathbf{M}$, $\mathbf{K}$, $\mathbf{C} = \beta\mathbf{K} + \mathbf{C}_b$ the FEM matrices of Section 6 (the fork inertia is the concentrated inertia of azimuth node 9). *Justification of J^{ax}.* Each axisymmetric part of the polar shaft has tensor $\mathrm{diag}(I_t, J, I_t)$ in any frame rotated about $\hat{J}$, and its centre of mass lies on $\hat{J}$ at distance y_c from o_2. Since $\hat{k} \perp \hat{J}$, its kinetic energy is $\frac{1}{2}(I_t + my_c^2)\dot{\psi}_L^2 + \frac{1}{2}J\dot{\theta}_j^2$, without mixed terms and without dependence on θ: the axial part is in $\mathbf{M}_\theta$, the transverse part is the constant $J^{ax} = \sum(I_t + my_c^2)$ (here $0.08237\,\mathrm{kg\,m^2}$). **The polar shafts therefore produce no gyroscopic coupling**: the dynamic coupling comes only from the telescope. In the current data the azimuth load (node 9) is a CAD export that already contains the polar shafts, so their contribution about $\hat{k}$, including the true my_c^2 terms, is already in the node inertia and J^{ax} is *not* added (the value computed from the polar elements, $0.0008348\,\mathrm{kg\,m^2}$, would count the shafts twice). The axial inertia of the shafts about $\hat{J}$ stays in the polar FEM.

Generalised forces and the reaction torque. The polar motor applies $+\tau_\theta\hat{J}$ to the rotor and $-\tau_\theta\hat{J}$ to the stator (fork); with $\delta\boldsymbol{\phi}_{rot} = \delta\psi_L\hat{k} + \delta\theta_m\hat{J}$ and $\delta\boldsymbol{\phi}_{stat} = \delta\psi_L\hat{k}$,

$$\delta W = \tau_\psi\,\delta\psi_m + \tau_\theta\,\hat{J}\cdot(\delta\boldsymbol{\phi}_{rot} - \delta\boldsymbol{\phi}_{stat}) = \tau_\psi\,\delta\psi_m + \tau_\theta\,\delta\theta_m, \qquad \mathbf{Q} = \mathbf{B}\boldsymbol{\tau} = \tau_\psi\,\mathbf{e}_{\psi_m} + \tau_\theta\,\mathbf{e}_{\theta_m}. \tag{8.4}$$

The reaction torque on the fork is along $\hat{J} \perp \hat{k}$: it does no work on ψ_L and is carried by the azimuth bearings as a bending moment (rigid fork, Q12).

Telescope terms. Only the telescope depends nonlinearly on the state. Its Lagrangian $T_p - V_g$, a function of $\mathbf{q}_p$ alone, is differentiated by SymPy, which also checks the matrix form against Lagrange's equations:

$$\mathbf{M}_p(\theta_p) = \begin{bmatrix} I_{II}\cos^2(\theta_p) + I_{IK}\sin(2\theta_p) + I_{KK}\sin^2(\theta_p) & I_{IJ}\cos(\theta_p) + I_{JK}\sin(\theta_p) \\ I_{IJ}\cos(\theta_p) + I_{JK}\sin(\theta_p) & I_{JJ} \end{bmatrix},$$

$$\mathbf{C}_p\dot{\mathbf{q}}_p = \begin{bmatrix} \dot{\theta}_p\left(\dot{\psi}_L\left(-I_{II}\sin(2\theta_p) + 2I_{IK}\cos(2\theta_p) + I_{KK}\sin(2\theta_p)\right) - \dot{\theta}_p\left(I_{IJ}\sin(\theta_p) - I_{JK}\cos(\theta_p)\right)\right) \\ \frac{\dot{\psi}_L^2(I_{II}\sin(2\theta_p) - 2I_{IK}\cos(2\theta_p) - I_{KK}\sin(2\theta_p))}{2} \end{bmatrix},$$

$$\mathbf{g}_p = \begin{bmatrix} 0 \\ g m_p\left(-d_I\sin(\theta_p) + d_K\cos(\theta_p)\right) \end{bmatrix}. \tag{8.5}$$

$J_{\psi,p}(\theta) = M_{p,11}$ is the telescope inertia about the azimuth axis. Lagrange's equations of the whole system are then

$$\big[\mathbf{M}_s + \mathbf{E}^{\mathsf{T}}\mathbf{M}_p(\theta_p)\mathbf{E}\big]\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} + \mathbf{E}^{\mathsf{T}}\big(\mathbf{C}_p\dot{\mathbf{q}}_p + \mathbf{g}_p\big) = \mathbf{B}\boldsymbol{\tau}, \qquad \mathbf{M}_s = \mathrm{blkdiag}(\mathbf{M}_\psi + J^{ax}\mathbf{e}_L\mathbf{e}_L^{\mathsf{T}},\ \mathbf{M}_\theta), \tag{8.6}$$

and the state dependence is only through θ_p and the velocities $\dot{\mathbf{q}}_p$.

9 Operating point and linearisation

Operating point. A steady tracking motion: $\theta_p = \theta_0$, rigid azimuth rotation at $\dot{\psi} = \Omega_0$, all $\dot{\theta} = 0$, $\ddot{\mathbf{q}} = \mathbf{0}$, $\psi_e = 0$ (cyclic). With θ_p fixed the residual of (8.6) is affine in the remaining coordinates, so the static torsions and the nominal torques $\boldsymbol{\tau}_0$ follow from a linear system (Section 13).

Linearised telescope terms. At the operating point the Jacobians of the telescope terms with respect to $\ddot{\mathbf{q}}_p$, $\dot{\mathbf{q}}_p$ and $\mathbf{q}_p$ are

$$\mathbf{M}_{p0} = \mathbf{M}_p(\theta_0), \qquad \mathbf{G}_{p0} = \begin{bmatrix} 0 & \gamma \\ -\gamma & 0 \end{bmatrix}, \qquad \mathbf{K}_{p0} = \begin{bmatrix} 0 & 0 \\ 0 & k_c + k_g \end{bmatrix}, \tag{9.1}$$

$$\gamma = \Omega_0\, J'_{\psi,p}(\theta_0), \qquad k_c = -\tfrac{1}{2}\Omega_0^2\, J''_{\psi,p}(\theta_0), \qquad k_g = -m_p g\,(d_K \sin\theta_0 + d_I \cos\theta_0), \tag{9.2}$$

(γ gyroscopic, k_c centrifugal, k_g gravitational stiffness; $\mathbf{G}_{p0}$ stays purely skew-symmetric also with products of inertia, which enter only through $M_{p,12}$). With zero products, $\gamma = (I_{KK} - I_{II})\,\Omega_0 \sin 2\theta_0$ and $k_c = -(I_{KK} - I_{II})\,\Omega_0^2 \cos 2\theta_0$. The linear model of the full system is

$$\big[\mathbf{M}_s + \mathbf{E}^{\mathsf{T}}\mathbf{M}_{p0}\mathbf{E}\big]\delta\ddot{\mathbf{q}} + \big[\mathbf{C} + \mathbf{E}^{\mathsf{T}}\mathbf{G}_{p0}\mathbf{E}\big]\delta\dot{\mathbf{q}} + \big[\mathbf{K} + \mathbf{E}^{\mathsf{T}}\mathbf{K}_{p0}\mathbf{E}\big]\delta\mathbf{q} = \mathbf{B}\,\delta\boldsymbol{\tau}; \tag{9.3}$$

its projection on the modal basis is the reduced linear model of Section 10. Table 6 lists the operating points ($k_g < 0$: the unbalanced telescope is statically unstable without control).

Table 6: Operating points: nominal polar torque and linearised telescope terms (N m, N m/rad, N m s/rad, kg m^2). *Result from placeholder parameters.*

θ_0 [deg]	Ω_0 [deg/s]	$\tau_{\theta 0}$	k_g	k_c	γ	$M_{p,12}$
−45	0	7.2105	8.314	0	0	1.31×10^{-6}
−45	10	7.2015	8.314	0.000457	0.103	1.31×10^{-6}
−15	0	10.401	3.595	0	0	9.83×10^{-7}
−15	10	10.397	3.595	0.0158	0.0491	9.83×10^{-7}
0	0	10.977	0.7802	0	0	7.13×10^{-7}
0	10	10.978	0.7802	0.0179	−0.00262	7.13×10^{-7}
15	0	10.805	−2.087	0	0	3.93×10^{-7}
15	10	10.81	−2.087	0.0153	−0.0536	3.93×10^{-7}
45	0	8.3139	−7.21	0	0	-3.02×10^{-7}
45	10	8.3229	−7.21	−0.000457	−0.103	-3.02×10^{-7}

10 Modal truncation (Gawronski): the default reduced model

Each axis is reduced by a modal truncation in the sense of Gawronski (*Modeling and Control of Antennas and Telescopes*, 2008, ch. 2 and 4; `reduction.method: modal`, module `modal.py`), built from the *full* FEM of each axis (9 azimuth nodes, 10 polar nodes with P): the retained modes keep their frequencies, shapes, modal masses and damping exactly, and the modes above the band of interest are dropped.

Modal basis. For each axis a, with the telescope inertia frozen at the attachment node in a reference configuration θ_{ref} and $k_g = 0$,

$$\bar{\mathbf{M}}_\psi = \mathbf{M}_\psi + \big[J^{ax} + J_{\psi,p}(\theta_{ref})\big]\mathbf{e}_L\mathbf{e}_L^{\mathsf{T}}, \qquad \bar{\mathbf{M}}_\theta = \mathbf{M}_\theta + I_{JJ}\,\mathbf{e}_P\mathbf{e}_P^{\mathsf{T}}, \qquad (\mathbf{K}_a - \omega^2\bar{\mathbf{M}}_a)\boldsymbol{\phi} = \mathbf{0}, \tag{10.1}$$

$$\boldsymbol{\Phi}_a = [\mathbf{1},\ \boldsymbol{\phi}_1,\ \ldots,\ \boldsymbol{\phi}_k], \qquad \boldsymbol{\Phi}_a^{\mathsf{T}}\bar{\mathbf{M}}_a\boldsymbol{\Phi}_a = \mathrm{diag}(J_{a,tot}, 1, \ldots, 1), \qquad \boldsymbol{\Phi}_a^{\mathsf{T}}\mathbf{K}_a\boldsymbol{\Phi}_a = \mathrm{diag}(0, \omega_1^2, \ldots, \omega_k^2). \tag{10.2}$$

The rigid mode $\mathbf{1}$ ($\mathbf{K}\mathbf{1} = \mathbf{0}$, Gawronski eq. 2.30) is not normalised, so η_0 is an angle (the inertia-weighted mean rotation of the chain) with modal mass $J_{a,tot} = \mathbf{1}^{\mathsf{T}}\bar{\mathbf{M}}_a\mathbf{1}$; the flexible modes are mass-normalised. The rows of $\boldsymbol{\Phi}$ at the encoder, the motor and the attachment node are Gawronski's output and input vectors (eq. 2.21–2.22), $\mathbf{c}_e$, $\mathbf{c}_m$, and the new $\mathbf{c}_L$ (azimuth node 9) and $\mathbf{c}_P$, all with first component 1. With $\mathbf{C} = \beta\mathbf{K} + \mathbf{C}_b$: $\zeta_k = \beta\omega_k/2$ exactly, and the bearing friction is kept as a full matrix, $\tilde{\mathbf{D}} = \mathrm{diag}(0, 2\zeta_k\omega_k) + \boldsymbol{\Phi}^{\mathsf{T}}\mathbf{C}_b\boldsymbol{\Phi}$.

Nonlinear equations in modal coordinates. The telescope is De Donatis' rigid body attached at L and P through $[\psi_L,\ \theta_p]^\mathsf{T} = \mathbf{E}_\eta \boldsymbol{\eta}$, $\mathbf{E}_\eta = \mathrm{blkdiag}(\mathbf{c}_L^\mathsf{T}, \mathbf{c}_P^\mathsf{T})$:

$$\tilde{\mathbf{M}}\ddot{\boldsymbol{\eta}} + \tilde{\mathbf{D}}\dot{\boldsymbol{\eta}} + \tilde{\mathbf{K}}\boldsymbol{\eta} + \mathbf{E}_\eta^\mathsf{T}\Big\{\big[\mathbf{M}_p(\theta_p) - \bar{\mathbf{M}}_p\big]\ddot{\mathbf{q}}_p + \mathbf{C}_p\dot{\mathbf{q}}_p + \mathbf{g}_p(\theta_p)\Big\} = \boldsymbol{\Phi}^\mathsf{T}\mathbf{B}\boldsymbol{\tau}, \tag{10.3}$$

with $\bar{\mathbf{M}}_p = \mathrm{diag}(J_{\psi,p}(\theta_{ref}), I_{JJ})$ the inertia already in the modes. $\mathbf{M}_p$, $\mathbf{C}_p$, $\mathbf{g}_p$ come from the Lagrangian of the telescope alone, derived by SymPy with the same kinematics as Section 8, and are compiled with `lambdify`. Keeping all modes, (10.3) is $\boldsymbol{\Phi}^\mathsf{T}$ times the equations of the coupled model with *all* FEM degrees of freedom (the exact reference, 9 + 10 coordinates, verified to 10^{-10} on random states, test T2).

Linear model. At each operating point the basis is rebuilt with $\theta_{ref} = \theta_0$, and the linearisation is the modal Gawronski model perturbed by three low-rank corrections built from the attachment vectors (document `MODELLO_unificato`, eq. 7.2):

$$\big[\tilde{\mathbf{M}} + M_{12}(\hat{\mathbf{c}}_L\hat{\mathbf{c}}_P^\mathsf{T} + \hat{\mathbf{c}}_P\hat{\mathbf{c}}_L^\mathsf{T})\big]\delta\ddot{\boldsymbol{\eta}} + \big[\tilde{\mathbf{D}} + \gamma(\hat{\mathbf{c}}_L\hat{\mathbf{c}}_P^\mathsf{T} - \hat{\mathbf{c}}_P\hat{\mathbf{c}}_L^\mathsf{T})\big]\delta\dot{\boldsymbol{\eta}} + \big[\tilde{\mathbf{K}} + (k_c + k_g)\hat{\mathbf{c}}_P\hat{\mathbf{c}}_P^\mathsf{T}\big]\delta\boldsymbol{\eta} = \boldsymbol{\Phi}^\mathsf{T}\mathbf{B}\,\delta\boldsymbol{\tau}. \tag{10.4}$$

$k_c + k_g$ is kept as a rank-1 coupling on θ_p, not inside the modes. In state space, $\mathbf{x}_i = [\omega_i\eta_i,\ \dot{\eta}_i]$ for the flexible modes and $[\eta_0,\ \dot{\eta}_0]$ for the rigid ones give Gawronski's diagonal blocks $\left[\begin{smallmatrix} 0 & \omega_i \\ -\omega_i & -2\zeta_i\omega_i \end{smallmatrix}\right]$ (eq. 2.25) and $\left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$ (eq. 2.32), plus off-diagonal terms from the corrections; with all modes the FRF equals the full model to 10^{-8} (test T3), and with the rigid modes alone it is the linearised two-DOF (De Donatis) model (test T4).

Choice of the modes. Dynamics above $f_{band} = 2000\,\mathrm{Hz}$ are not of interest. The rigid modes are always kept (infinite norm, Gawronski eq. 4.11–4.12); flexible modes are kept if $f_i \le 2\,f_{band}$ and, optionally, if their Hankel norm for the position output exceeds a fraction of the largest,

$$\|G_i\|_h \simeq \frac{|\phi_i(m)\,\phi_i(e)|}{4\zeta_i\omega_i^2} \qquad \text{(Gawronski eq. 4.4–4.6, } \boldsymbol{\Phi} \text{ mass-normalised)}. \tag{10.5}$$

A validation loop then compares the truncated linear model with the full one at every operating point (torque→encoder, up to f_{band}): direct channels within 1 dB and 5° outside ±2 % bands around the reference zeros, each excluded zero within 1 % in frequency (near a narrow antiresonance the FRF ratio is hypersensitive and is not used alone), and the 2×2 matrix within 12 % in the 2-norm. If a point fails, the discarded mode with the largest Hankel norm is added and the loop repeats. Here 1 mode(s) were added: azimuth mode 2 (4595.1 Hz). The model has 16 states.

Table 7: Modes of the full FEM of each axis at θ_{ref} (consistent mass, telescope frozen, $k_g = 0$): frequency, damping, participation at encoder, motor and attachment node, Hankel norm (rad/(N m)), and whether the mode is kept. Modal mass: J_{tot} for the rigid mode, 1 for the flexible ones. *Result from placeholder parameters.*

axis	mode	f [Hz]	ζ	c_e	c_m	c_L/c_P	$\|G_i\|_h$	kept
azimuth	0	0	0	1	1	1	∞	rigid
	1	752.184	0.01	52.33	33.54	−0.00893	0.00196	cut-off
	2	4595.13	0.0611	−12.59	147	−0.001153	9.09×10^{-6}	validation
	3	21343.7	0.284	0.1758	−12.45	−0.001637	1.07×10^{-10}	no
	4	29518.9	0.392	1416	0.09145	1.082×10^{-7}	2.4×10^{-9}	no
	5	47898.7	0.637	0.0255	3.527	−0.001475	3.9×10^{-13}	no
polar	0	0	0	1	1	1	∞	rigid
	1	109.467	0.01	6.53	11.28	−0.1564	0.00389	cut-off
	2	173.103	0.0158	−3.164	22.07	0.02966	0.000933	cut-off
	3	1592.54	0.145	238.4	−0.0323	2.51×10^{-5}	1.32×10^{-7}	cut-off
	4	2784.78	0.254	0.01962	142.2	1.429×10^{-6}	8.95×10^{-9}	cut-off
	5	19595.1	1.79	0.0001561	−27.67	5.084×10^{-9}	3.98×10^{-14}	no

Table 8: Truncated modal model against the full-DOF model, torque→encoder up to 2000 Hz: worst direct-channel error (dB, degrees) outside the zero bands, worst relative error of the excluded zeros, worst relative 2-norm error of the 2×2 matrix. *Result from placeholder parameters.*

		modal			
θ_0	Ω_0	dB	deg	zeros	2×2
−45	0	0.0086	0.00062	2.3×10^{-5}	0.00022
−45	10	0.0086	0.00062	2.3×10^{-5}	0.00022
−15	0	0.0095	0.00062	2.3×10^{-5}	0.00022
−15	10	0.0095	0.00062	2.3×10^{-5}	0.00022
0	0	0.0095	0.00062	2.3×10^{-5}	0.00022
0	10	0.0095	0.00062	2.3×10^{-5}	0.00022
15	0	0.0087	0.00062	2.3×10^{-5}	0.00022
15	10	0.0087	0.00062	2.3×10^{-5}	0.00022
45	0	0.0099	0.00062	2.3×10^{-5}	0.00022
45	10	0.0099	0.00062	2.3×10^{-5}	0.00022

Residual flexibility of the discarded modes. A modal truncation keeps the poles of the retained modes but moves the zeros, which depend on the static flexibility of the discarded ones. For each axis the report gives the collocated and non-collocated residual flexibility $\sum_{i\ \text{disc.}} \phi_i(m)^2/\omega_i^2$ and $\sum_{i\ \text{disc.}} \phi_i(e)\phi_i(m)/\omega_i^2$ and the zero shift they cause on the isolated axis. The option `static_correction: residual_vectors` adds, for every loaded node (motor and attachment), the residual static vector $\mathbf{r}_j = \sum_{i\ \text{disc.}} \boldsymbol{\phi}_i \phi_i(j)/\omega_i^2$ (residual flexibility with inertia relief, the only one defined for a free chain), orthogonalises it and solves the eigenproblem in the enlarged space: the model stays a valid ODE also in the nonlinear case, and the retained frequencies do not move.

Table 9: Zeros of the isolated axes (Hz): full FEM, pure truncation, truncation plus static correction; residual flexibilities in rad/(N m). *Result from placeholder parameters.*

axis	transfer function	full	truncated	+ static corr.
azimuth	$\tau \to$ encoder	12.2558	12.256	12.2558
azimuth	$\tau \to$ motor	12.2558	12.2615	12.2558
azimuth	$\tau \to$ motor	1257.06	1258.03	1257.06
		$F_{coll} = 7.14 \times 10^{-8}$, $F_{non\text{-}coll} = 3.44 \times 10^{-9}$		
polar	$\tau \to$ encoder	16.7815	16.7815	16.7815
polar	$\tau \to$ encoder	504.488	504.488	504.488
polar	$\tau \to$ motor	6.12751	6.12799	6.12751
polar	$\tau \to$ motor	124.481	124.483	124.481
polar	$\tau \to$ motor	504.488	504.864	504.488
		$F_{coll} = 1.17 \times 10^{-7}$, $F_{non\text{-}coll} = -2.58 \times 10^{-13}$		

Mesh check. For the retained modes the lumped and consistent mass give azimuth 1: −0.02 %, azimuth 2: −0.48 %, polar 1: −0.00 %, polar 2: −0.00 %, polar 3: −0.00 %, polar 4: −0.36 % (a warning is raised above 2 %).

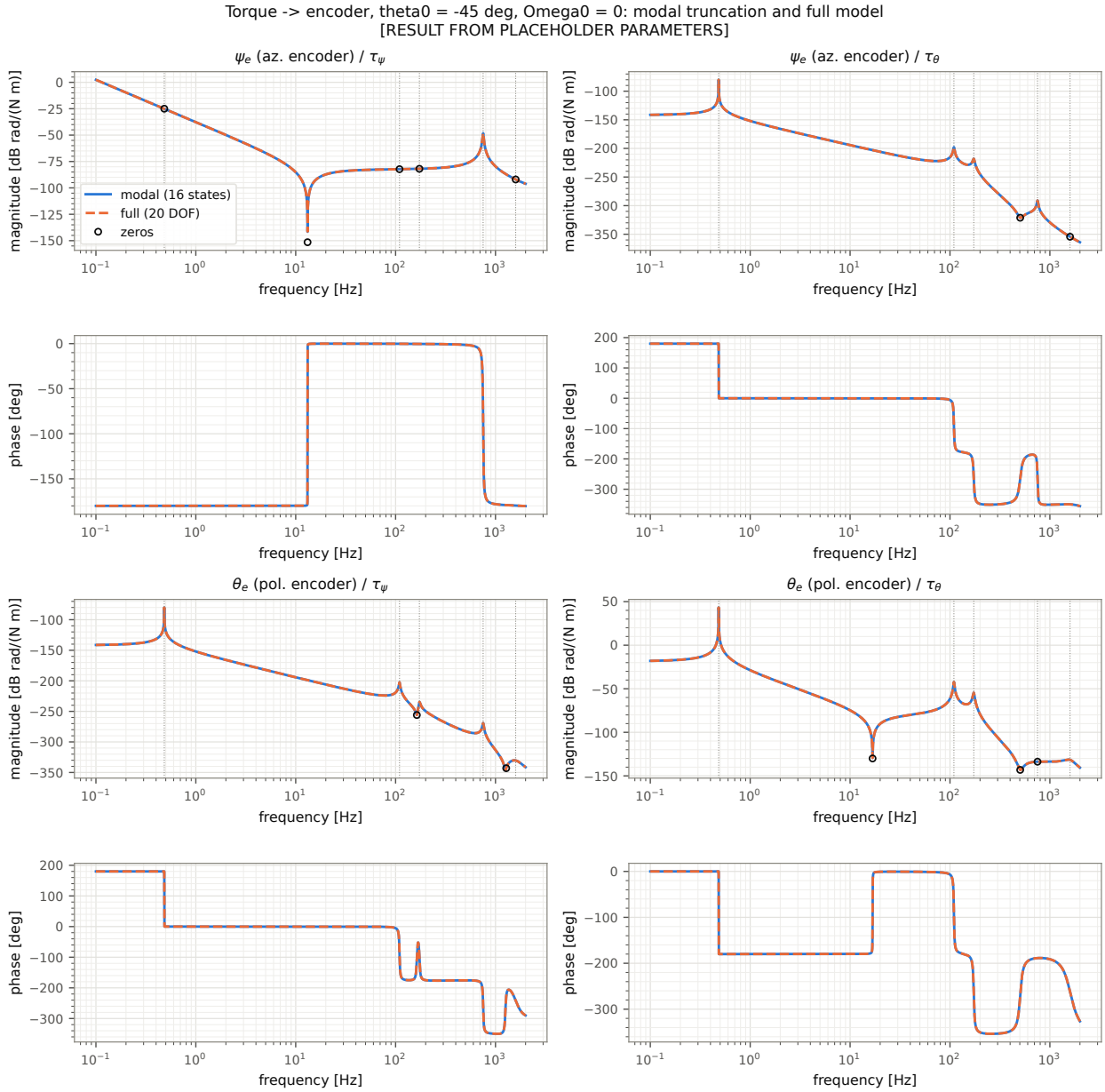


Figure 5: Torque→encoder 2×2 matrix up to the band of interest: truncated modal model (with its poles, dotted, and invariant zeros, circles) and full-DOF reference. *Result from placeholder parameters.*

10.1 Results of the reduced linear model

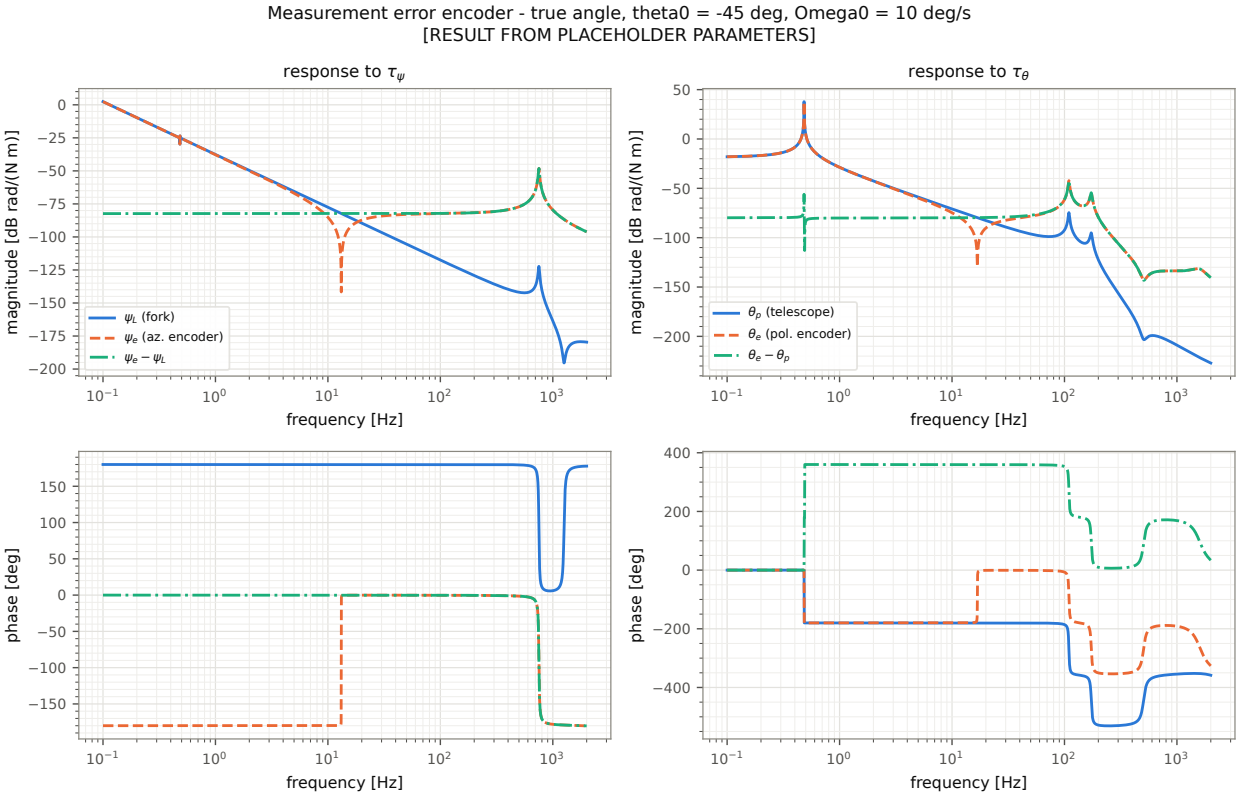


Figure 6: Torque to true pointing (ψ_L, θ_p), to the encoders and to the measurement error (encoder minus true angle), modal model. *Result from placeholder parameters.*

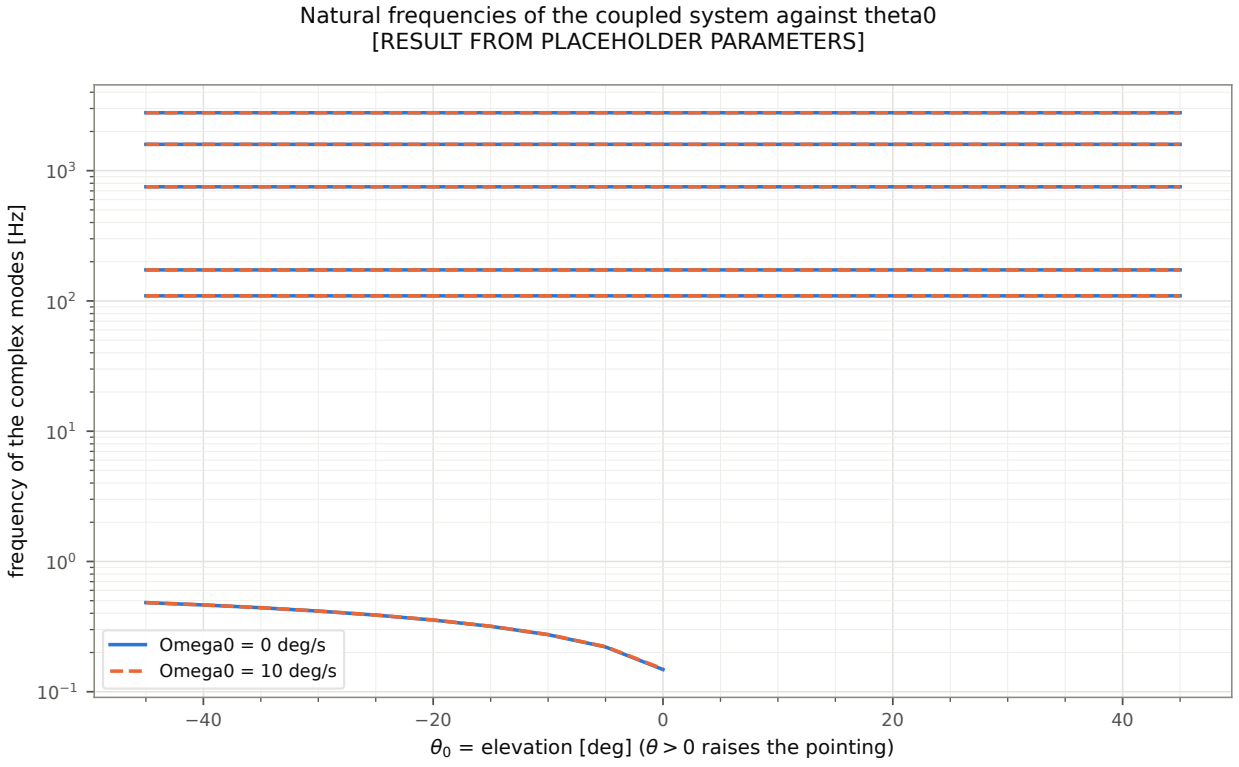


Figure 7: Natural frequencies of the coupled modal model as a function of θ_0. *Result from placeholder parameters.*

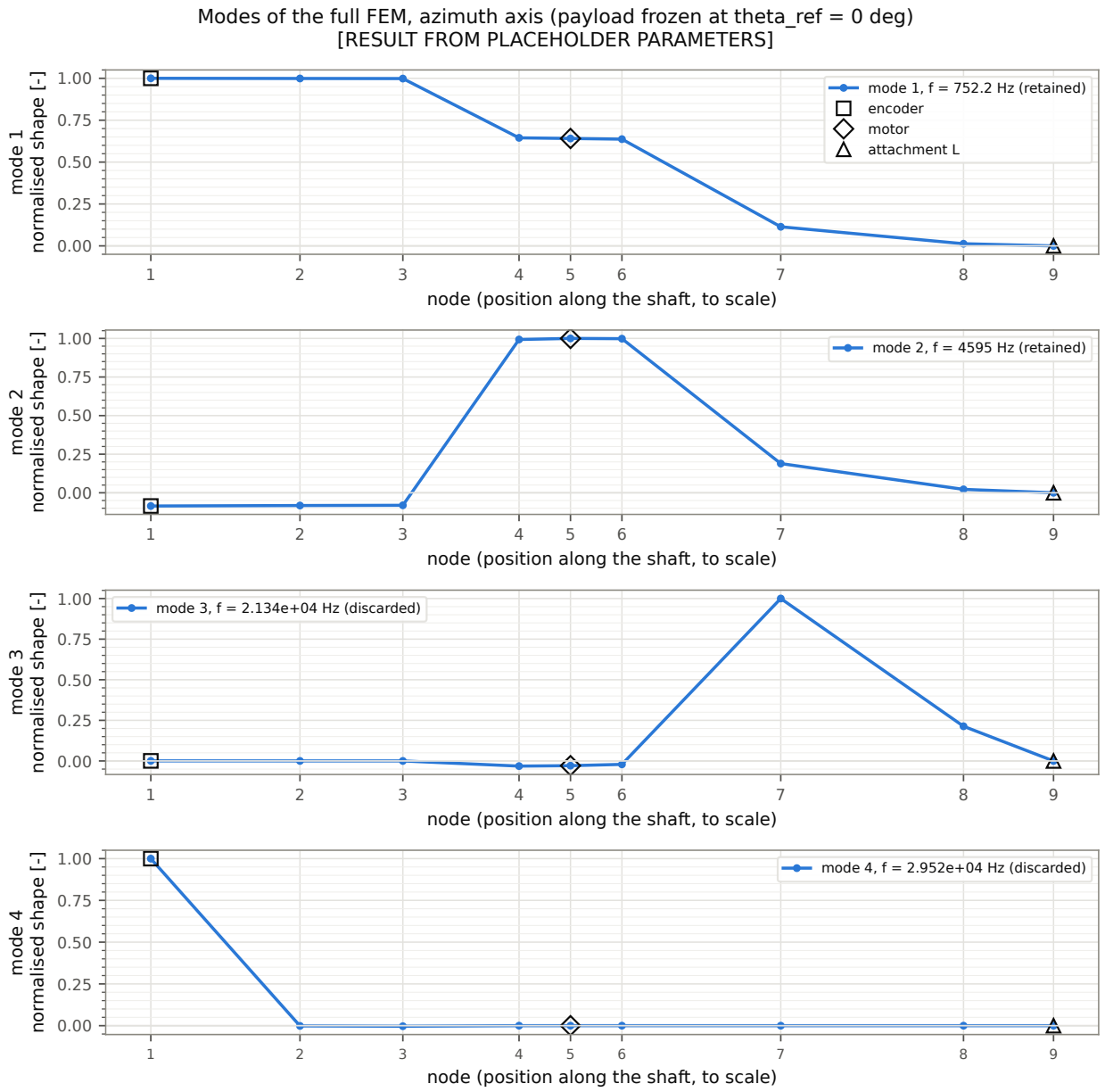


Figure 8: First modes of the full FEM of the azimuth axis (basis of the truncation), retained or discarded. *Result from placeholder parameters.*

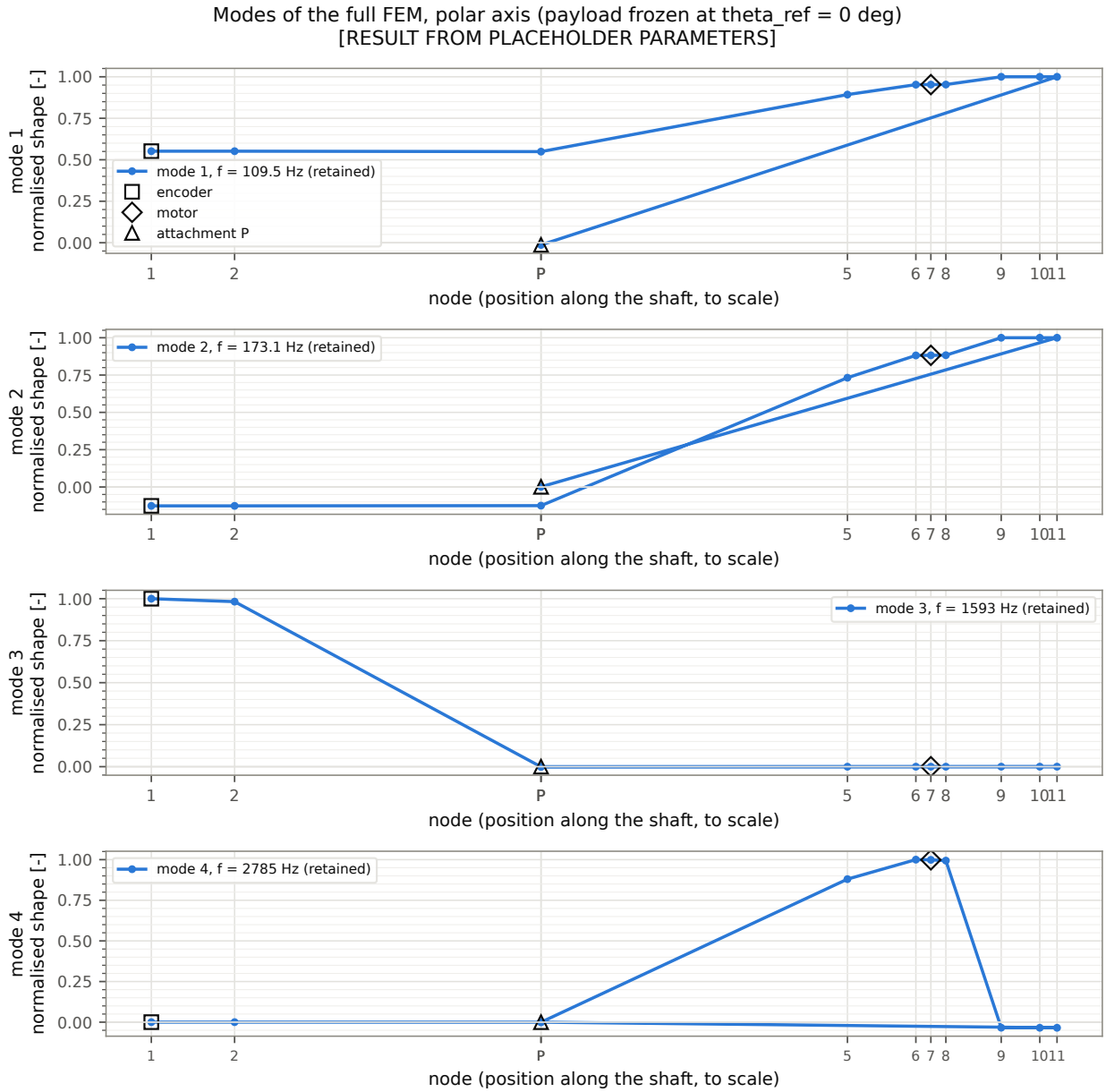


Figure 9: First modes of the full FEM of the polar axis (basis of the truncation), retained or discarded. *Result from placeholder parameters.*

Table 10: Poles and invariant zeros of the torque→encoder matrix at $\theta_0 = -45°$, $\Omega_0 = 0°/\mathrm{s}$ (only roots with Im ≥ 0).

quantity	Re [rad/s]	Im [rad/s]	$\lvert\lambda\rvert/2\pi$ [Hz]	type
pole	-1.2×10^{-5}	0	1.9093×10^{-6}	reale_stabile
pole	1.2×10^{-5}	0	1.9093×10^{-6}	reale_instabile
pole	-5.965×10^{-11}	3.0367	0.4833	complesso
pole	−6.878	687.77	109.47	complesso
pole	−17.2	1087.5	173.1	complesso
pole	−47.26	4725.9	752.19	complesso
pole	−1456	9899.8	1592.5	complesso
pole	−4451	16922	2784.8	complesso
pole	−1764	28818	4595.1	complesso
zero $G_{\psi_e \tau_\psi}$	-8.701×10^{-11}	3.0367	0.4833	complesso
zero $G_{\psi_e \tau_\psi}$	−0.01442	82.555	13.139	complesso
zero $G_{\psi_e \tau_\psi}$	−6.878	687.77	109.47	complesso

quantity	Re [rad/s]	Im [rad/s]	$\lvert\lambda\rvert/2\pi$ [Hz]	type
zero $G_{\psi_e\tau_\psi}$	-17.2	1087.5	173.1	complesso
zero $G_{\psi_e\tau_\psi}$	-1456	9899.8	1592.5	complesso
zero $G_{\psi_e\tau_\psi}$	-4451	16922	2784.8	complesso
zero $G_{\psi_e\tau_\psi}$	-9.448×10^{4}	0	15038	reale_stabile
zero $G_{\psi_e\tau_\psi}$	1.574×10^{5}	0	25056	reale_instabile
zero $G_{\psi_e\tau_\theta}$	1.799×10^{-15}	0	0	nullo
zero $G_{\psi_e\tau_\theta}$	0.0005344	0	8.5045×10^{-5}	reale_instabile
zero $G_{\psi_e\tau_\theta}$	-146	3166.9	504.57	complesso
zero $G_{\psi_e\tau_\theta}$	-1454	9895.5	1591.8	complesso
zero $G_{\psi_e\tau_\theta}$	-2.94×10^{4}	0	4679.7	reale_stabile
zero $G_{\psi_e\tau_\theta}$	-4.085×10^{4}	8991.7	6656.7	complesso
zero $G_{\psi_e\tau_\theta}$	1.082×10^{5}	0	17219	reale_instabile
zero $G_{\psi_e\tau_\theta}$	2.062×10^{5}	0	32810	reale_instabile
zero $G_{\theta_e\tau_\psi}$	-1.051×10^{-10}	0	0	nullo
zero $G_{\theta_e\tau_\psi}$	-0.01368	0	0.002177	reale_stabile
zero $G_{\theta_e\tau_\psi}$	-15.28	1025.1	163.17	complesso
zero $G_{\theta_e\tau_\psi}$	-85.19	8042.9	1280.1	complesso
zero $G_{\theta_e\tau_\psi}$	-4451	16921	2784.7	complesso
zero $G_{\theta_e\tau_\psi}$	-3.161×10^{4}	0	5031	reale_stabile
zero $G_{\theta_e\tau_\psi}$	-3.303×10^{4}	0	5256.7	reale_stabile
zero $G_{\theta_e\tau_\psi}$	-7.552×10^{4}	0	12019	reale_stabile
zero $G_{\theta_e\tau_\psi}$	1.11×10^{5}	0	17664	reale_instabile
zero $G_{\theta_e\tau_\psi}$	5.292×10^{5}	2.8624×10^{5}	95756	complesso
zero $G_{\theta_e\tau_\psi}$	-5.272×10^{8}	0	8.3899×10^{7}	reale_stabile
zero $G_{\theta_e\tau_\theta}$	-9.351×10^{-12}	0	0	nullo
zero $G_{\theta_e\tau_\theta}$	2.058×10^{-10}	0	0	nullo
zero $G_{\theta_e\tau_\theta}$	-0.1616	105.49	16.789	complesso
zero $G_{\theta_e\tau_\theta}$	-146.1	3166.4	504.49	complesso
zero $G_{\theta_e\tau_\theta}$	-47.26	4725.9	752.19	complesso
zero $G_{\theta_e\tau_\theta}$	-1764	28818	4595.1	complesso
zero $G_{\theta_e\tau_\theta}$	-3.066×10^{4}	2702.5	4898.3	complesso
zero $G_{\theta_e\tau_\theta}$	1.735×10^{5}	1.5055×10^{5}	36559	complesso
zero $G_{\theta_e\tau_\theta}$	9.662×10^{9}	0	1.5378×10^{9}	reale_instabile
zero $G_{\theta_e\tau_\theta}$	-9.663×10^{9}	0	1.5378×10^{9}	reale_stabile

11 Lumped-parameter model (3 + 5 DOF)

Topology. A reduced model with the nodes of the original topology only:

$$\text{azimuth: } 1 \overset{e_{\psi,1}}{\longleftrightarrow} 5 \overset{e_{\psi,2}}{\longleftrightarrow} 9, \qquad \text{polar: } 1 \overset{e_{\theta,1}}{\longleftrightarrow} \mathrm{H} \overset{e_{\theta,2}}{\longleftrightarrow} 7 \overset{e_{\theta,3}}{\longleftrightarrow} 11, \quad \mathrm{H} \overset{e_{\theta,4}}{\longleftrightarrow} \mathrm{P}, \tag{11.1}$$

with coordinates $\mathbf{q} = [\psi_e,\ \psi_m,\ \psi_L,\ \theta_e,\ \theta_h,\ \theta_m,\ \theta_r,\ \theta_p]^{\mathsf{T}}$ (encoder, motor, load / fork; encoder, hub centre, motor, rear end, telescope). In the hub model the polar chain is a tree: the telescope node P hangs from the hub centre H through the attachment spring $e_{\theta,4}$ ($k_{\theta,4} = k_b = 9.8\times10^{3}\,\mathrm{N\,m/rad}$), outside the path encoder–motor. Each equivalent element condenses the FEM elements between two consecutive nodes (module `concentrato.py`):

$$J_{eq} = \sum_i J_i, \qquad \frac{1}{k_{eq}} = \sum_i \frac{1}{k_i} \quad \text{(springs in series)}, \tag{11.2}$$

and any concentrated inertia of an intermediate node is added to J_{eq}. The model is a lumped one: each node carries its own concentrated inertia plus half of the inertia of the adjacent equivalent elements, and the elements are massless springs with the structural damper $c = \beta k$ in parallel (β tuned on the first flexible mode of each isolated lumped axis, as for the other models). The dynamic coupling between the axes is the one of the coupled model (Section 8): the fork turns with ψ_L, the telescope is the rigid body attached at P, which hangs from the hub centre H through k_b, and the polar shafts add the constant J^{ax} about $\hat{k}$ (here $0.08237\,\mathrm{kg\,m^2}$).

Table 11: Equivalent elements: condensed FEM elements, inertia $J_{eq} = \sum J_i$ (kg m^2), series stiffness $1/k_{eq} = \sum 1/k_i$ (N m/rad) and structural damping $c = \beta k$ (N m s/rad). *Result from placeholder parameters.*

axis	element	nodes	FEM elements	J_{eq}	k_{eq}	c
ψ	1	1–5	1, 2, 3, 4	0.0003699	9547.6	0.03926
ψ	2	5–9	5, 6, 7, 8	5.46×10^{-5}	13091	0.05383
θ	1	1–H	1, 2	1.788×10^{-5}	1742.7	0.05052
θ	2	H–7	4, 5, 6	2.509×10^{-5}	1535.5	0.04451
θ	3	7–11	7, 8, 9, 10	0.001288	9694.2	0.281
θ	4	H–P	b	0	9800	0.2841

Table 12: Lumped node inertias $J = J^n + \frac{1}{2}\sum_{e \ni \text{node}} J_{eq,e}$ (kg m^2), with J^n the concentrated inertia of the node (fork at azimuth node 9), and bearing friction b (N m s/rad). The telescope is not included: it enters through its own terms. *Result from placeholder parameters.*

symbol	node	J	J^n	b
$J_{\psi,e}$	1	0.00018494	0	0
$J_{\psi,m}$	5	0.00021224	0	0
$J_{\psi,L}$	9	1.2522	1.252	0
$J_{\theta,e}$	1	8.9416×10^{-6}	0	0
$J_{\theta,h}$	H	0.01883	0.01881	0
$J_{\theta,m}$	7	0.00065669	0	0
$J_{\theta,r}$	11	0.00064415	0	0
$J_{\theta,p}$	P	0	0	0

Table 13: Flexible modes of the isolated lumped axes against the full FEM (telescope and fork included, $\theta_0 = 0$, $k_g = 0$ for the polar axis). *Result from placeholder parameters.*

axis	mode	lumped [Hz]	full FEM [Hz]	difference
azimuth	1	774.044	752.184	2.91 %
	2	1846.74	4595.13	−59.8 %
polar	1	109.811	109.467	0.314 %
	2	179.216	173.103	3.53 %
	3	886.405	1592.54	−44.3 %
	4	2222.44	2784.78	−20.2 %

Halving the inertia of an equivalent element puts part of a heavy disc on the node at the other end (polar element 2 on the encoder node, element 9 on the rear end): this, and the missing intermediate masses, is why the lumped frequencies differ from the full FEM. The lumped stiffness chain is the exact static condensation (the same $\mathbf{K}_r$ as a static reduction on these nodes).

11.1 Newton–Euler equations

Each degree of freedom gets a moment balance on its node: the rate of change of the angular momentum equals the external moments about the axis of the node. The torque transmitted by an equivalent element, positive on its second node, is

$$T_{\psi,1} = k_{\psi,1}(\psi_e - \psi_m) + c_{\psi,1}(\dot\psi_e - \dot\psi_m), \qquad T_{\psi,2} = k_{\psi,2}(\psi_m - \psi_L) + c_{\psi,2}(\dot\psi_m - \dot\psi_L), \tag{11.3}$$

$$T_{\theta,1} = k_{\theta,1}(\theta_e - \theta_h) + c_{\theta,1}(\dot\theta_e - \dot\theta_h), \qquad T_{\theta,2} = k_{\theta,2}(\theta_h - \theta_m) + c_{\theta,2}(\dot\theta_h - \dot\theta_m), \tag{11.4}$$

$$T_{\theta,3} = k_{\theta,3}(\theta_m - \theta_r) + c_{\theta,3}(\dot\theta_m - \dot\theta_r), \qquad T_{\theta,4} = k_{\theta,4}(\theta_h - \theta_p) + c_{\theta,4}(\dot\theta_h - \dot\theta_p), \tag{11.5}$$

and $b\,\dot q$ is the viscous friction of a bearing (azimuth: to the ground; polar: to the fork).

Chain nodes. For the azimuth nodes the axis $\hat k$ is fixed. The polar nodes are axisymmetric about $\hat J$, which turns with the fork: $\dot{\hat J} = \dot\psi_L \hat k \times \hat J \perp \hat J$, so the axial component of Euler's equation of each polar node is simply $J\ddot\theta$ (its gyroscopic terms are orthogonal to $\hat J$ and are carried by the bearings).

Telescope. Both rotation axes pass through o_2, a fixed point of the telescope, so its Euler equation can be written about o_2 with the tensor $\mathbf{I}_{p,o_2}$: $\dot{\mathbf{H}} = \mathbf{M}_{o_2}$, $\mathbf{H} = \mathbf{I}_{p,o_2}\boldsymbol{\omega}_2$, with $\boldsymbol{\omega}_2$ of Section 8 and, in body components, $\dot{\mathbf{H}}|_N = \dot{\mathbf{H}}|_2 + \boldsymbol{\omega}_2 \times \mathbf{H}$. *Azimuth node L* (fork, polar shafts and telescope, rotating about the fixed $\hat{z}$): only the component along $\hat{z}$ is needed, and the reaction of the polar motor and of the polar bearings is orthogonal to $\hat{z}$, so

$$\frac{d}{dt}\Big[(J_{\psi,L} + J^{ax})\dot{\psi}_L + \hat{z}\cdot\mathbf{H}\Big] = T_{\psi,2} - b_{\psi,L}\dot{\psi}_L, \tag{11.6}$$

$$\hat{z}\cdot\mathbf{H} = \left[I_{II}\cos^2(\theta_p) + I_{IK}\sin(2\theta_p) + I_{KK}\sin^2(\theta_p)\right]\dot{\psi}_L + \left[I_{IJ}\cos(\theta_p) + I_{JK}\sin(\theta_p)\right]\dot{\theta}_p. \tag{11.7}$$

Polar node P (telescope, off the shaft chain): the component of Euler's equation along $\hat{J}$ contains the torque $T_{\theta,4}$ of the attachment spring and the moment of gravity about o_2, $\mathbf{M}_g = \mathbf{r}_G \times (-m_p g\,\hat{z})$; the moments orthogonal to $\hat{J}$ are carried by the bearings in the fork.

Equations of motion, one per degree of freedom. With

$$J_{\psi,p}(\theta_p) = I_{II}\cos^2(\theta_p) + I_{IK}\sin(2\theta_p) + I_{KK}\sin^2(\theta_p), \qquad M_{12}(\theta_p) = I_{IJ}\cos(\theta_p) + I_{JK}\sin(\theta_p), \tag{11.8}$$

$$J'_{\psi,p}(\theta_p) = -I_{II}\sin(2\theta_p) + 2I_{IK}\cos(2\theta_p) + I_{KK}\sin(2\theta_p), \qquad M'_{12}(\theta_p) = -I_{IJ}\sin(\theta_p) + I_{JK}\cos(\theta_p), \tag{11.9}$$

$$M_{g,J}(\theta_p) = -g m_p\left(-d_I\sin(\theta_p) + d_K\cos(\theta_p)\right), \tag{11.10}$$

the nonlinear equations are

$$(\psi_e)\quad J_{\psi,e}\,\ddot{\psi}_e = -T_{\psi,1} - b_{\psi,e}\dot{\psi}_e, \tag{11.11}$$

$$(\psi_m)\quad J_{\psi,m}\,\ddot{\psi}_m = \tau_\psi + T_{\psi,1} - T_{\psi,2} - b_{\psi,m}\dot{\psi}_m, \tag{11.12}$$

$$(\psi_L)\quad \left[J_{\psi,L} + J^{ax} + J_{\psi,p}(\theta_p)\right]\ddot{\psi}_L + M_{12}(\theta_p)\,\ddot{\theta}_p + J'_{\psi,p}(\theta_p)\,\dot{\psi}_L\dot{\theta}_p + M'_{12}(\theta_p)\,\dot{\theta}_p^2 = T_{\psi,2} - b_{\psi,L}\dot{\psi}_L, \tag{11.13}$$

$$(\theta_e)\quad J_{\theta,e}\,\ddot{\theta}_e = -T_{\theta,1} - b_{\theta,e}\dot{\theta}_e, \tag{11.14}$$

$$(\theta_h)\quad J_{\theta,h}\,\ddot{\theta}_h = T_{\theta,1} - T_{\theta,2} - T_{\theta,4} - b_{\theta,h}\dot{\theta}_h, \tag{11.15}$$

$$(\theta_p)\quad \left[J_{\theta,p} + I_{JJ}\right]\ddot{\theta}_p + M_{12}(\theta_p)\,\ddot{\psi}_L - \tfrac{1}{2}J'_{\psi,p}(\theta_p)\,\dot{\psi}_L^2 = T_{\theta,4} + M_{g,J}(\theta_p) - b_{\theta,p}\dot{\theta}_p, \tag{11.16}$$

$$(\theta_m)\quad J_{\theta,m}\,\ddot{\theta}_m = \tau_\theta + T_{\theta,2} - T_{\theta,3} - b_{\theta,m}\dot{\theta}_m, \tag{11.17}$$

$$(\theta_r)\quad J_{\theta,r}\,\ddot{\theta}_r = T_{\theta,3} - b_{\theta,r}\dot{\theta}_r. \tag{11.18}$$

The only nonlinearities are the trigonometric functions of θ_p in $J_{\psi,p}$, M_{12}, $M_{g,J}$ and the products of velocities: $J'_{\psi,p}\dot{\psi}_L\dot{\theta}_p$ (Coriolis, on the azimuth), $-\frac{1}{2}J'_{\psi,p}\dot{\psi}_L^2$ (centrifugal, on the polar axis) and $M'_{12}\dot{\theta}_p^2$ (from the products of inertia). Written out with the trigonometric functions, (11.13) and (11.16) read

$$\Big[J_{\psi,L} + J^{ax} + I_{II}\cos^2(\theta_p) + I_{IK}\sin(2\theta_p) + I_{KK}\sin^2(\theta_p)\Big]\ddot{\psi}_L + \Big[I_{IJ}\cos(\theta_p) + I_{JK}\sin(\theta_p)\Big]\ddot{\theta}_p$$
$$+ \Big[-I_{II}\sin(2\theta_p) + 2I_{IK}\cos(2\theta_p) + I_{KK}\sin(2\theta_p)\Big]\dot{\psi}_L\dot{\theta}_p + \Big[-I_{IJ}\sin(\theta_p) + I_{JK}\cos(\theta_p)\Big]\dot{\theta}_p^2 = T_{\psi,2} - b_{\psi,L}\dot{\psi}_L, \tag{11.19}$$

$$\left[J_{\theta,p} + I_{JJ}\right]\ddot{\theta}_p + \Big[I_{IJ}\cos(\theta_p) + I_{JK}\sin(\theta_p)\Big]\ddot{\psi}_L + \Big[\frac{I_{II}\sin(2\theta_p)}{2} - I_{IK}\cos(2\theta_p) - \frac{I_{KK}\sin(2\theta_p)}{2}\Big]\dot{\psi}_L^2$$
$$= T_{\theta,4} - g m_p\left(-d_I\sin(\theta_p) + d_K\cos(\theta_p)\right) - b_{\theta,p}\dot{\theta}_p. \tag{11.20}$$

Verification. The equations above are derived by SymPy from the moment balances (function `newton_eulero` of `concentrato.py`), rewritten in the compact form (11.11)–(11.18) and checked term by term against both the expanded form and Lagrange's equations of the coupled model (Section 8) with the lumped chain: the differences are identically zero. In matrix form they are

$$\mathbf{M}(\theta_p)\ddot{\mathbf{q}} + \mathbf{C}(\theta_p,\dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{C}_d\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} - M_{g,J}(\theta_p)\,\mathbf{e}_{\theta_p} = \mathbf{E}\boldsymbol{\tau}, \tag{11.21}$$

where $\mathbf{M}$ differs from $\mathrm{diag}(J_{\psi,e},\ldots,J_{\theta,p})$ only in the entries (ψ_L,ψ_L), (ψ_L,θ_p), (θ_p,θ_p).

11.2 Transfer functions

Symbolic transfer functions of the isolated chains ($\Omega_0 = 0$, no damping, zero products of inertia). For the lumped model the chain is exactly the inertia–spring–inertia system, so $\mathbf{P}(s) = \mathrm{diag}(J_i)s^2 + \mathbf{K}$ and

$G_{im}(s) = [\operatorname{adj}\mathbf{P}(s)]_{im}/\det\mathbf{P}(s)$ hold without approximation. Azimuth (torque on ψ_m; $J^*_{\psi,L} = J_{\psi,L} + J^{ax} + J_{\psi,p}(\theta_0)$ written $J_{\psi,L}$):

$$\mathbf{P}_\psi(s) = \begin{bmatrix} J_{\psi,e}s^2 + k_{\psi,1} & -k_{\psi,1} & 0 \\ -k_{\psi,1} & J_{\psi,m}s^2 + k_{\psi,1} + k_{\psi,2} & -k_{\psi,2} \\ 0 & -k_{\psi,2} & J_{\psi,L}s^2 + k_{\psi,2} \end{bmatrix}, \qquad \Delta_\psi(s) = \det\mathbf{P}_\psi(s), \tag{11.22}$$

$$\frac{\Psi_e}{T_\psi} = \frac{k_{\psi,1}\left(J_{\psi,L}s^2 + k_{\psi,2}\right)}{\Delta_\psi}, \qquad \frac{\Psi_m}{T_\psi} = \frac{\left(J_{\psi,L}s^2 + k_{\psi,2}\right)\left(J_{\psi,e}s^2 + k_{\psi,1}\right)}{\Delta_\psi}, \qquad \frac{\Psi_L}{T_\psi} = \frac{k_{\psi,2}\left(J_{\psi,e}s^2 + k_{\psi,1}\right)}{\Delta_\psi}, \tag{11.23}$$

$$\Delta_\psi(s) = s^2\left(a_4 s^4 + a_2 s^2 + a_0\right), \qquad a_4 = J_{\psi,L}J_{\psi,e}J_{\psi,m}, \tag{11.24}$$

$$a_2 = J_{\psi,L}J_{\psi,e}k_{\psi,1} + J_{\psi,L}J_{\psi,e}k_{\psi,2} + J_{\psi,L}J_{\psi,m}k_{\psi,1} + J_{\psi,e}J_{\psi,m}k_{\psi,2}, \qquad a_0 = k_{\psi,1}k_{\psi,2}\left(J_{\psi,L} + J_{\psi,e} + J_{\psi,m}\right), \tag{11.25}$$

so the azimuth has the rigid double pole and two flexible poles, ω^2 being the roots of the quadratic Δ_ψ/s^2 in $-s^2$. Polar (torque on θ_m; $J_{\theta,p}$ includes I_{JJ}; k_g acts on the telescope node):

$$\mathbf{P}_\theta(s) = \begin{bmatrix} J_{\theta,e}s^2 + k_{\theta,1} & -k_{\theta,1} & 0 & 0 & 0 \\ -k_{\theta,1} & J_{\theta,h}s^2 + k_{\theta,1} + k_{\theta,2} + k_{\theta,4} & -k_{\theta,2} & 0 & -k_{\theta,4} \\ 0 & -k_{\theta,2} & J_{\theta,m}s^2 + k_{\theta,2} + k_{\theta,3} & -k_{\theta,3} & 0 \\ 0 & 0 & -k_{\theta,3} & J_{\theta,r}s^2 + k_{\theta,3} & 0 \\ 0 & -k_{\theta,4} & 0 & 0 & J_{\theta,p}s^2 + k_g + k_{\theta,4} \end{bmatrix},$$

$$\frac{\Theta_e}{T_\theta} = \frac{k_{\theta,1}k_{\theta,2}\left(J_{\theta,r}s^2 + k_{\theta,3}\right)\left(J_{\theta,p}s^2 + k_g + k_{\theta,4}\right)}{\Delta_\theta}, \qquad \frac{\Theta_p}{T_\theta} = \frac{k_{\theta,2}k_{\theta,4}\left(J_{\theta,e}s^2 + k_{\theta,1}\right)\left(J_{\theta,r}s^2 + k_{\theta,3}\right)}{\Delta_\theta}, \tag{11.26}$$

$$\frac{\Theta_m}{T_\theta} = \frac{\left(J_{\theta,r}s^2 + k_{\theta,3}\right)\Delta_{ehp|m}(s)}{\Delta_\theta}, \tag{11.27}$$

$$\Delta_{ehp|m}(s) = \det\begin{bmatrix} J_{\theta,e}s^2 + k_{\theta,1} & -k_{\theta,1} & 0 \\ -k_{\theta,1} & J_{\theta,h}s^2 + k_{\theta,1} + k_{\theta,2} + k_{\theta,4} & -k_{\theta,4} \\ 0 & -k_{\theta,4} & J_{\theta,p}s^2 + k_g + k_{\theta,4} \end{bmatrix}, \tag{11.28}$$

where $\Delta_{ehp|m}$ is the determinant of the encoder–hub–telescope piece with the motor clamped. The zeros follow by inspection (Figure 10). In the azimuth the load is beyond the motor with respect to the encoder, so Ψ_e/T_ψ vanishes when the load resonates on $k_{\psi,2}$ with the motor clamped. In the hub model the telescope hangs from the hub centre through $k_{\theta,4} = k_b$, outside the path encoder–motor, so Θ_e/T_θ vanishes both when the rear end resonates with the motor clamped and when the telescope resonates on k_b with the hub clamped. Table 14 compares these closed forms, evaluated with the lumped parameters at $\theta_0 = -45°$, with the invariant zeros of the coupled linearised model.

Table 14: Antiresonances of the lumped model: closed form against the invariant zeros of the coupled linearised model (Hz). *Result from placeholder parameters.*

zero	closed form	coupled model
Ψ_e/T_ψ: $\sqrt{k_{\psi,2}/J^*_{\psi,L}}$	13.1388	13.1388
Ψ_m/T_ψ: $\sqrt{k_{\psi,1}/J_{\psi,e}}$	1143.53	1143.53
Ψ_m/T_ψ: $\sqrt{k_{\psi,2}/J^*_{\psi,L}}$	13.1388	13.1388
Θ_e/T_θ: $\sqrt{k_{\theta,3}/J_{\theta,r}}$	617.424	617.424
Θ_e/T_θ: $\sqrt{(k_{\theta,4}+k_g)/J_{\theta,p}}$	16.7886	16.7886

Reading poles and zeros from the chain. Figure 10 draws the lumped polar chain of the hub model at $\theta_0 = 0°$ for the torque→encoder and torque→motor channels. The poles belong to the whole free chain and are the same in every channel, 0.1481 Hz, 109.8 Hz, 179.2 Hz, 886.4 Hz, 2222 Hz. The zeros are the natural frequencies of the branches outside the path from the motor to the sensor, each with its attachment node clamped; the springs on the path give only a constant factor, so a chain of n inertias has $n - 1 -$

(springs on the path) zeros: two for the encoder, four for the motor. At the encoder the telescope, hanging from the hub through $k_b = 9.8 \times 10^3$ N m/rad, gives the zero at 16.78 Hz with the hub clamped, and the rear end gives the zero at 617.4 Hz with the motor clamped. At the motor, clamping the motor leaves the encoder–hub–telescope piece and the rear end, with zeros 6.129 Hz, 124.4 Hz, 617.4 Hz, 2222 Hz.

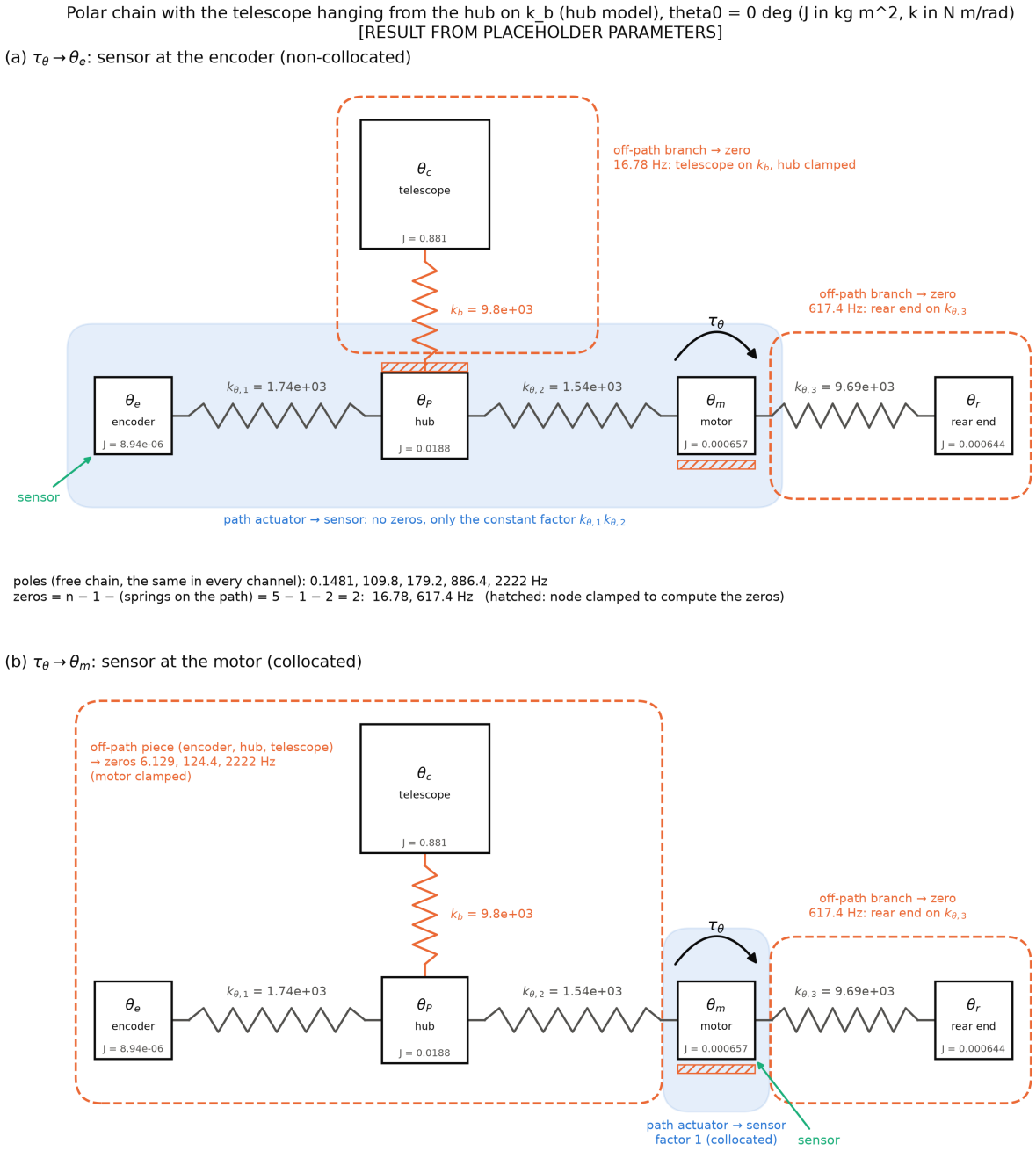


Figure 10: Lumped polar chain of the hub model drawn for the two channels: (a) torque→encoder, (b) torque→motor. Blue: path from the actuator (torque τ_θ on the motor) to the sensor, whose springs create no zero. Orange, dashed: branches outside the path, each with the zeros given by its natural frequencies with the attachment node clamped (hatched). The poles, listed under each panel, are those of the whole free chain. *Result from placeholder parameters.*

Linearised coupled model. The lumped model is linearised at every operating point exactly as the other models (Section 9): $\mathbf{M}_0$, $\mathbf{C}_0$, $\mathbf{K}_0$ from the Jacobians of the residual, FRF from $\mathbf{P}(j\omega)^{-1}$, poles from eig($\mathbf{A}$), invariant zeros from the Rosenbrock pencil. Table 15 applies the criteria of Section 10 against the full-DOF model: the lumped model reproduces the rigid and low-frequency behaviour, including the load antiresonance,

but not the structural modes, whose frequencies it shifts (Table 13).

Table 15: Lumped model against the full-DOF model, torque→encoder up to the band of interest: worst direct-channel error outside the zero bands, worst relative error of the zeros, worst relative 2-norm error of the 2×2 matrix. *Result from placeholder parameters.*

θ_0	Ω_0	dB	deg	zeros	2×2
−45	0	25.4	160	0.4	19
−45	10	25.4	160	0.4	19
−15	0	25.4	160	0.4	19
−15	10	25.4	160	0.4	19
0	0	25.4	160	0.4	19
0	10	25.4	160	0.4	19
15	0	25.4	160	0.4	19
15	10	25.4	160	0.4	19
45	0	25.4	160	0.4	19
45	10	25.4	160	0.4	19

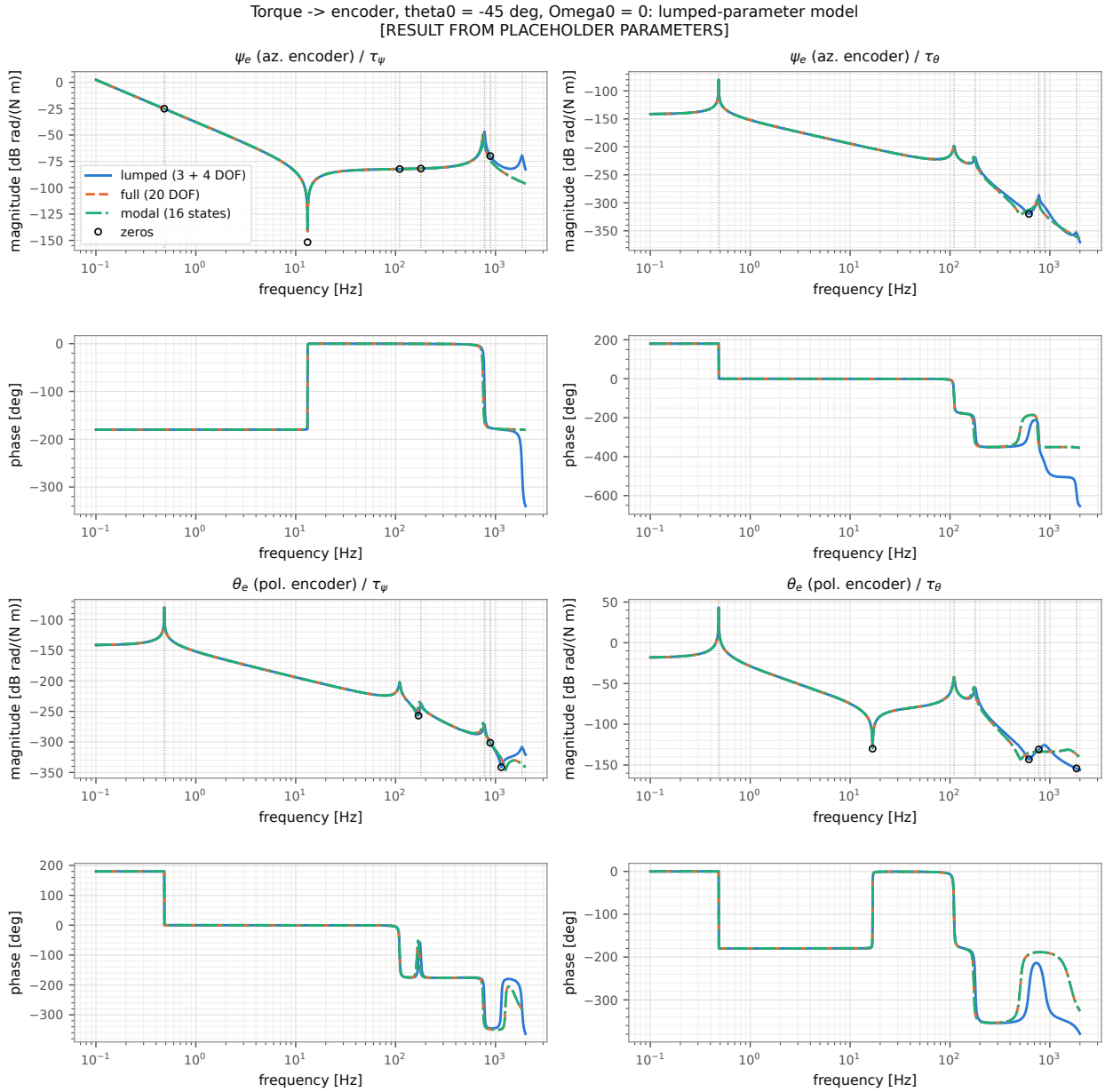


Figure 11: Torque→encoder 2×2 matrix: lumped model (with its poles, dotted, and invariant zeros, circles), full-DOF reference and modal model. *Result from placeholder parameters.*

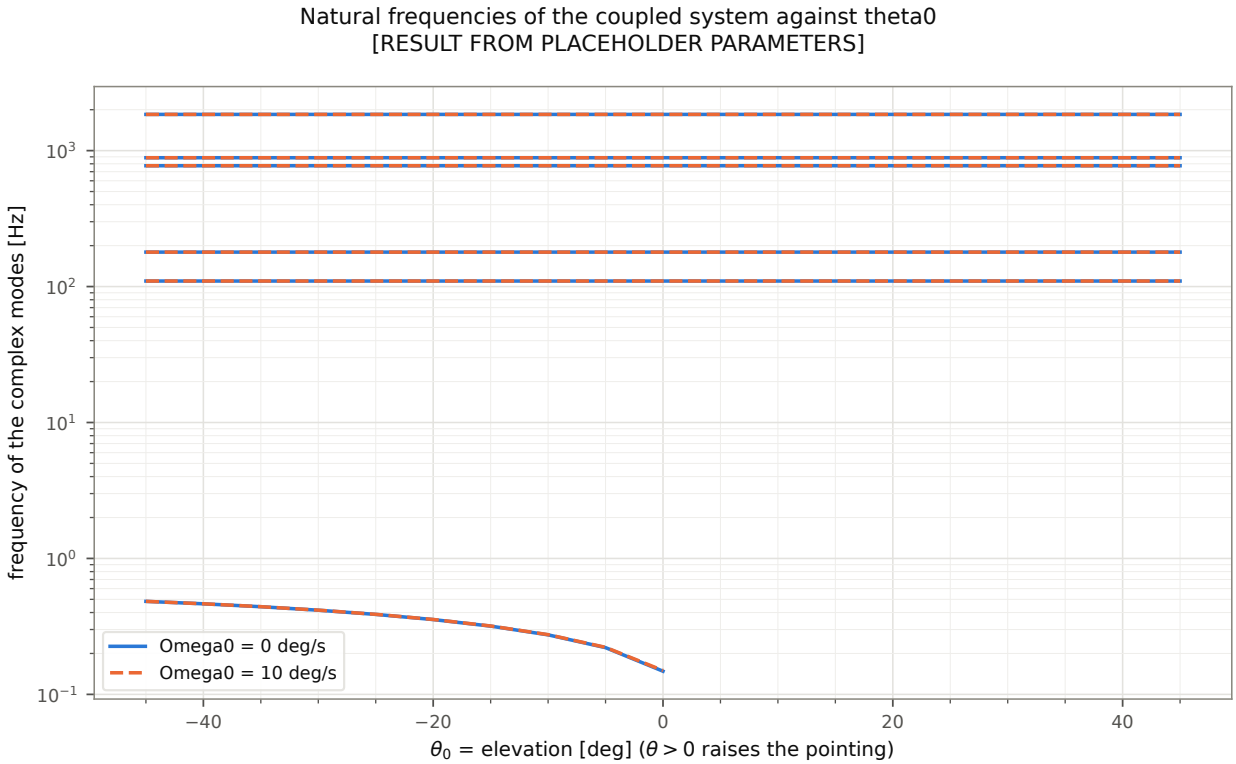


Figure 12: Natural frequencies of the coupled lumped model as a function of θ_0. *Result from placeholder parameters.*

Table 16: Poles and invariant zeros of the torque→encoder matrix at $\theta_0 = -45°$, $\Omega_0 = 0°/\mathrm{s}$ (only roots with Im ≥ 0).

quantity	Re [rad/s]	Im [rad/s]	$\lvert\lambda\rvert/2\pi$ [Hz]	type
pole	1.949×10^{-13}	2.1973×10^{-6}	3.4971×10^{-7}	complesso
pole	-5.814×10^{-11}	3.0367	0.4833	complesso
pole	−6.9	689.93	109.81	complesso
pole	−18.38	1125.9	179.22	complesso
pole	−48.64	4863.3	774.05	complesso
pole	−449.6	5551.3	886.4	complesso
pole	−276.8	11600	1846.7	complesso
pole	−2826	13675	2222.4	complesso
zero $G_{\psi_e \tau_\psi}$	-5.712×10^{-9}	3.0367	0.4833	complesso
zero $G_{\psi_e \tau_\psi}$	−0.01401	82.553	13.139	complesso
zero $G_{\psi_e \tau_\psi}$	−6.9	689.93	109.81	complesso
zero $G_{\psi_e \tau_\psi}$	−18.38	1125.9	179.22	complesso
zero $G_{\psi_e \tau_\psi}$	−449.6	5551.3	886.4	complesso
zero $G_{\psi_e \tau_\psi}$	−2826	13675	2222.4	complesso
zero $G_{\psi_e \tau_\psi}$	-2.432×10^5	0	38702	reale_stabile
zero $G_{\psi_e \tau_\theta}$	4.814×10^{-12}	0	0	nullo
zero $G_{\psi_e \tau_\theta}$	−0.001467	0	0.00023353	reale_stabile
zero $G_{\psi_e \tau_\theta}$	−216.1	3873.7	617.48	complesso
zero $G_{\psi_e \tau_\theta}$	−3020	13402	2186.5	complesso
zero $G_{\psi_e \tau_\theta}$	-2.924×10^4	8155.6	4831.9	complesso
zero $G_{\theta_e \tau_\psi}$	2.177×10^{-11}	2.1094×10^{-6}	3.3571×10^{-7}	complesso
zero $G_{\theta_e \tau_\psi}$	−16.44	1065	169.52	complesso
zero $G_{\theta_e \tau_\psi}$	−449.5	5551	886.36	complesso
zero $G_{\theta_e \tau_\psi}$	−106.1	7184.2	1143.5	complesso
zero $G_{\theta_e \tau_\psi}$	-3.45×10^4	0	5490.5	reale_stabile
zero $G_{\theta_e \tau_\psi}$	-3.45×10^4	0	5490.5	reale_stabile

quantity	Re [rad/s]	Im [rad/s]	$\lvert\lambda\rvert/2\pi$ [Hz]	type
zero $G_{\theta_e\tau_\psi}$	-2.432×10^{5}	0	38702	reale_stabile
zero $G_{\theta_e\tau_\theta}$	-4.631×10^{-14}	0	0	nullo
zero $G_{\theta_e\tau_\theta}$	-3.651×10^{-13}	0	0	nullo
zero $G_{\theta_e\tau_\theta}$	-0.1611	105.49	16.789	complesso
zero $G_{\theta_e\tau_\theta}$	-218.1	3873.2	617.42	complesso
zero $G_{\theta_e\tau_\theta}$	-48.64	4863.3	774.05	complesso
zero $G_{\theta_e\tau_\theta}$	-276.8	11600	1846.7	complesso
zero $G_{\theta_e\tau_\theta}$	-3.45×10^{4}	0.00018612	5490.5	reale_stabile

11.3 Nonlinear simulator

The simulator (module `simulatore_concentrato.py`) integrates the Newton–Euler equations (11.11)–(11.18) as they are, with all the trigonometric functions of θ_p. Writing their residual as $\mathbf{r}(\mathbf{q},\dot{\mathbf{q}},\ddot{\mathbf{q}},\boldsymbol{\tau})$,

$$\mathbf{M}(\theta_p)=\frac{\partial\mathbf{r}}{\partial\ddot{\mathbf{q}}},\qquad \mathbf{h}(\mathbf{q},\dot{\mathbf{q}})=\mathbf{r}\big|_{\ddot{\mathbf{q}}=\mathbf{0},\ \boldsymbol{\tau}=\mathbf{0}},\qquad \dot{\mathbf{x}}=\begin{bmatrix}\dot{\mathbf{q}}\\ \mathbf{M}(\theta_p)^{-1}\big[\mathbf{E}\boldsymbol{\tau}(t)-\mathbf{h}(\mathbf{q},\dot{\mathbf{q}})\big]\end{bmatrix},\qquad \mathbf{x}=[\mathbf{q},\ \dot{\mathbf{q}}], \tag{11.29}$$

where $\mathbf{h}$ collects the gyroscopic and centrifugal terms ($J'_{\psi,p}\dot\psi_L\dot\theta_p$, $-\frac12 J'_{\psi,p}\dot\psi_L^2$, $M'_{12}\dot\theta_p^2$), the spring and damper torques, the bearing friction and the gravity moment $-M_{g,J}(\theta_p)$. Both are compiled from SymPy with `lambdify`; the linear system for $\ddot{\mathbf{q}}$ is solved at every call. Integration: `solve_ivp`, DOP853, relative tolerance 1×10^{-10}, absolute 1×10^{-12}, split at the discontinuities of the input. Energy, p_ψ and the linearisation are taken from the equivalent Lagrangian model, whose accelerations the simulator reproduces to 10^{-16} (test suite).

Exported code. `simulatore_concentrato.esporta_codice` writes the same model as stand-alone code: a Python module (NumPy only: `massa(q)`, `termini(q, dq)`, `rhs(t, x, tau)` for `solve_ivp`) and a MATLAB/Octave function for `ode45`, with the parameters of Tables 11 and 12 and the telescope in the components of frame 2 ($I_{II},\dots,d_K$). The Python version is checked against the simulator to 10^{-16}; the MATLAB file is generated by the same expressions but has not been run (no MATLAB or Octave available here). The exported files (`ne_concentrato.py`, `ne_concentrato.m`, `ne_concentrato_parametri.m`) are in `results_hub/simulatore_concentrato`.

Scenarios. S1–S3 are those of Section 13. S4 starts from the steady tracking motion at (θ_0,Ω_0) with the nominal torques applied and the whole polar axis rotated by $\Delta\theta$: the telescope swings as a pendulum about θ_0, with the restoring moment $M_{g,J}(\theta_p)$, and the azimuth responds through $J_{\psi,p}(\theta_p)$ and the Coriolis term. The same initial condition is integrated with the linearised model; the error $e(y)=\max_t|y_{nl}-y_{lin}|/\max_t|y_{lin}|$ grows with $\Delta\theta$, which measures the effect of the trigonometric nonlinearities.

Table 17: Simulations of the lumped model (relative error of the nonlinear against the linearised response; conservation drifts for S3). *Result from placeholder parameters.*

id	θ_0 [deg]	Ω_0 [deg/s]	forcing	result
S1	0	0	doublet on τ_ψ	$e(\psi_e)=3.1\times10^{-10}$, $e(\psi_L)=3\times10^{-12}$, $e(\theta_p)=0.88$
S3	0	10	$\mathbf{C}=\mathbf{0}$, $\boldsymbol{\tau}=\mathbf{0}$	$\lvert\Delta E\rvert/E=1.6\times10^{-13}$, $\lvert\Delta p_\psi\rvert/p_\psi=1.7\times10^{-15}$
S4a	-45	10	polar offset 2°, $\boldsymbol{\tau}=\boldsymbol{\tau}_0$	$e(\psi_e)=0.0044$, $e(\psi_L)=0.0044$, $e(\theta_p)=0.012$
S4b	-45	10	polar offset 10°, $\boldsymbol{\tau}=\boldsymbol{\tau}_0$	$e(\psi_e)=0.041$, $e(\psi_L)=0.041$, $e(\theta_p)=0.085$
S4c	0	10	polar offset 10°, $\boldsymbol{\tau}=\boldsymbol{\tau}_0$	$e(\psi_e)=16$, $e(\psi_L)=16$, $e(\theta_p)=4.7$

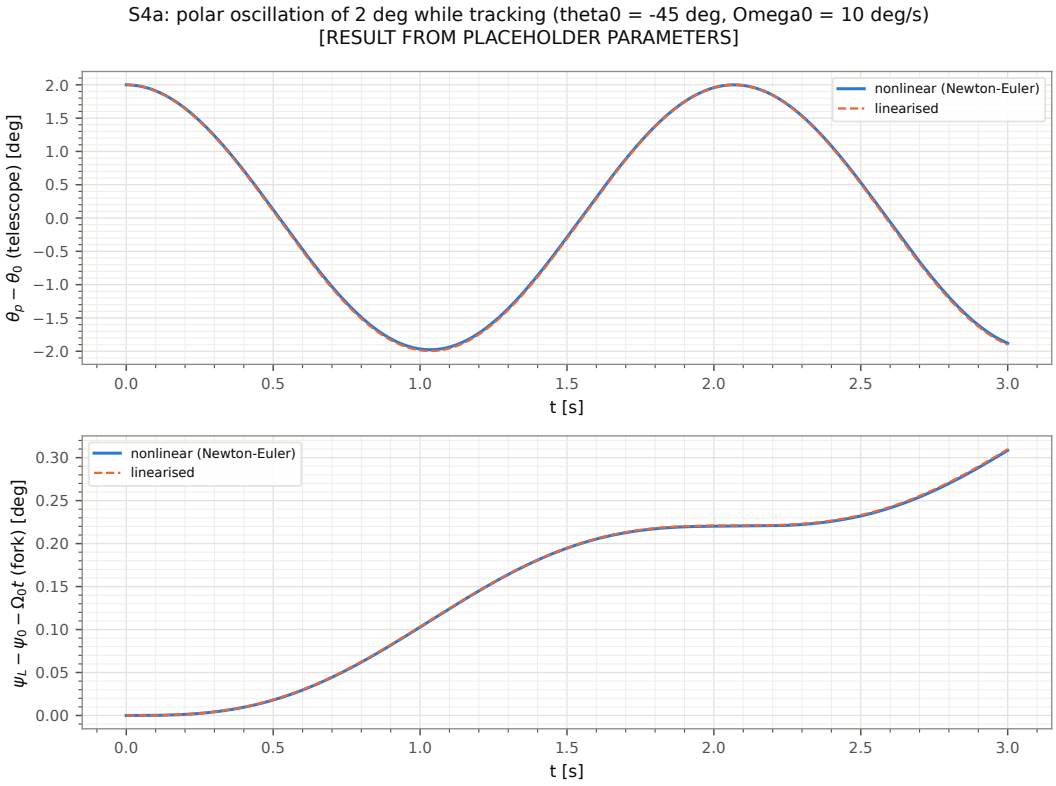


Figure 13: S4 at small amplitude: nonlinear (solid) and linearised (dashed) responses almost coincide. *Result from placeholder parameters.*

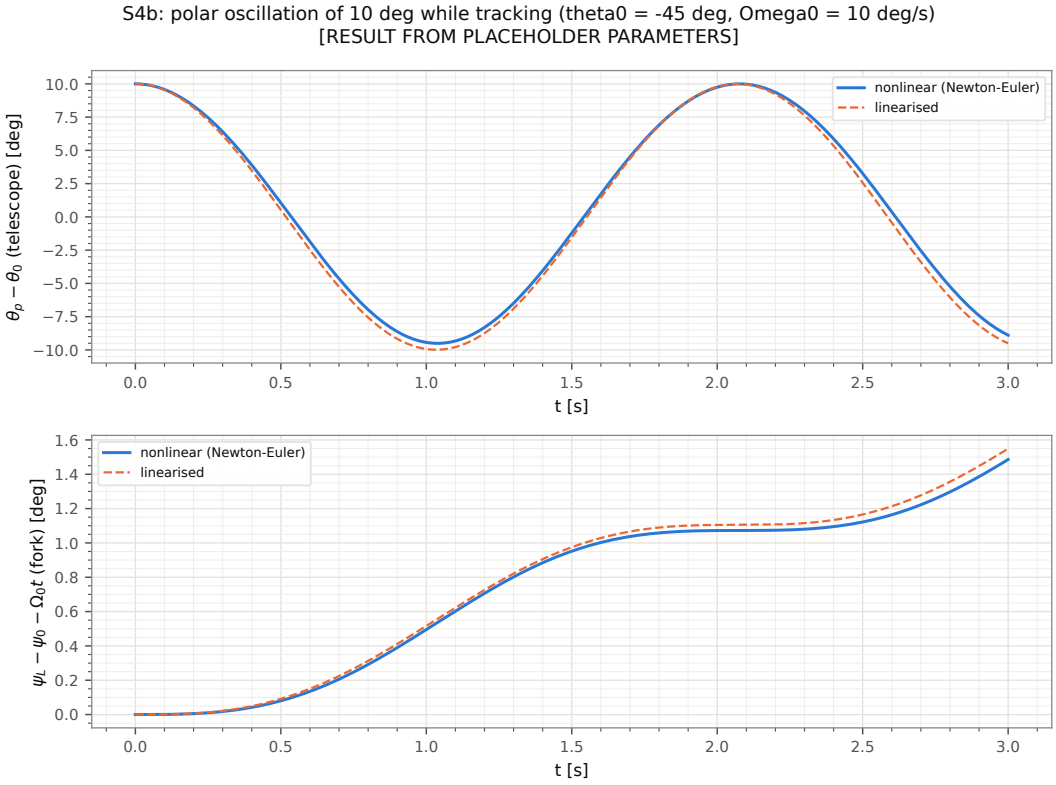


Figure 14: S4 at larger amplitude: the trigonometric terms detune the polar pendulum and modulate the azimuth inertia $J_{\psi,p}(\theta_p)$; the linear model drifts away. *Result from placeholder parameters.*

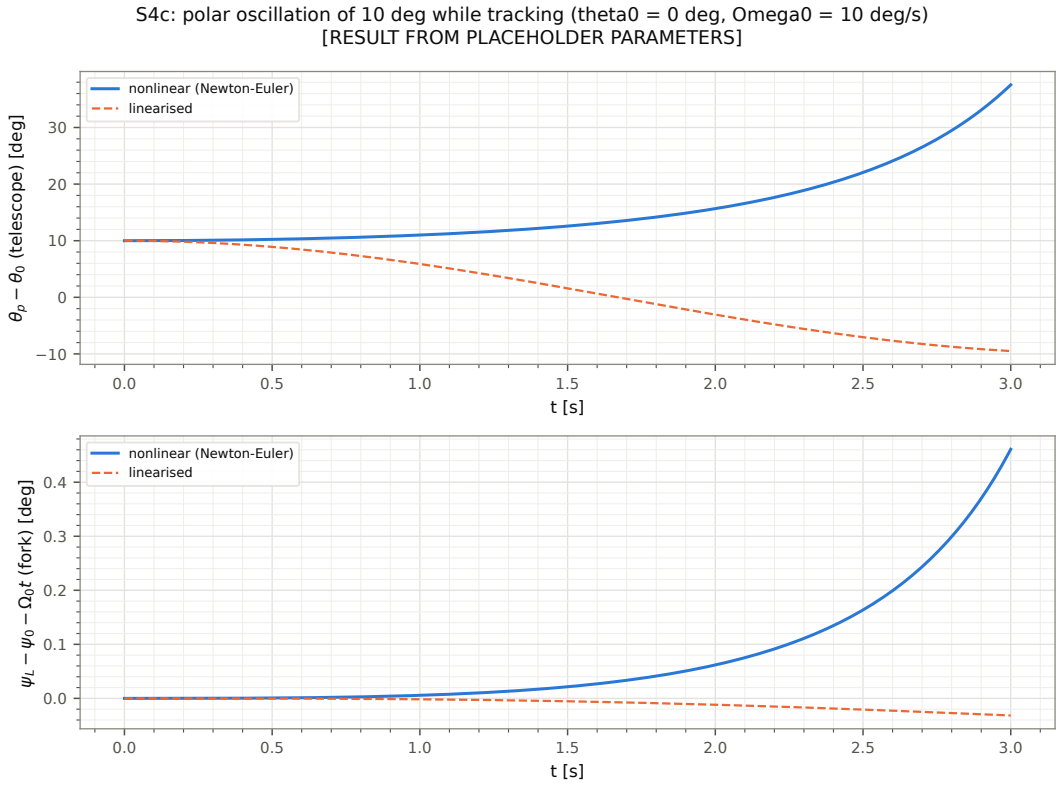


Figure 15: S4 near the stability limit of the unbalanced telescope: the nonlinear response leaves the potential well that the linear model assumes. *Result from placeholder parameters.*

12 Comparison with experimental frequency responses

Measurements. Plant measurements exported by Kollmorgen WorkBench (folder `LucyIdentification`; the files of the sub-folder `Validation` are not used). The drive injects a stepped-sine current (amplitude in Table 18, frequency step of 1.03–1.1 %, 8 kHz sampling, 4096-point FFT) with the loops closed, and estimates the plant as the ratio of the feedback velocity to the current, in (rad/s)/A. The drives close their loops on the axis encoders, so the measured plant is the encoder velocity over the current; the model is also evaluated at the motor node for reference. The units are confirmed by the rigid-body line of the elevation, which matches $K_t/(J\omega)$ of the model within about 1 dB. For the elevation the measurement of 16 October 2026, extended to 2 kHz, replaces `Elevation_BodeMeasurement.csv`, which stopped at about 277 Hz (`Elevation_new_BodeMeasurement.csv` holds the same data as the latter); neither is used.

Table 18: Measurements used: axis, elevation θ_0 (deg), telescope mounted, current amplitude (A) and frequency range (Hz).

file	axis	θ_0	telescope	I	range
`Azimuth_BodeMeasurement_Po_0.csv`	azimuth	0	yes	3.5	1–1399
`Azimuth_BodeMeasurement_Po_0_dep.csv`	azimuth	0	yes	2.36	1–1399
`Azimuth_BodeMeasurement_Po_30.csv`	azimuth	30	yes	3.5	1–1456
`Elevation_BodeMeasurement_bode_New_16_10_26.csv`	polar	0	yes	3.5	1–1991
`BodeMeasurement_NoTelescope.csv`	polar	0	no	2.5	1–968.5

Model. Only the 7-DOF lumped model of Section 11 is used (below 1 kHz it has the dynamics of interest). It is linearised at the elevation of each test with $\Omega_0 = 0$ and converted to the units of the measurement:

$$G_{meas}(j\omega) = \frac{\dot\theta(j\omega)}{I(j\omega)} = j\omega\, K_t\, \frac{\theta(j\omega)}{\tau(j\omega)}, \qquad K_t = 0.258 \text{ N m/A (TBM(S)-7646-A, 25°C)}, \tag{12.1}$$

at the encoder and at the motor node of the excited axis. For the test without the telescope the payload is removed (polar shafts only). The phase is shown wrapped to $(-180°, 180°]$ as in the WorkBench export.

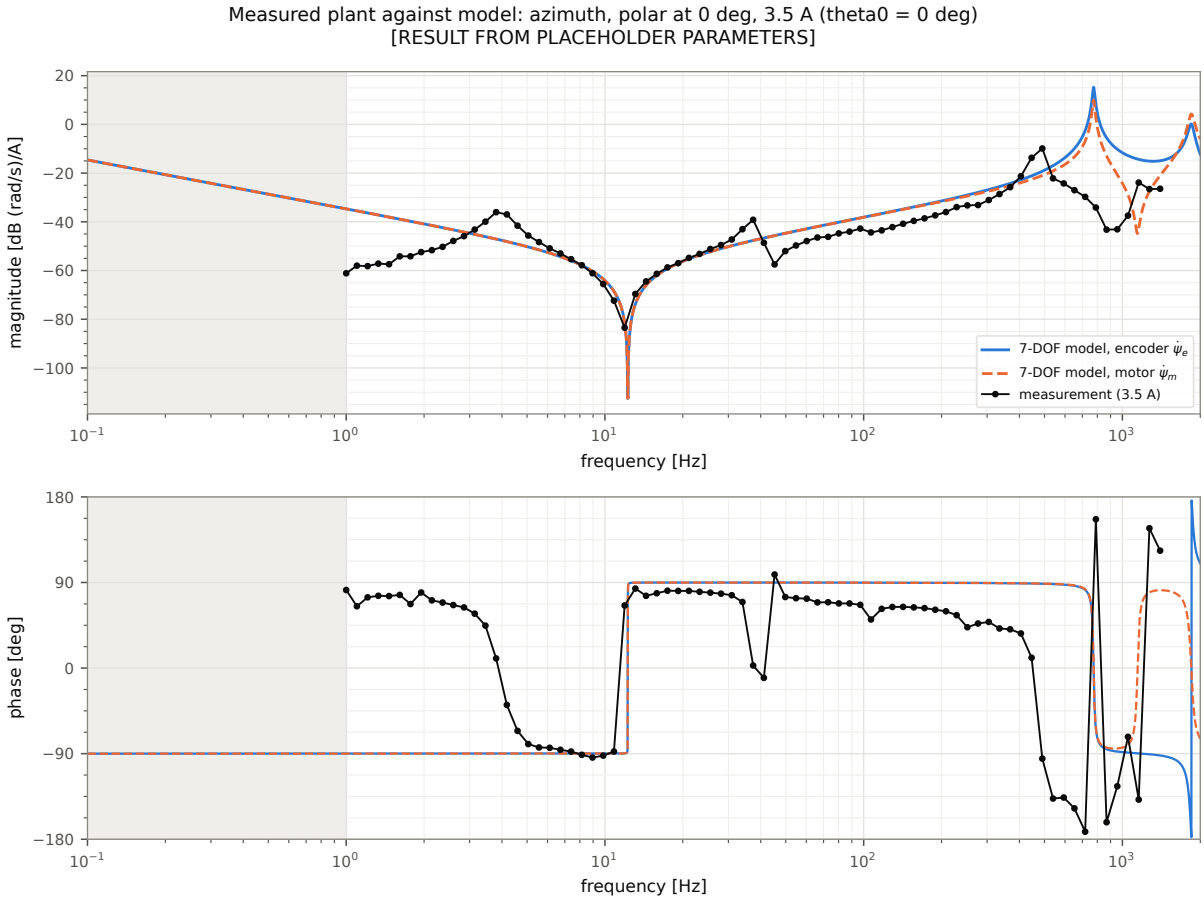


Figure 16: Measured plant (azimuth, polar at 0 deg, 3.5 A, black) against the 7-DOF lumped model at $\theta_0 = 0°$, velocity at the encoder (blue: the measured channel, the drive feedback being the encoder) and at the motor node (orange, dashed, for reference). The grey band marks the range without data (the measurement starts at 1 Hz). *Result from placeholder parameters.*

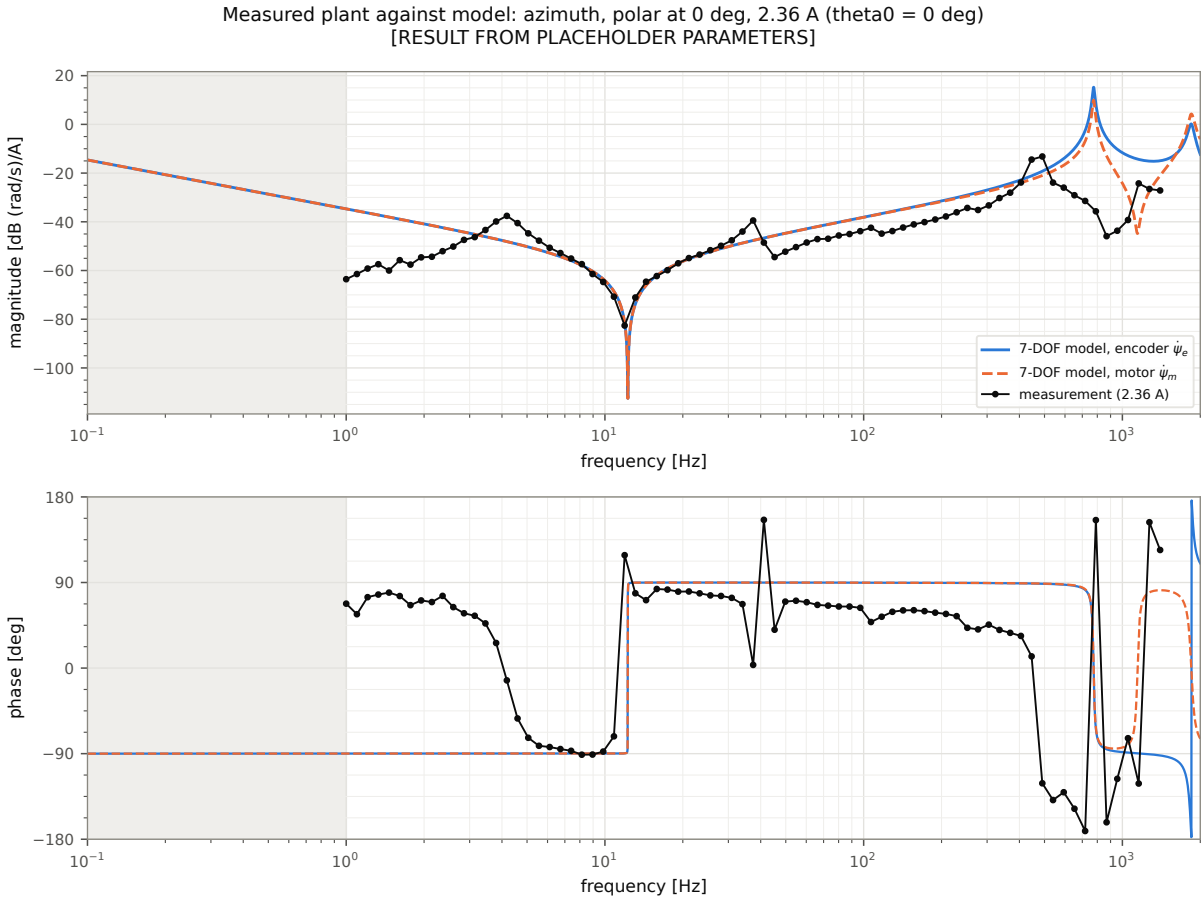


Figure 17: Measured plant (azimuth, polar at 0 deg, 2.36 A, black) against the 7-DOF lumped model at $\theta_0 = 0°$, velocity at the encoder (blue: the measured channel, the drive feedback being the encoder) and at the motor node (orange, dashed, for reference). The grey band marks the range without data (the measurement starts at 1 Hz). *Result from placeholder parameters.*

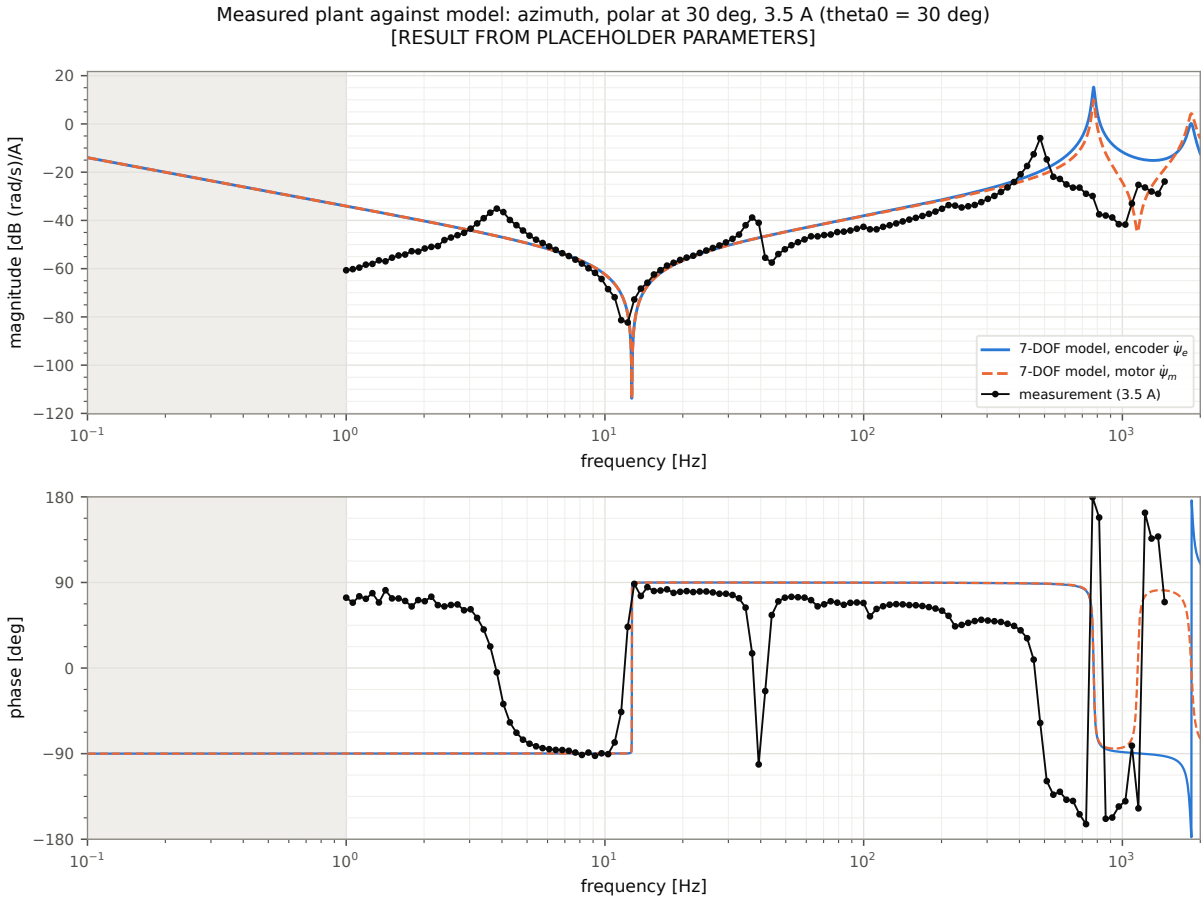


Figure 18: Measured plant (azimuth, polar at 30 deg, 3.5 A, black) against the 7-DOF lumped model at $\theta_0 = 30°$, velocity at the encoder (blue: the measured channel, the drive feedback being the encoder) and at the motor node (orange, dashed, for reference). The grey band marks the range without data (the measurement starts at 1 Hz). *Result from placeholder parameters.*

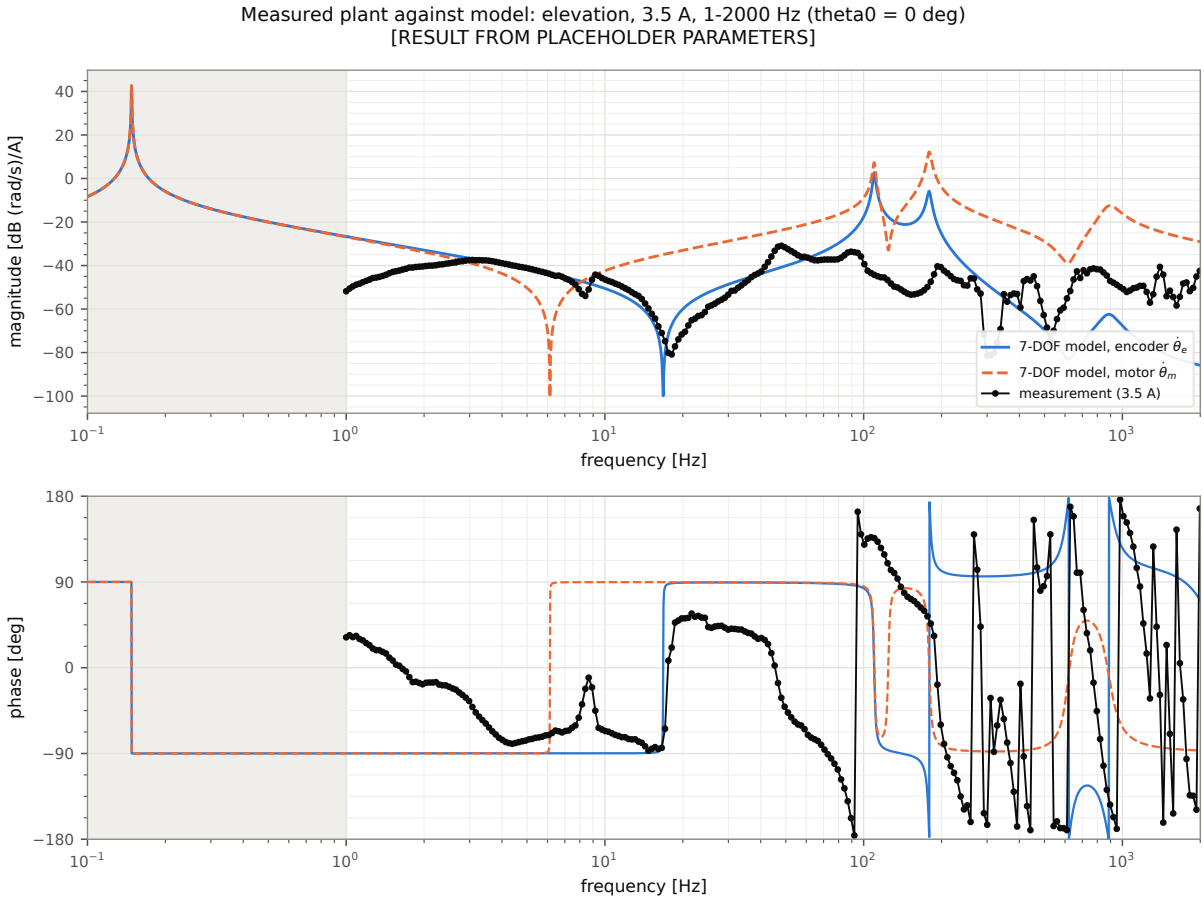


Figure 19: Measured plant (elevation, 3.5 A, 1-2000 Hz, black) against the 7-DOF lumped model at $\theta_0 = 0°$, velocity at the encoder (blue: the measured channel, the drive feedback being the encoder) and at the motor node (orange, dashed, for reference). The grey band marks the range without data (the measurement starts at 1 Hz). *Result from placeholder parameters.*

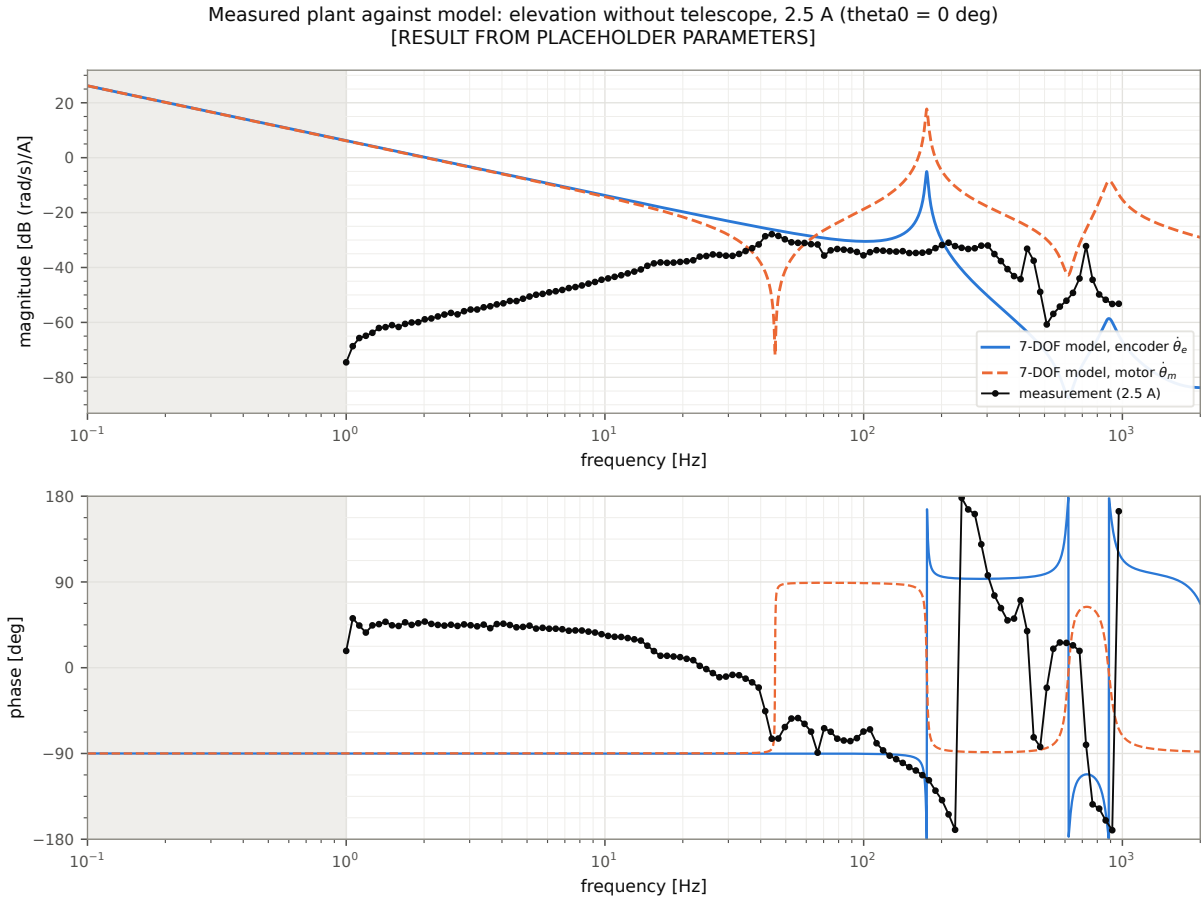


Figure 20: Measured plant (elevation without telescope, 2.5 A, black) against the 7-DOF lumped model at $\theta_0 = 0°$, velocity at the encoder (blue: the measured channel, the drive feedback being the encoder) and at the motor node (orange, dashed, for reference). The grey band marks the range without data (the measurement starts at 1 Hz). *Result from placeholder parameters.*

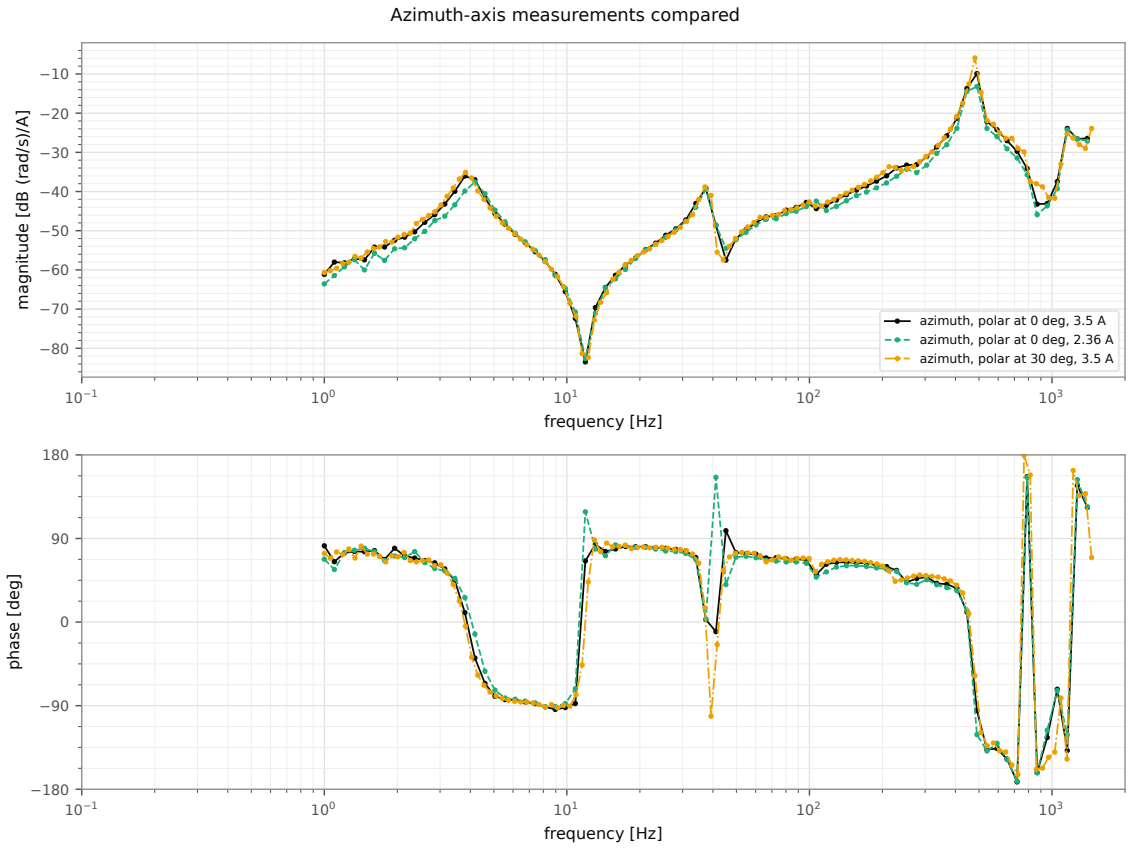


Figure 21: The three azimuth measurements: effect of the excitation amplitude (3.5 and 2.36 A) and of the elevation (0 and 30°).

Measured features against the model. Peaks and notches of the measured magnitude with a local prominence of at least 8 dB, paired with the nearest mode of the excited axis (resonances) or invariant zero of the encoder channel, the measured one (antiresonances), within a factor 2 in frequency; "–" means no counterpart in the model.

Table 19: Measured resonances and antiresonances against the 7-DOF lumped model (Hz). *Result from placeholder parameters.*

measurement	feature	measured	model	channel	ratio
azimuth, polar at 0 deg, 3.5 A	resonance	3.797	–	–	–
azimuth, polar at 0 deg, 3.5 A	antiresonance	11.92	12.26	$\dot{\psi}_e$	0.972
azimuth, polar at 0 deg, 3.5 A	resonance	37.4	–	–	–
azimuth, polar at 0 deg, 3.5 A	antiresonance	45.26	–	–	–
azimuth, polar at 0 deg, 3.5 A	resonance	490.4	774	$\dot{\psi}_e$	0.634
azimuth, polar at 0 deg, 2.36 A	resonance	4.177	–	–	–
azimuth, polar at 0 deg, 2.36 A	antiresonance	11.92	12.26	$\dot{\psi}_e$	0.972
azimuth, polar at 0 deg, 2.36 A	resonance	37.4	–	–	–
azimuth, polar at 0 deg, 2.36 A	antiresonance	45.26	–	–	–
azimuth, polar at 0 deg, 2.36 A	resonance	490.4	774	$\dot{\psi}_e$	0.634
azimuth, polar at 30 deg, 3.5 A	resonance	3.82	–	–	–
azimuth, polar at 30 deg, 3.5 A	antiresonance	12.25	12.71	$\dot{\psi}_e$	0.964
azimuth, polar at 30 deg, 3.5 A	resonance	37.06	–	–	–
azimuth, polar at 30 deg, 3.5 A	antiresonance	44.14	–	–	–
azimuth, polar at 30 deg, 3.5 A	resonance	481.3	774	$\dot{\psi}_e$	0.622
elevation, 3.5 A, 1-2000 Hz	antiresonance	8.4	16.78	$\dot{\theta}_e$	0.501
elevation, 3.5 A, 1-2000 Hz	resonance	9.179	–	–	–
elevation, 3.5 A, 1-2000 Hz	antiresonance	18.12	16.78	$\dot{\theta}_e$	1.08
elevation, 3.5 A, 1-2000 Hz	antiresonance	152.2	–	–	–

measurement	feature	measured	model	channel	ratio
elevation, 3.5 A, 1-2000 Hz	resonance	192.8	179.2	$\dot{\theta}_e$	1.08
elevation, 3.5 A, 1-2000 Hz	antiresonance	300.3	–	–	–
elevation, 3.5 A, 1-2000 Hz	resonance	348.2	179.2	$\dot{\theta}_e$	1.94
elevation, 3.5 A, 1-2000 Hz	antiresonance	358.6	617.4	$\dot{\theta}_e$	0.581
elevation, 3.5 A, 1-2000 Hz	resonance	380.4	–	–	–
elevation, 3.5 A, 1-2000 Hz	antiresonance	403.6	617.4	$\dot{\theta}_e$	0.654
elevation, 3.5 A, 1-2000 Hz	resonance	428.2	–	–	–
elevation, 3.5 A, 1-2000 Hz	resonance	454.3	886.4	$\dot{\theta}_e$	0.512
elevation, 3.5 A, 1-2000 Hz	antiresonance	526.6	617.4	$\dot{\theta}_e$	0.853
elevation, 3.5 A, 1-2000 Hz	resonance	667.1	886.4	$\dot{\theta}_e$	0.753
elevation, 3.5 A, 1-2000 Hz	resonance	707.7	886.4	$\dot{\theta}_e$	0.798
elevation, 3.5 A, 1-2000 Hz	resonance	773.3	886.4	$\dot{\theta}_e$	0.872
elevation, 3.5 A, 1-2000 Hz	antiresonance	1039	774	$\dot{\theta}_e$	1.34
elevation without telescope, 2.5 A	antiresonance	404.1	617.4	$\dot{\theta}_e$	0.655
elevation without telescope, 2.5 A	resonance	428.4	–	–	–
elevation without telescope, 2.5 A	antiresonance	510.2	617.4	$\dot{\theta}_e$	0.826

12.1 Discrepancies and their likely causes

1. **Low frequency (< 5 Hz): stiffness to the ground.** Every azimuth measurement has a resonance at 3.8 Hz (azimuth, polar at 0 deg, 3.5 A), 4.18 Hz (azimuth, polar at 0 deg, 2.36 A), 3.82 Hz (azimuth, polar at 30 deg, 3.5 A) and rises at +20 dB/dec below it, with phase near +90°: the axis behaves as an inertia on a spring to the ground. The model has a free rigid-body mode (phase −90°, −20 dB/dec). With J_{tot} of the model the measured resonance requires $k_0 \approx 1.26 \times 10^3$ N m/rad. The resonance moves up when the excitation is reduced (3.5 A → 2.36 A), i.e. the stiffness depends on the amplitude: this is the signature of pre-sliding (Dahl-type) friction in the bearings and seals, which acts as a spring for motions smaller than the break-away displacement. The model has only viscous bearing friction, currently zero. The elevation measurement shows the same behaviour below a few hertz.

2. **Load antiresonance: reproduced.** The deep antiresonance of the azimuth, where the fork oscillates on $k_{\psi,2}$ with motor and encoder still, is measured at 11.92 against 12.26 Hz (azimuth, polar at 0 deg, 3.5 A), 11.92 against 12.26 Hz (azimuth, polar at 0 deg, 2.36 A), 12.25 against 12.71 Hz (azimuth, polar at 30 deg, 3.5 A); its shift with the elevation (through $J_{\psi,p}(\theta_0)$) is also reproduced. The measured $k_{\psi,2}/J_L^*$ is about −5.4 % off: a joint compliance in series with the shafts (Q5) or a heavier fork.

3. **Level between the antiresonance and the first resonance.** In this band the encoder velocity is that of the motor and encoder inertias on the compliance $1/k_{\psi,2}$, $|G| \approx K_t\omega/k_{\psi,2}$. The measurement is −4.9 dB below the model. A stiffer $k_{\psi,2}$ would contradict the antiresonance, which indicates a slightly softer one; the first suspect is the torque scale: K_t refers to A rms at 25°C, and a current read in A peak gives −3 dB, a hot winding −0.8 dB. The current units of the WorkBench export should be checked before adjusting stiffnesses.

4. **First structural resonance of the azimuth.** Measured 490.4 against 774 Hz (azimuth, polar at 0 deg, 3.5 A), 490.4 against 774 Hz (azimuth, polar at 0 deg, 2.36 A), 481.3 against 774 Hz (azimuth, polar at 30 deg, 3.5 A) (about 37 % lower). It is the motor and encoder side oscillating on $k_{\psi,2}$ against the heavy fork, $\omega_1^2 \approx k_{\psi,2}/(J_{\psi,e} + J_{\psi,m})$: with the model stiffness it requires $J_{\psi,e} + J_{\psi,m} \approx 0.00099\,\mathrm{kg\,m^2}$ instead of $0.000397\,\mathrm{kg\,m^2}$. Part of the difference is known: the azimuth configuration has no rotor inertia at the motor node (TBM(S)-7646 rotor $8.19 \times 10^{-5}\,\mathrm{kg\,m^2}$) and no encoder disc at node 1 (warning `NODI_SENZA_INERZIA_AZIMUT`); the rest points to a softer drive-side connection (shaft–rotor and shaft–hub joints) than the shaft geometry alone.

5. **Unmodelled mode at 37.4/45.3 Hz (azimuth).** A resonance–antiresonance pair, present at both elevations and amplitudes, with no counterpart in the model: a degree of freedom outside the torsional chains (fork bending, assumed rigid in Q12; base or pier; cable chain).

6. **Elevation: antiresonance at the encoder (hub model).** Below the antiresonance the measurement follows the rigid-body line of the model within 0.052 dB (5–11 Hz), which confirms the units and the total inertia $J_{tot} = 0.902\,\mathrm{kg\,m^2}$. It then has a deep antiresonance at 18.12 Hz followed by a broad resonance at about 48 Hz; the anti-resonance filters of the drive are tuned at 50 and 96 Hz. In the hub

model the telescope hangs from the hub centre through k_b, outside the encoder–motor path (Section 5): the encoder channel has the zero at 16.78 Hz (telescope on k_b with the hub clamped), against 18.12 Hz measured, and the first flexible mode of the polar chain is at 109.8 Hz, against 48 Hz measured. The zero is set by k_b, here the two-mass estimate from the measurement: the fit hangs only $J_c \approx 0.77\,\mathrm{kg\,m^2}$ on k_b, while the model hangs the whole telescope, hence the slightly lower model zero. The resonance depends on the stiffness of the path, dominated by the support ring taken alone (Table 4), and on the inertia on the hub side. The two-mass fit of the measurement gives $J_1 \approx 0.13\,\mathrm{kg\,m^2}$ on the shaft side, $J_c \approx 0.77\,\mathrm{kg\,m^2}$ beyond the attachment and $k_b \approx 1 \times 10^4\,\mathrm{N\,m/rad}$. Above the resonance the measurement is -27.9 dB from the encoder channel of the hub model (90.6–362 Hz). Between 72 Hz and 1000 Hz the measurement has 8 resonances against 3 modes of the polar chain. The elevation of this test is not recorded and is assumed to be 0°.

7. **Elevation without the telescope.** The model (polar shafts only, $0.0201\,\mathrm{kg\,m^2}$) is a nearly free rigid inertia and lies far above the measurement, which rises like a spring up to its first peak. From the low-frequency slope $|G| \approx K_t\omega/k_0$, $k_0 \approx 2.75 \times 10^3\,\mathrm{N\,m/rad}$, and from the peak at 44.1 Hz an inertia of about $0.0358\,\mathrm{kg\,m^2}$: the parts left on the polar axis without the telescope (flanges, clamps) are much heavier than the shafts of the model, and the friction stiffness dominates the light axis.

8. **High frequency.** Above a few hundred hertz the measured phase falls faster than the model: the current loop (1.5 kHz bandwidth) and the sampling of the drive add a delay that the mechanical model does not contain.

Summary. The 7-DOF model reproduces the structure of the azimuth response between about 5 Hz and 300 Hz: rigid-body line, load antiresonance and its dependence on the elevation, compliance line. The main discrepancies are a missing amplitude-dependent stiffness to the ground (bearing friction) below 5 Hz, a first azimuth structural mode about one third lower than predicted (missing drive-side inertias and joint compliances), a few dB of level (torque scale to be checked), and modes outside the torsional chains (rigid fork and telescope). On the elevation the hub model, with the telescope hanging from the hub through k_b, has the antiresonance at the encoder; its frequency is set by the estimated k_b, while the following resonance depends on the stiffness of the hub (Section 5).

13 How the model works and how it is simulated

13.1 Computational workflow

Everything is computed from `config.yaml` by `run_all.py`; Figure 22 shows the chain of stages, each with its module and numerical tool, and Table 20 lists the equations solved at each stage.

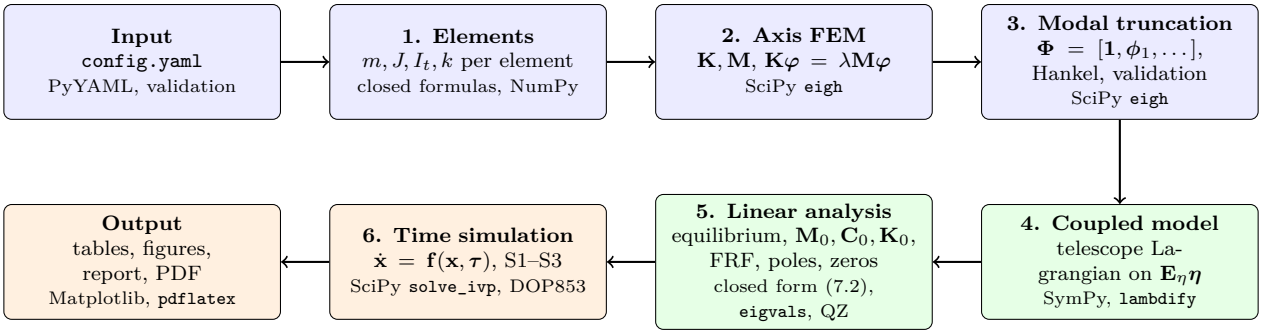


Figure 22: Computational workflow. Blue: structural stages (Sections 4–7 and 10); green: coupled model and its analysis; orange: simulation and output.

Table 20: Equations solved at each stage and numerical tools.

	equations	tool
1	m, J, I_t, J_p of every layer; k_s in parallel, $1/k_e$ in series (Section 4)	closed formulas in NumPy
2	assembly of $\mathbf{K}, \mathbf{M}^{\ell}, \mathbf{M}^{c}$; $\mathbf{K}\boldsymbol{\varphi} = \lambda\mathbf{M}\boldsymbol{\varphi}$ (Section 7)	`scipy.linalg.eigh` (LAPACK `dsygvd`)
3	$(\mathbf{K} - \omega^2\bar{\mathbf{M}})\boldsymbol{\phi} = \mathbf{0}$ of the full FEM; Hankel norms; validation against the full model (Section 10)	`scipy.linalg.eigh`, `numpy.linalg.solve`
4	Lagrange's equations of the telescope, (10.3) (this section)	SymPy for the derivation; `lambdify` to NumPy functions
5	steady state $\mathbf{r}(\mathbf{q}_0) = \mathbf{E}\boldsymbol{\tau}_0$; Jacobians; $\mathbf{G}(j\omega)$; poles; zeros	NumPy `solve`, `eigvals`; SymPy `jacobian`; `scipy.linalg.eig` (QZ)
6	$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\tau})$ and its linearisation	`scipy.integrate.solve_ivp`, method DOP853
7	tables (CSV), figures, `.npz`/`.mat`/JSON exports, this document	Matplotlib, `scipy.io.savemat`, `pdflatex`

13.2 Equations of the reduced model

The default model has $n = 8$ modal coordinates $\boldsymbol{\eta}$ (Section 10) and obeys (10.3). Its mass matrix

$$\mathbf{M}(\boldsymbol{\eta}) = \tilde{\mathbf{M}} + \mathbf{E}_\eta^{\mathsf{T}}\big[\mathbf{M}_p(\theta_p) - \bar{\mathbf{M}}_p\big]\mathbf{E}_\eta, \qquad \theta_p = \mathbf{c}_P^{\mathsf{T}}\boldsymbol{\eta}_\theta, \tag{13.1}$$

depends on the state only through θ_p, and the force vector is $\tilde{\mathbf{D}}\dot{\boldsymbol{\eta}} + \tilde{\mathbf{K}}\boldsymbol{\eta} + \mathbf{E}_\eta^{\mathsf{T}}(\mathbf{C}_p\dot{\mathbf{q}}_p + \mathbf{g}_p)$. The 2×2 payload blocks $\mathbf{M}_p$, $\mathbf{C}_p\dot{\mathbf{q}}_p$ and $\mathbf{g}_p$ of Section 8 are derived once by SymPy (which also checks them against Lagrange's equations of $T_p - V_g$) and compiled with `lambdify`; the structural matrices are constant projections $\mathbf{T}^{\mathsf{T}}\mathbf{M}\mathbf{T}$, $\mathbf{T}^{\mathsf{T}}\mathbf{K}\mathbf{T}$, $\mathbf{T}^{\mathsf{T}}\mathbf{C}\mathbf{T}$ of the full FEM. In first-order form, with $\mathbf{x} = [\boldsymbol{\eta},\ \dot{\boldsymbol{\eta}}]$,

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\tau}) = \begin{bmatrix} \dot{\boldsymbol{\eta}} \\ \mathbf{M}(\boldsymbol{\eta})^{-1}\Big[\boldsymbol{\Phi}^{\mathsf{T}}\mathbf{B}\boldsymbol{\tau} - \tilde{\mathbf{D}}\dot{\boldsymbol{\eta}} - \tilde{\mathbf{K}}\boldsymbol{\eta} - \mathbf{E}_\eta^{\mathsf{T}}(\mathbf{C}_p\dot{\mathbf{q}}_p + \mathbf{g}_p)\Big] \end{bmatrix}, \tag{13.2}$$

where the inverse is never formed: the code solves $\mathbf{M}\ddot{\boldsymbol{\eta}} = \dots$ with `numpy.linalg.solve` at every right-hand-side call. The same code, with the identity as basis, is the full-DOF reference model used for validation.

13.3 Operating point and linearisation

An operating point is a steady tracking motion: $\theta_p = \theta_0$, rigid azimuth rotation at Ω_0 ($\dot{\eta}_{\psi 0} = \Omega_0$, all other velocities zero), $\ddot{\boldsymbol{\eta}} = \mathbf{0}$ and $\psi_e = 0$ (cyclic). With θ_p fixed the residual is affine in $\boldsymbol{\eta}$, so the static deformation and the nominal torques solve the square linear system

$$\begin{bmatrix} \tilde{\mathbf{K}} & -\boldsymbol{\Phi}^{\mathsf{T}}\mathbf{B} \\ \hat{\mathbf{c}}_P^{\mathsf{T}} & \mathbf{0} \\ \hat{\mathbf{c}}_e^{\mathsf{T}} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{\eta}_0 \\ \boldsymbol{\tau}_0 \end{bmatrix} = \begin{bmatrix} -\mathbf{r}_0 \\ \theta_0 \\ 0 \end{bmatrix}, \tag{13.3}$$

with $\mathbf{r}_0$ the payload and bearing forces of the steady motion, solved by `numpy.linalg.solve`; the code verifies that the residual is below 10^{-8} of its scale. The linear matrices are the closed form of Section 10, with the payload Jacobians of Section 9; the test suite checks them against finite differences of the nonlinear residual. The linear model is

$$\mathbf{M}_0\,\delta\ddot{\boldsymbol{\eta}} + \mathbf{C}_0\,\delta\dot{\boldsymbol{\eta}} + \mathbf{K}_0\,\delta\boldsymbol{\eta} = \boldsymbol{\Phi}^{\mathsf{T}}\mathbf{B}\,\delta\boldsymbol{\tau}, \qquad \delta\dot{\mathbf{x}} = \mathbf{A}\,\delta\mathbf{x} + \mathbf{B}\,\delta\boldsymbol{\tau}, \qquad \mathbf{y} = \mathbf{C}_q\,\delta\boldsymbol{\eta}, \tag{13.4}$$

with outputs ψ_e, θ_e (encoders), ψ_m, θ_m (motors), ψ_L, θ_p (true pointing), θ_r and the measurement errors $\psi_e - \psi_L$, $\theta_e - \theta_p$, all obtained from the rows of the modal basis at the corresponding nodes. Frequency responses, poles and zeros of this model are computed as in Section 7.

13.4 Time-domain simulation

Three scenarios are run on the default reduced model, the modal truncation of Section 10 with a single θ_{ref} (`reduction.modal.theta_ref_deg`), for which (10.3) is exact for any θ_p (Table 21). In S1 and S2 the system

starts from the equilibrium of the operating point with the nominal torque applied as a feedforward, so that the perturbation is only the doublet

$$\tau_\psi(t) = \tau_{\psi 0} + a\left[\,\chi_{[t_i,\,t_i+T)}(t) - \chi_{[t_i+T,\,t_i+2T)}(t)\,\right], \qquad a = 0.05\ \mathrm{N\,m},\ T = 20\ \mathrm{ms},\ t_i = 0.1\,T, \tag{13.5}$$

of zero mean so that it excites the dynamics without shifting the operating point. The nonlinear system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{\tau})$ and the linear one $\delta\dot{\mathbf{x}} = \mathbf{A}\delta\mathbf{x} + \mathbf{B}\delta\boldsymbol{\tau}$ are integrated with the same solver and compared as $\mathbf{q} - \mathbf{q}_0$ against $\delta\mathbf{q}$ (the rigid drift $\Omega_0 t$ of the ψ's is removed). In S3 all damping and torques are removed, the state starts from a rigid motion ($\dot{\psi}_i = \Omega_{max}$, $\dot{\theta}_i$ small, unstrained chains) and the conserved quantities $E = T + V$ and $p_\psi = \sum_i \partial\mathcal{L}/\partial\dot{\psi}_i$ are monitored.

Integrator. `scipy.integrate.solve_ivp` with method `DOP853`: an explicit Runge–Kutta scheme of order 8 (Dormand–Prince 8(5,3)) with adaptive step and embedded error control, relative tolerance 10^{-10} and absolute tolerance 10^{-12}. Since the doublet is discontinuous, the integration is split at t_i, $t_i + T$, $t_i + 2T$, so that the adaptive solver never steps across a jump (which would degrade its order). The step is limited by the highest natural frequency of the reduced chains, a few kHz, which sets the cost of a run (tens of seconds); the explicit method is adequate because the damping is light and the horizon short. The agreement is measured by $e(y) = \max_t |y_{nl}(t) - y_{lin}(t)| / \max_t |y_{lin}(t)|$ for each output.

Full-DOF reference. S1 is also run on the coupled model with all FEM degrees of freedom. Its modes reach hundreds of kHz, so the explicit DOP853 step is limited by stability and the reference is integrated with relative tolerance 10^{-8} and absolute tolerance 10^{-10}; the largest difference from the modal model on ψ_e, ψ_m, ψ_L is 0.00035 of the peak response.

Table 21: Simulation scenarios and results (relative errors of the nonlinear against the linearised response, and conservation drifts).

id	θ_0 [deg]	Ω_0 [deg/s]	forcing	result
S1	0	0	doublet on τ_ψ	$e(\psi_e) = 2.2 \times 10^{-7}$, $e(\psi_L) = 1.6 \times 10^{-10}$
S2	0	10	doublet on τ_ψ	$e(\psi_e) = 1.2 \times 10^{-6}$, $e(\psi_L) = 9.9 \times 10^{-9}$
S3	0	10	conservation, $\mathbf{C} = \mathbf{0}$, $\boldsymbol{\tau} = \mathbf{0}$	$\lvert\Delta E\rvert/E = 1.8 \times 10^{-14}$, $\lvert\Delta p_\psi\rvert/p_\psi = 2.5 \times 10^{-12}$

In S1 the polar channels respond only through the products of inertia, so their linear response is of the same order as the quadratic term $\frac{1}{2}J'_{\psi,p}\dot{\psi}^2$ that the linearisation drops: their relative error is therefore not an index of the quality of the linearisation, and only the azimuth channels are used for it. Their amplitude (about 10^{-12} rad) is also comparable with the absolute tolerance of the integrator, so the isolated spikes visible on the nonlinear polar curves are integration noise, not dynamics.

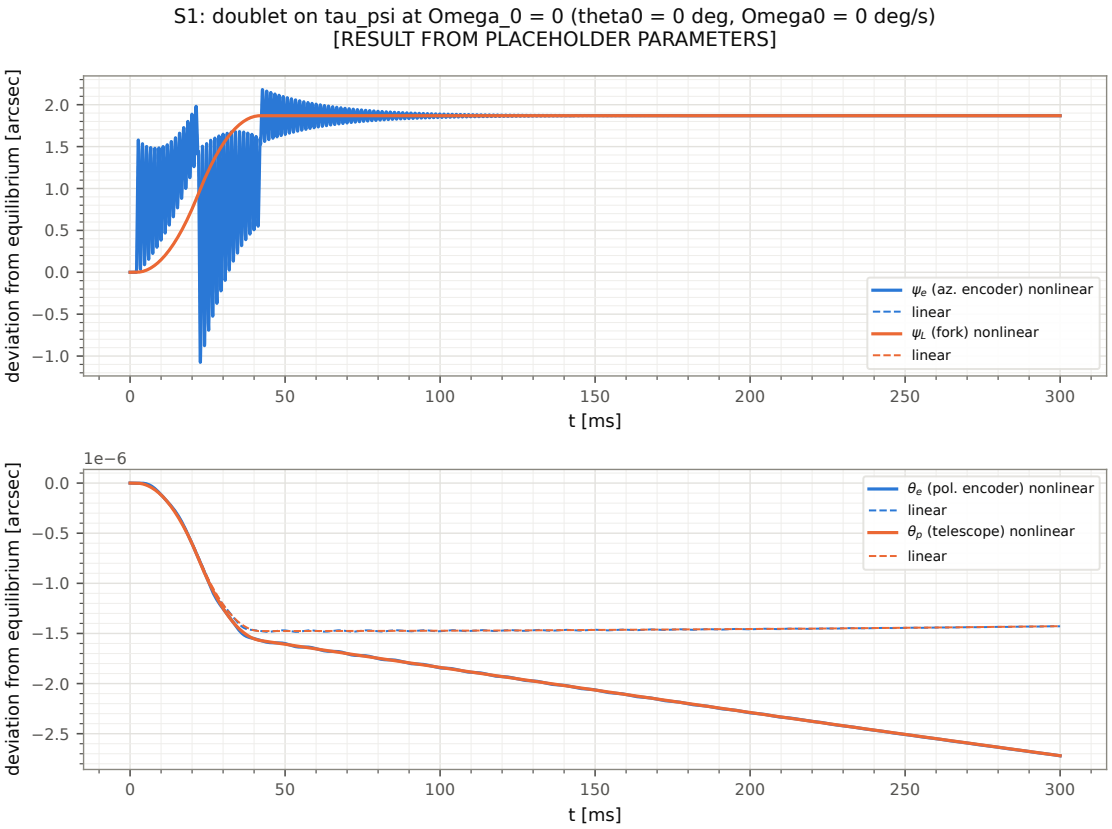


Figure 23: S1: nonlinear (solid) and linearised (dashed) response to the τ_ψ doublet at $\Omega_0 = 0$. *Result from placeholder parameters.*

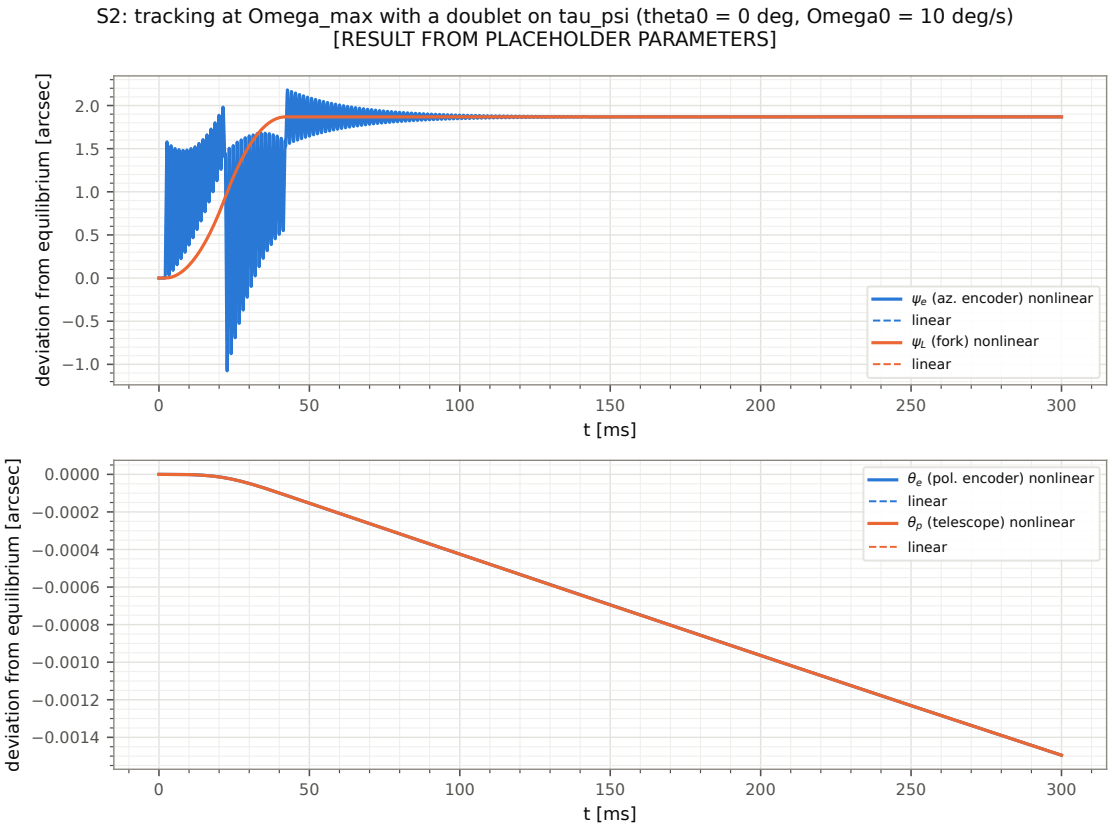


Figure 24: S2: the same doublet while tracking at Ω_{max}; the cross response on θ_e and θ_p is the gyroscopic coupling. *Result from placeholder parameters.*

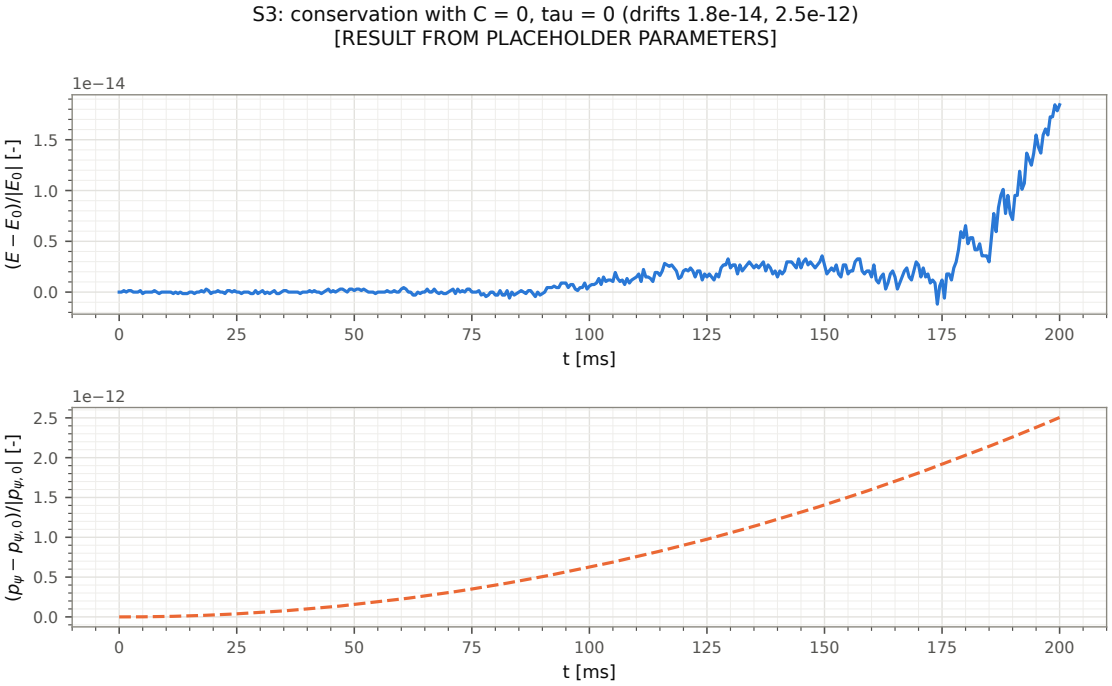


Figure 25: S3: drift of $E = T + V$ and p_ψ in the conservative case. *Result from placeholder parameters.*

13.5 Software

Table 22: Software used to produce this document.

Python	3.12.3
NumPy	2.5.4
SciPy	1.18.1 (LAPACK through `scipy.linalg`)
SymPy	1.14.0
Matplotlib, PyYAML, pytest	figures, input parsing, test suite
`pdflatex`	this document (TikZ for the drawings)

All results are reproducible with `python run_all.py`; the symbolic derivation is deterministic and the numerical tolerances above are fixed in `simulate.py`.

14 Assumptions and placeholders

code	status	description
`Q6_MATERIALI`	placeholder	Q6: placeholder materials ['steel', 'aluminium'] (steel: rho = 8000 kg/mˆ3, G = 77 GPa, aluminium: rho = 2810 kg/mˆ3, G = 26.9 GPa).
`TELESCOPIO_SENZA_ALBERO`	note	Polar element 3 in mode: telescope: no shaft under the telescope. In the hub model its nodes 3 and 4 coincide with the hub centre H; the telescope is the node P attached to H by k_b, and its inertia enters only through the payload section.
`MOZZO_TELESCOPIO`	note	Hub model (polar.hub.enabled: true): the hub (two side hubs, tube + flange, and the support ring) replaces polar elements [2, 4] and its centre H takes the nodes of element 3; the telescope is a node P off the chain, attached to H by k_b = 9800 N m/rad.
`MOZZO_K_B`	placeholder	k_b = 9800 N m/rad (telescope to hub) is a placeholder (placeholder: two-mass estimate from the antiresonance (18.1 Hz) and resonance (~48 Hz) measured at the elevation encoder): it sets the antiresonance at the encoder.
`MOZZO_INERZIA_AZIMUT`	placeholder	hub.inertia.in_azimuth_load = false: the inertia of the hub about z_hat is added to Jˆax. Check whether the azimuth load exported from Onshape already contains the hub parts.
`NODI_SENZA_INERZIA_AZIMUT`	placeholder	Azimuth axis: no concentrated inertia at encoder (node 1), motor (node 5) (rotor/disc missing): those nodes carry only the shaft mass and are almost inertia-free. This affects the zeros and the high modes (typically strongly) and the choice of the truncation modes.
`Q5_GIUNZIONI_AZIMUT`	placeholder	Q5: azimuth axis without joint compliances (shrink fits, keys, flanges): the stiffnesses are those of the shafts alone and probably overestimate the real one.
`Q8_CUSCINETTI_AZIMUT`	placeholder	Q8: zero viscous bearing friction on the azimuth axis.
`NODI_SENZA_INERZIA_POLARE`	placeholder	Polar axis: no concentrated inertia at encoder (node 1), motor (node 7) (node_inertias entirely commented out): those nodes carry only the shaft mass and are almost inertia-free. This affects the zeros and the high modes (typically strongly) and the choice of the truncation modes.

code	status	description
Q5_GIUNZIONI_POLARE	placeholder	Q5: polar axis without joint compliances (shrink fits, keys, flanges): the stiffnesses are those of the shafts alone and probably overestimate the real one.
Q7_POSIZIONE_POLARE	note	Q7: y_node1_from_o2_mm = null. The contribution of the polar shaft to the azimuth inertia (J^ax) is computed from the transverse moments of the parts about their own centres only, without Steiner terms about o_2. Irrelevant: the azimuth load includes the polar shafts (includes_polar_shafts: true), so J^ax is not added.
Q8_CUSCINETTI_POLARE	placeholder	Q8: zero viscous bearing friction on the polar axis.
TERNE_ORIGINI	note	Origins on z_hat: dist(O_N, o_1) = 59.695 mm, dist(O_N, o_2) = 548.865 mm (o_2 = centre of rotation of the polar shaft). O_N is the origin of the Onshape assembly (centre of the base of the azimuth shaft) and x_hat, y_hat, z_hat are its axes.
RIDUZIONE_MODALE	note	Reduction: Gawronski modal truncation of the full FEM of each axis (band 2000 Hz, cut-off 4000 Hz, static correction: none).
Q8_SMORZAMENTO	placeholder	Q8: stiffness-proportional damping, C = beta K, with zeta = 0.01 on the first flexible mode of each axis. Placeholder: the height of the FRF peaks and notches is not representative.
Q11_OMEGA_MAX	placeholder	Q11: tracking rates [0.0, 10.0] deg/s; Omega_max = 10 deg/s placeholder. It sets the gyroscopic coupling.
Q12_FORCELLA_RIGIDA	note	Q12: rigid fork between azimuth node 9 and the polar bearings. The reaction of tau_theta does no work on psi_L (section 9.4).
MISURA_THETA0_3	placeholder	Elevation_BodeMeasurement_bode_New_16_10_26.csv: elevation during the test not recorded in the file; theta0 = 0 deg assumed.
MISURE_SPERIMENTALI	note	Comparison with 5 plant measurements (Kollmorgen WorkBench, LucyIdentification); K_t = 0.258 N m/A. The Validation folder is not used.
JAX_NEL_CARICO	note	The azimuth load 'load (fork + polar shafts)' (node 9) includes the polar shafts (includes_polar_shafts: true): J^ax = 8.3476e-04 kg m^2 computed from the polar elements is not added, because their moment about the azimuth axis, with the real distances from the axis, is already in the CAD value.
PAYLOAD_SEGNO_PRODOTTI	placeholder	payload.products_sign = 'tensor': the sign convention of the Onshape products of inertia is not verified on a test body; it affects the products I_IJ, I_IK, I_JK, not the diagonal moments.
Q3_TELESCOPIO	note	Q3: telescope from the tensor entered in config.yaml (frame: cad_cg, Onshape, Telescopio LAMERA <1>, without mate connector), without subtractions (subtract_shaft_parts: false): the tensor must contain the telescope only, not the parts of the polar axis already modelled in the elements.